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Suiter et al.

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(54) **AIRCRAFT FLIGHT OPERATIONS
DATALINK**

(2025.01); **G08G 5/58** (2025.01); **G08G 5/76**
(2025.01); **G08G 5/26** (2025.01)

(71) Applicant: **Otto Aero Company**, Omaha, NE (US)

(58) **Field of Classification Search**

CPC .. **G08G 5/0056**; **G08G 5/0025**; **G08G 5/0039**;
G08G 5/0013

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(US)

See application file for complete search history.

(73) Assignee: **Otto Aero Company**, Omaha, NE (US)

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

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(21) Appl. No.: **18/203,402**

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(22) Filed: **May 30, 2023**

Flying, Issue 942, Oct. 2023 "We Fly: GARMIN Autoland for the
Beechcraft King Air 200" pp. 37-47.

(Continued)

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(63) Continuation of application No. 17/544,645, filed on
Dec. 7, 2021, which is a continuation of application
No. 17/492,448, filed on Oct. 1, 2021, now Pat. No.
11,657,721, and a continuation of application No.
17/478,807, filed on Sep. 17, 2021, now Pat. No.
11,551,564, and a continuation of application No.
(Continued)

Primary Examiner — Imran K Mustafa

(74) *Attorney, Agent, or Firm* — Suiter Swantz IP

(51) **Int. Cl.**

G08G 5/55 (2025.01)
B64D 45/08 (2006.01)
G08G 5/21 (2025.01)
G08G 5/26 (2025.01)
G08G 5/30 (2025.01)
G08G 5/34 (2025.01)
G08G 5/54 (2025.01)
G08G 5/58 (2025.01)
G08G 5/76 (2025.01)

(52) **U.S. Cl.**

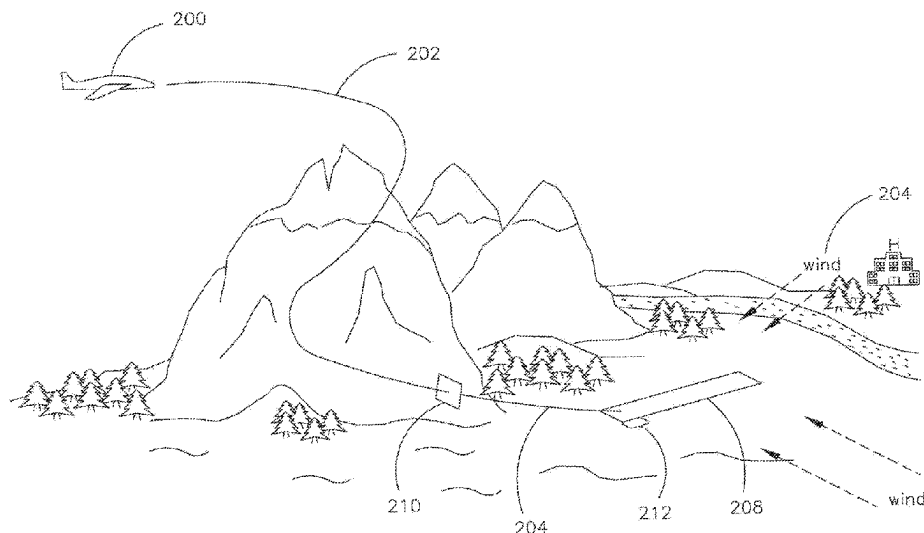
CPC **G08G 5/55** (2025.01); **B64D 45/08**
(2013.01); **G08G 5/21** (2025.01); **G08G 5/30**
(2025.01); **G08G 5/34** (2025.01); **G08G 5/54**

(57)

ABSTRACT

Aircraft flight operations datalink for preparing an aircraft
for flight, by a particular pilot (crew) for a particular flight
plan via a portable computing device. The aircraft flight
operations datalink includes a wireless data link; computer
processor; data bus; an association with a flight sensor for
collecting aircraft position, weather, traffic, terrain, aircraft
systems status, aircraft flight envelope parameters, pilot
(crew) status/condition; a display system; a ground-to-air
data link; and a database of aircraft and pilot data; a flight
plan; and flight conditions. In flight operations, the datalink
may be utilized to relieve a pilot (crew) from performing
tasks necessary to mitigate an emergency or reduce a flight
operation urgency.

30 Claims, 50 Drawing Sheets



Related U.S. Application Data

17/478,821, filed on Sep. 17, 2021, now Pat. No. 11,699,351, said application No. 17/492,448 is a continuation of application No. 16/673,725, filed on Nov. 4, 2019, said application No. 17/478,821 is a continuation of application No. 16/673,725, filed on Nov. 4, 2019, said application No. 17/478,807 is a continuation of application No. 16/673,725, filed on Nov. 4, 2019, which is a continuation of application No. 15/152,437, filed on May 11, 2016, now Pat. No. 10,467,913, said application No. 17/492,448 is a continuation of application No. 14/741,283, filed on Jun. 16, 2015, now Pat. No. 9,310,222, said application No. 15/152,437 is a continuation of application No. 14/142,390, filed on Dec. 27, 2013, now abandoned, and a continuation of application No. 13/831,398, filed on Mar. 14, 2013, now abandoned.

- (60) Provisional application No. 62/024,723, filed on Jul. 15, 2014, provisional application No. 62/012,734, filed on Jun. 16, 2014, provisional application No. 61/900,199, filed on Nov. 5, 2013, provisional application No. 61/870,125, filed on Aug. 26, 2013, provisional application No. 61/754,522, filed on Jan. 18, 2013, provisional application No. 61/750,286, filed on Jan. 8, 2013, provisional application No. 61/747,051, filed on Dec. 28, 2012.

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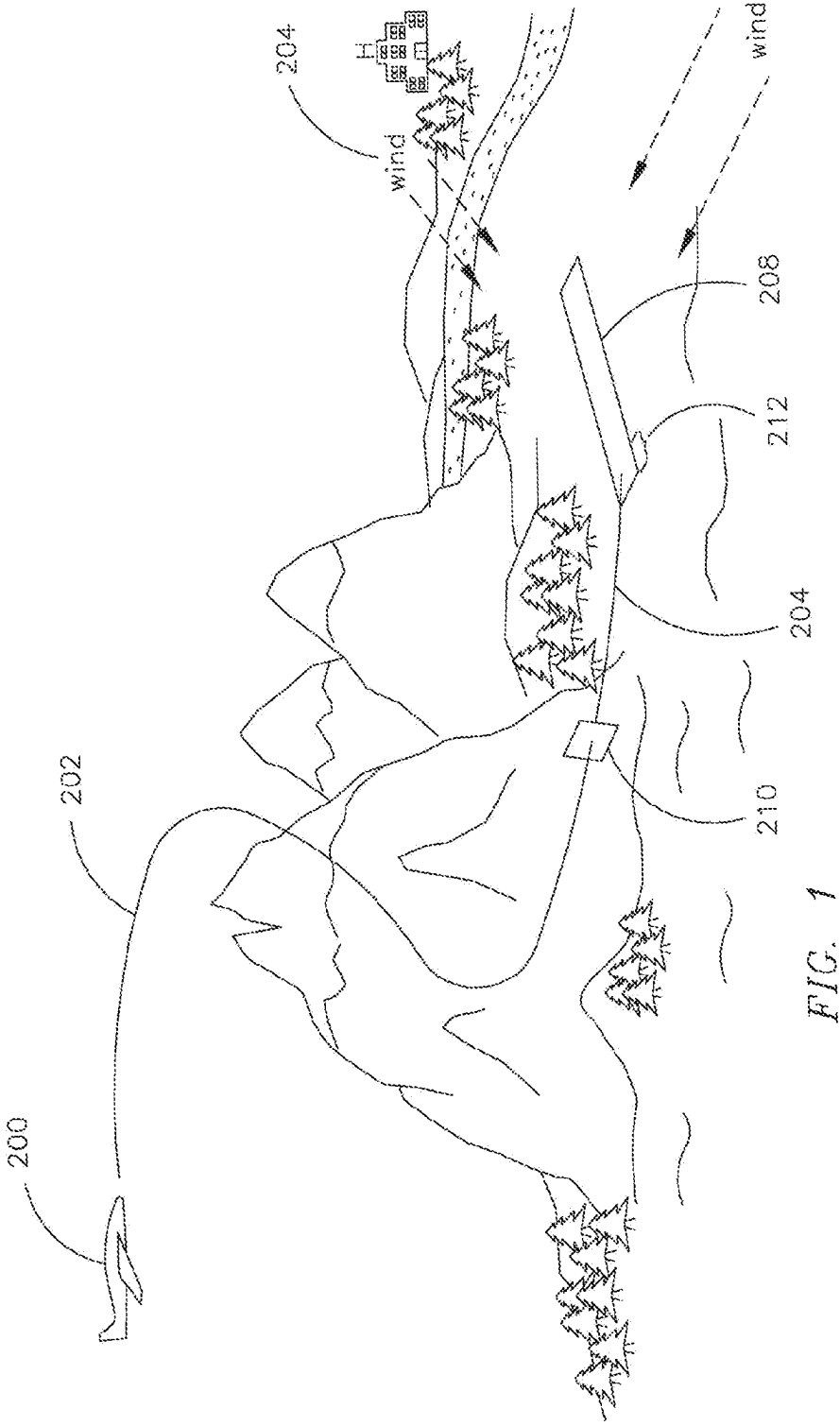
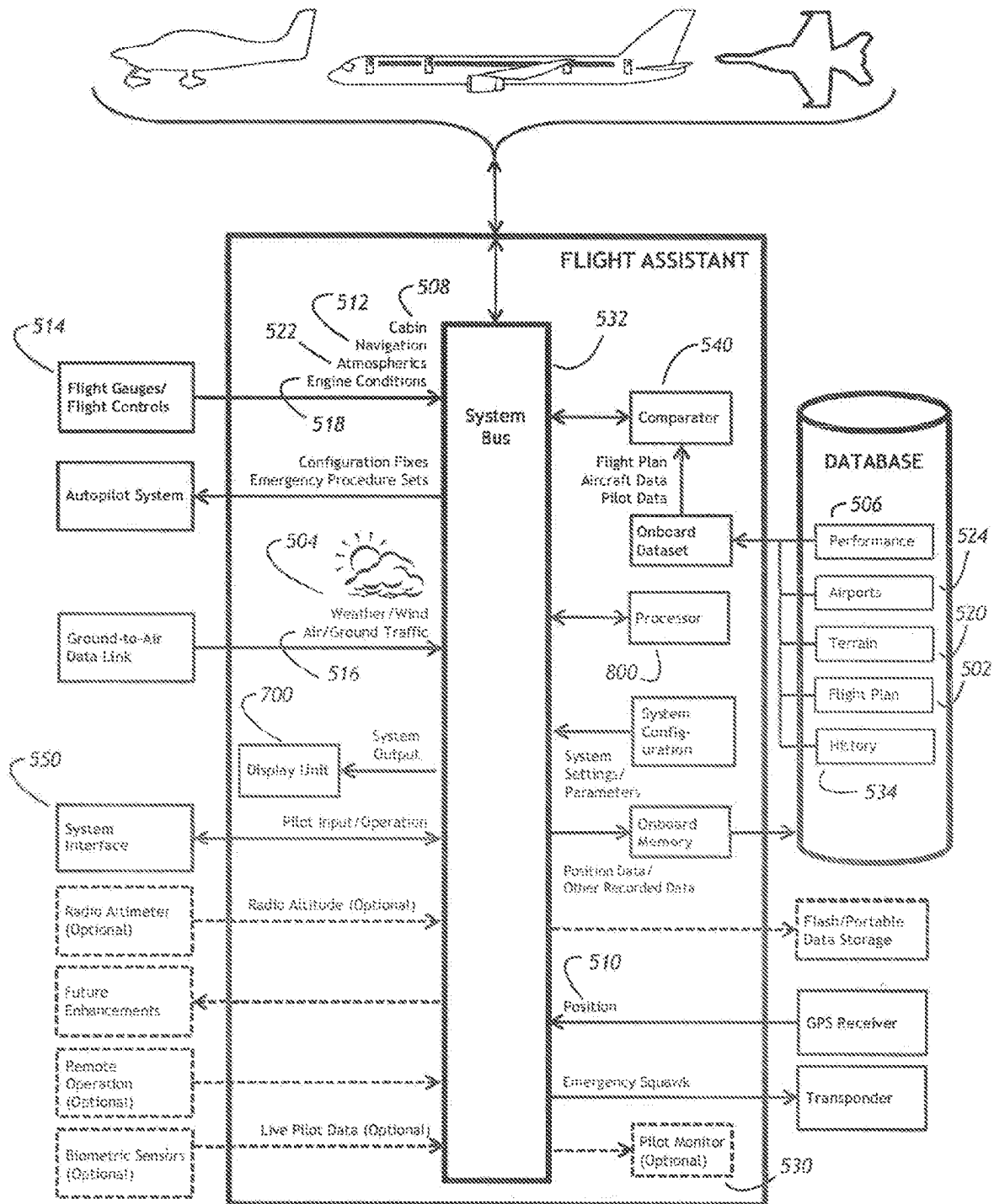


FIG. 2A



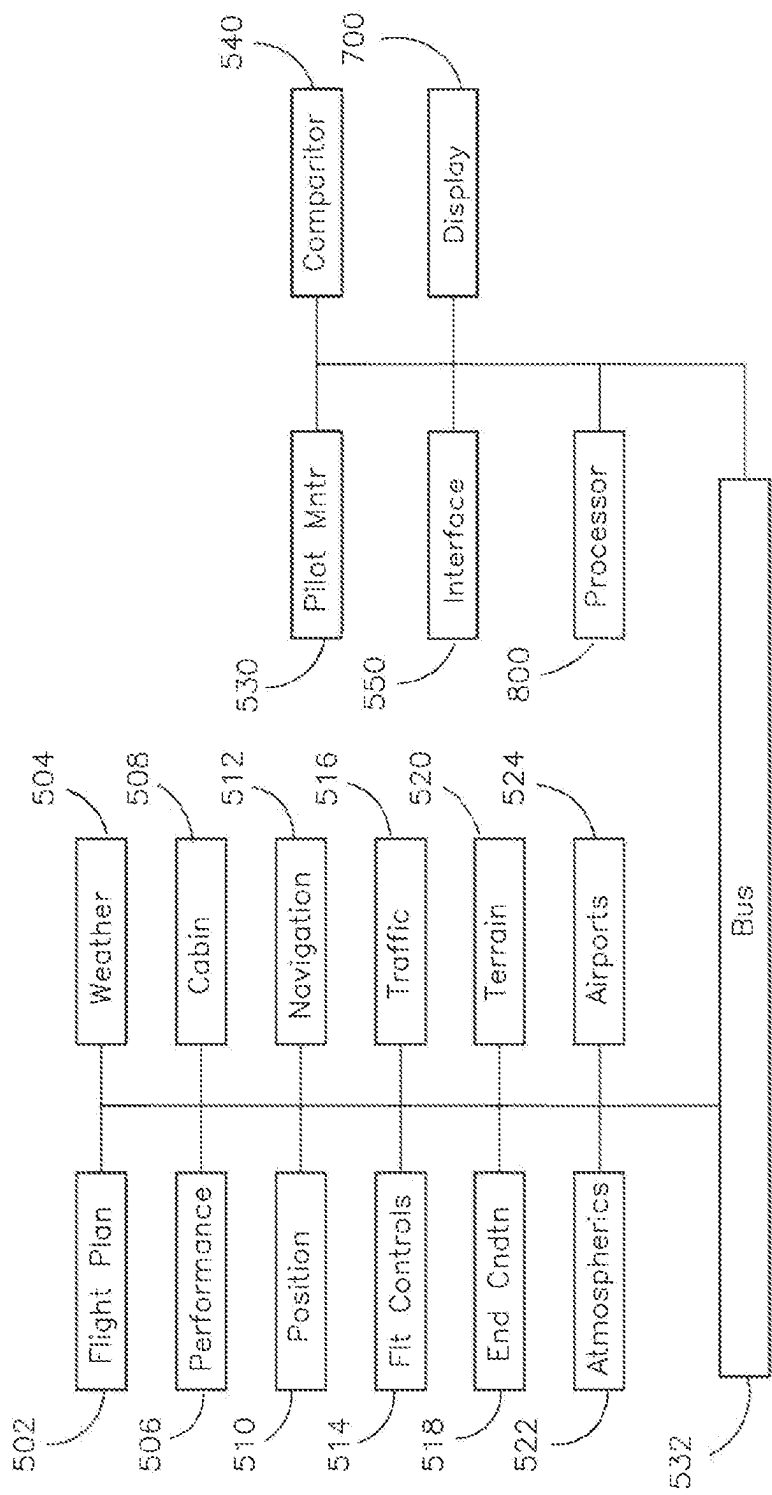


FIG. 2B

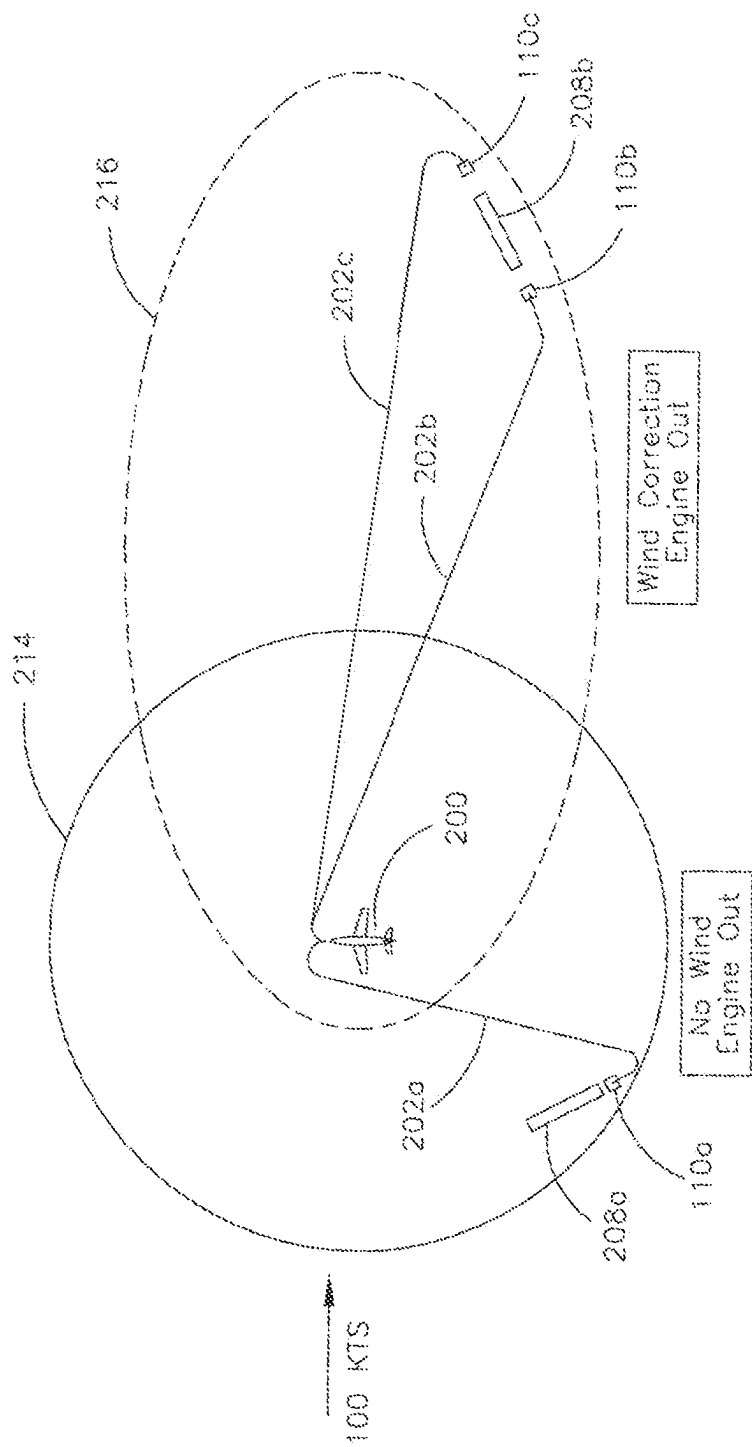


FIG. 3

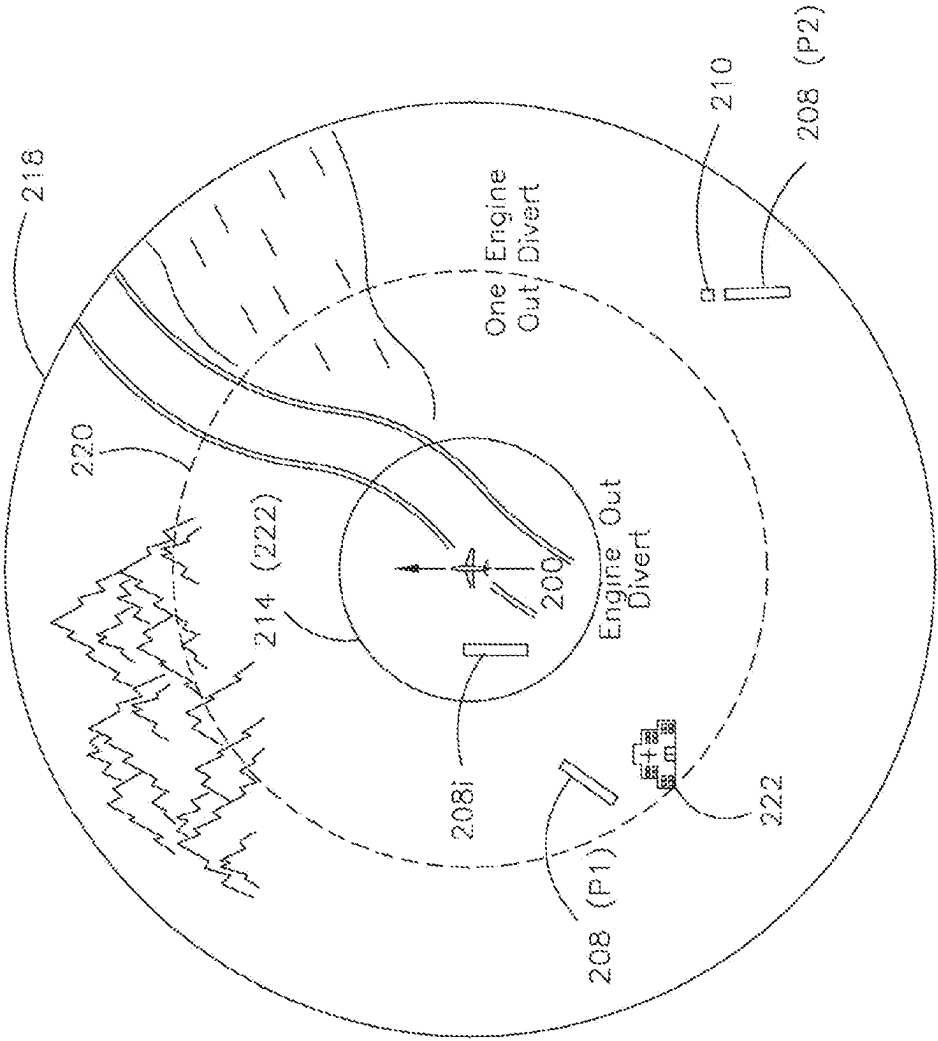
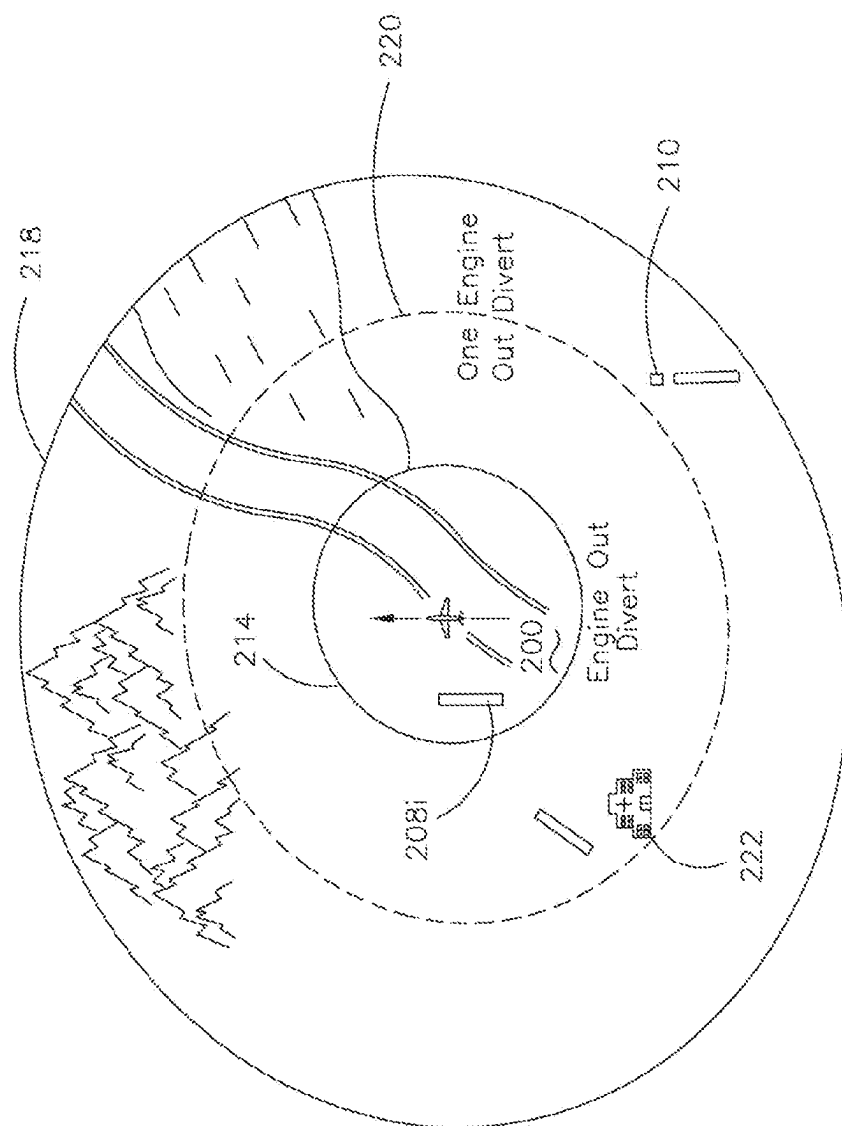


FIG. 4A



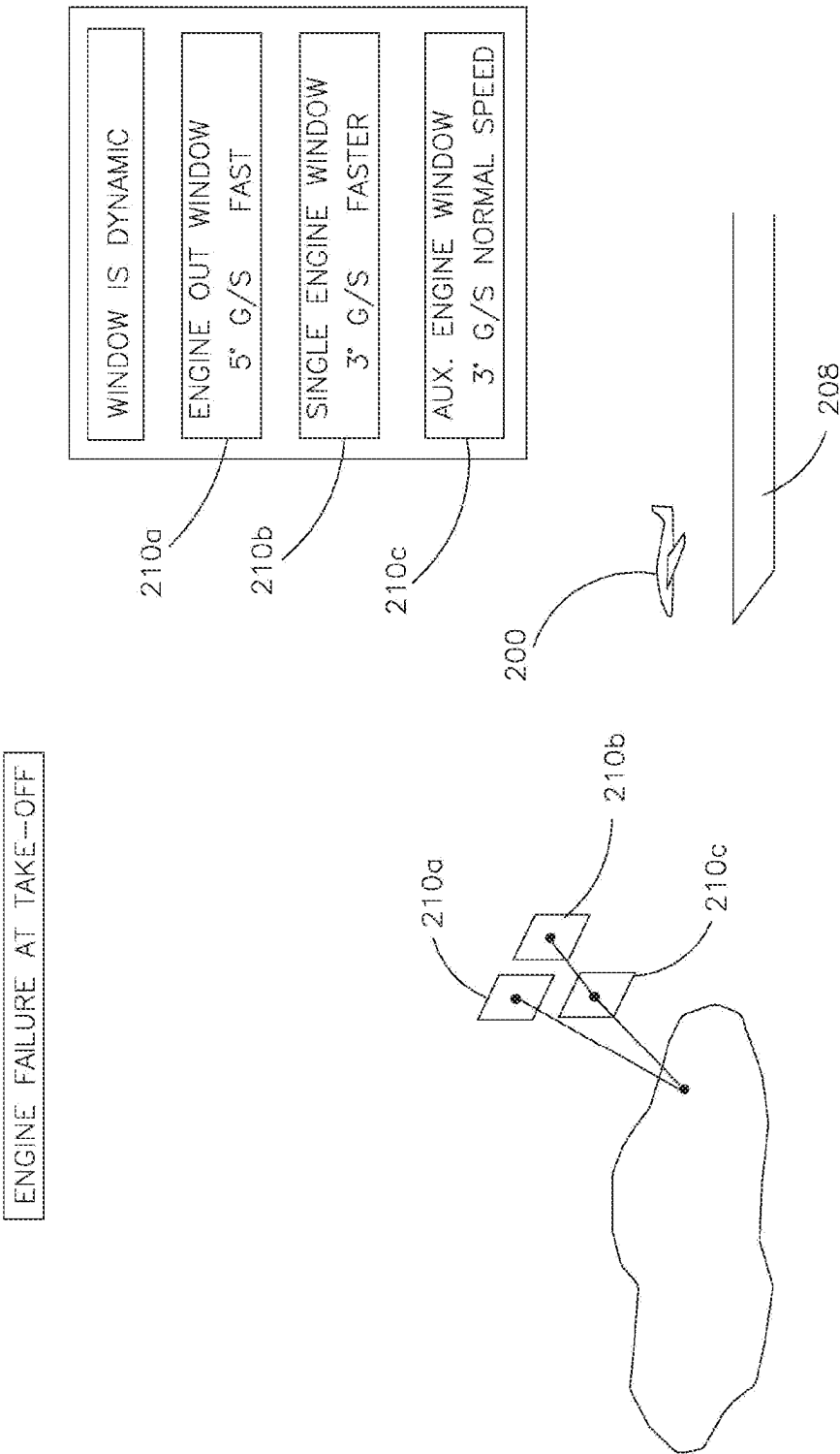


FIG. 5

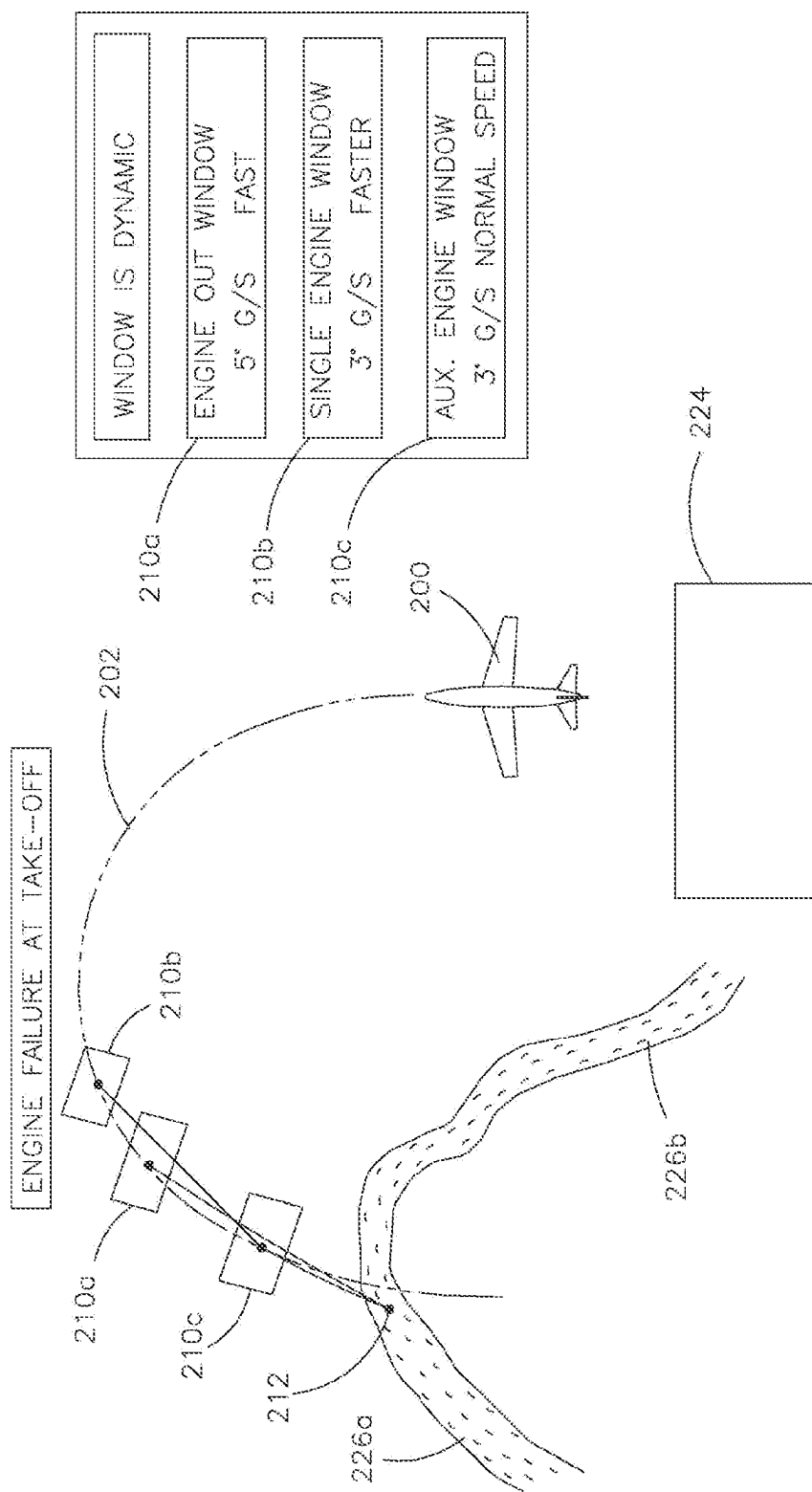


FIG. 6

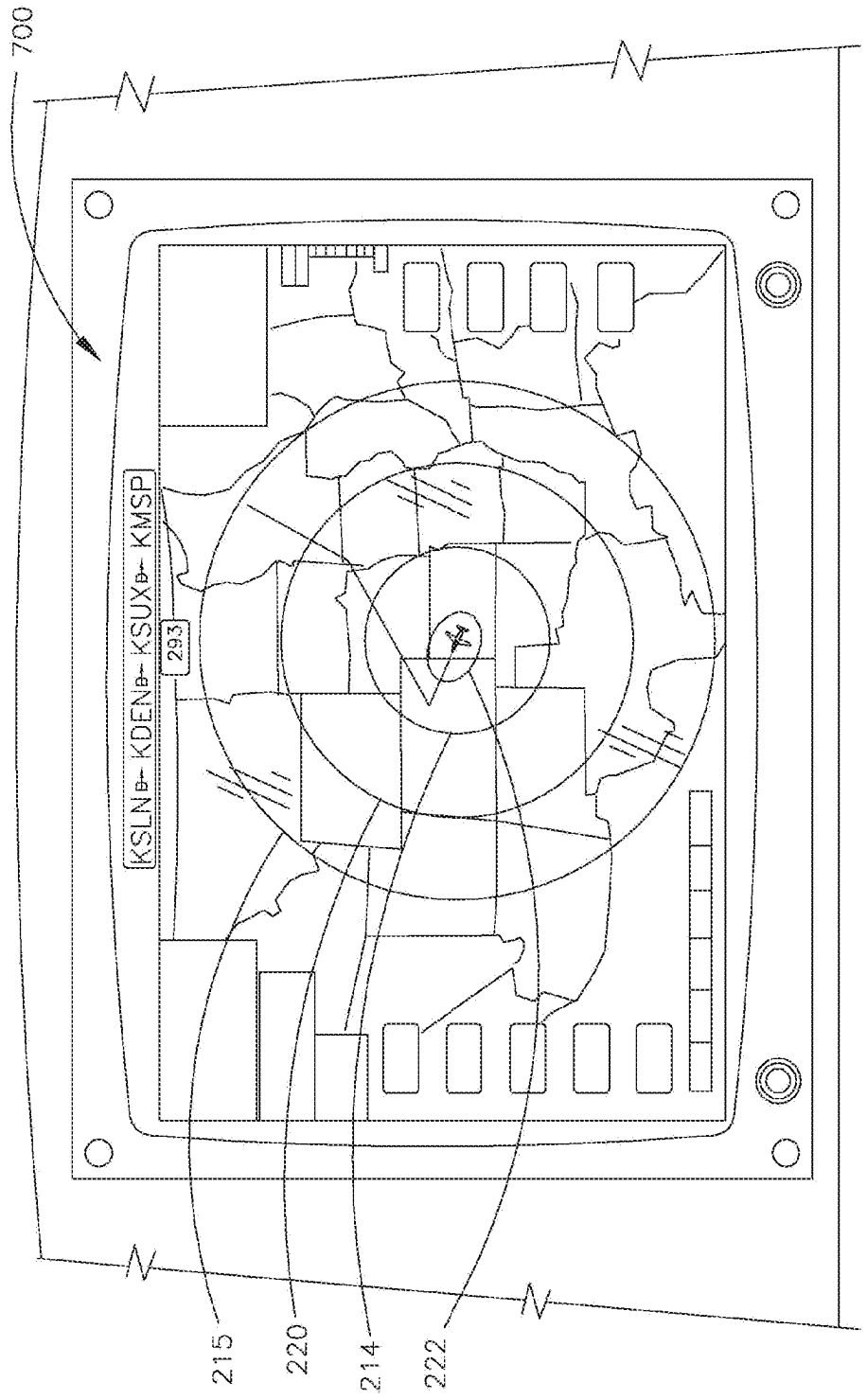


FIG. 7

FIG. 8A

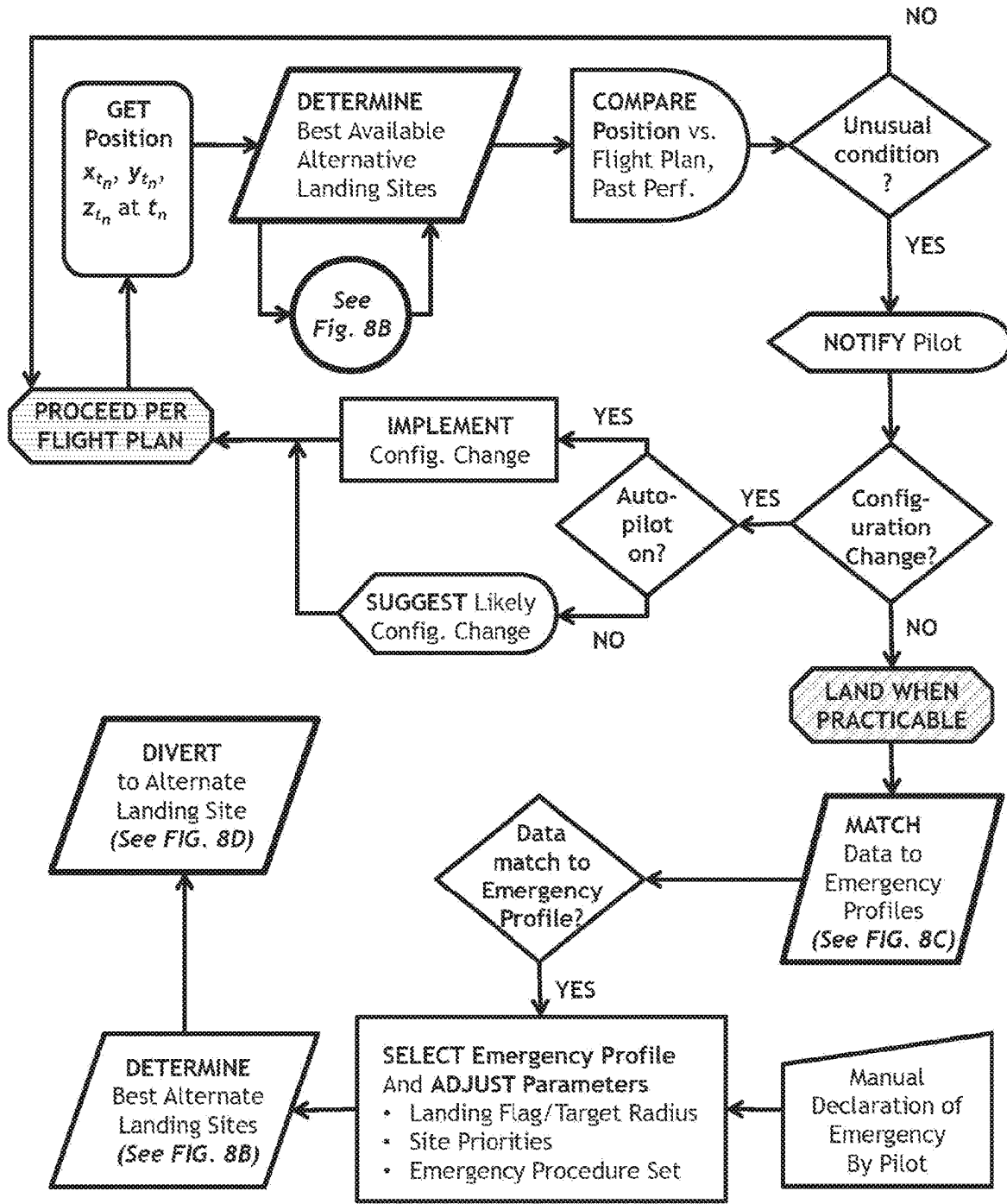


FIG. 8B

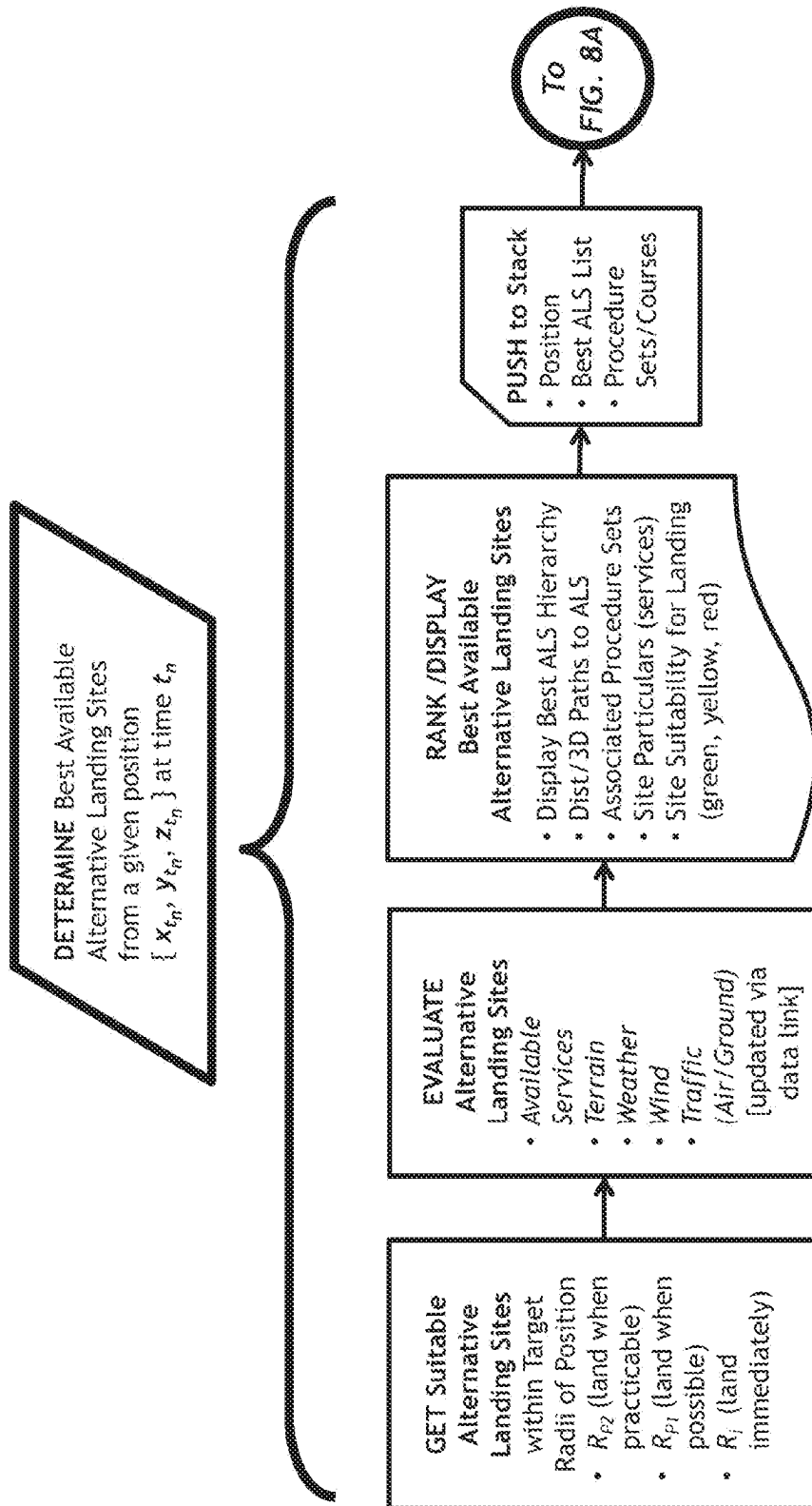
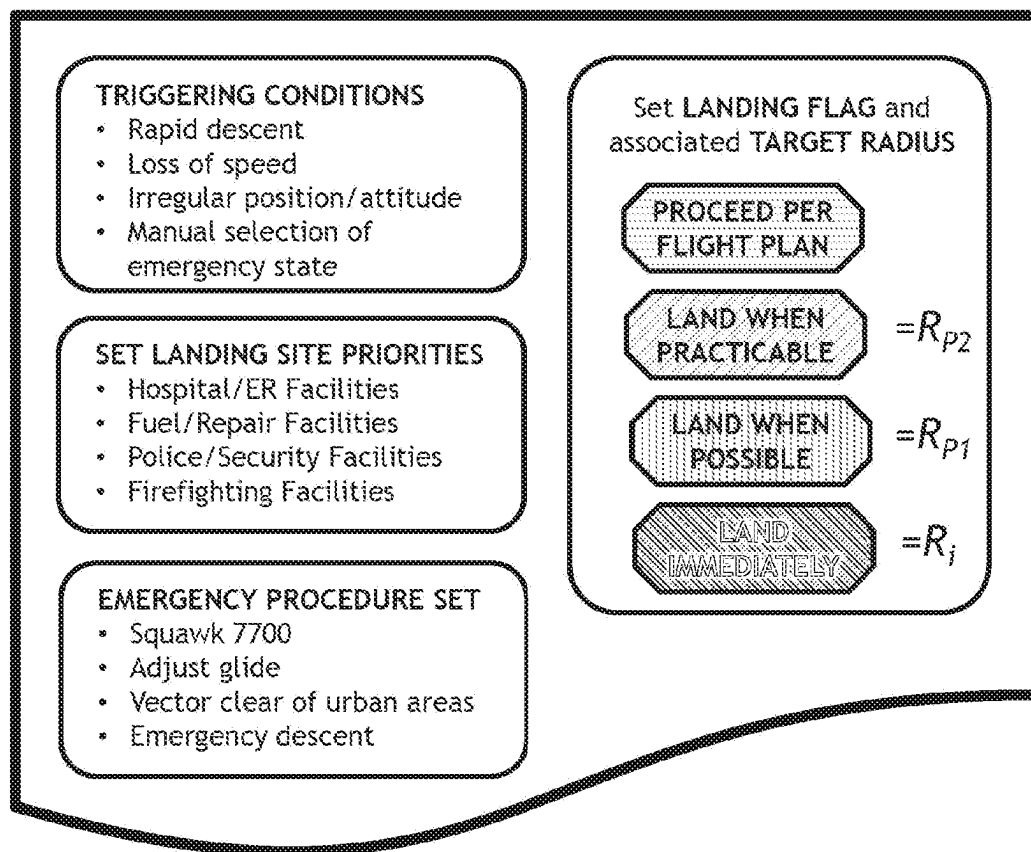


FIG. 8C

EMERGENCY PROFILE



- | | |
|--------------|-----------------|
| ▪ 1 ENG | ▪ WEATHER/OTHER |
| ▪ MEDICAL | ▪ FIRE/SMOKE |
| ▪ THREAT | ▪ PRESSURE |
| ▪ AUX ENGINE | ▪ SYS FAIL |
| ▪ ENG OUT | |

FIG. 8D

DIVERT OPTION

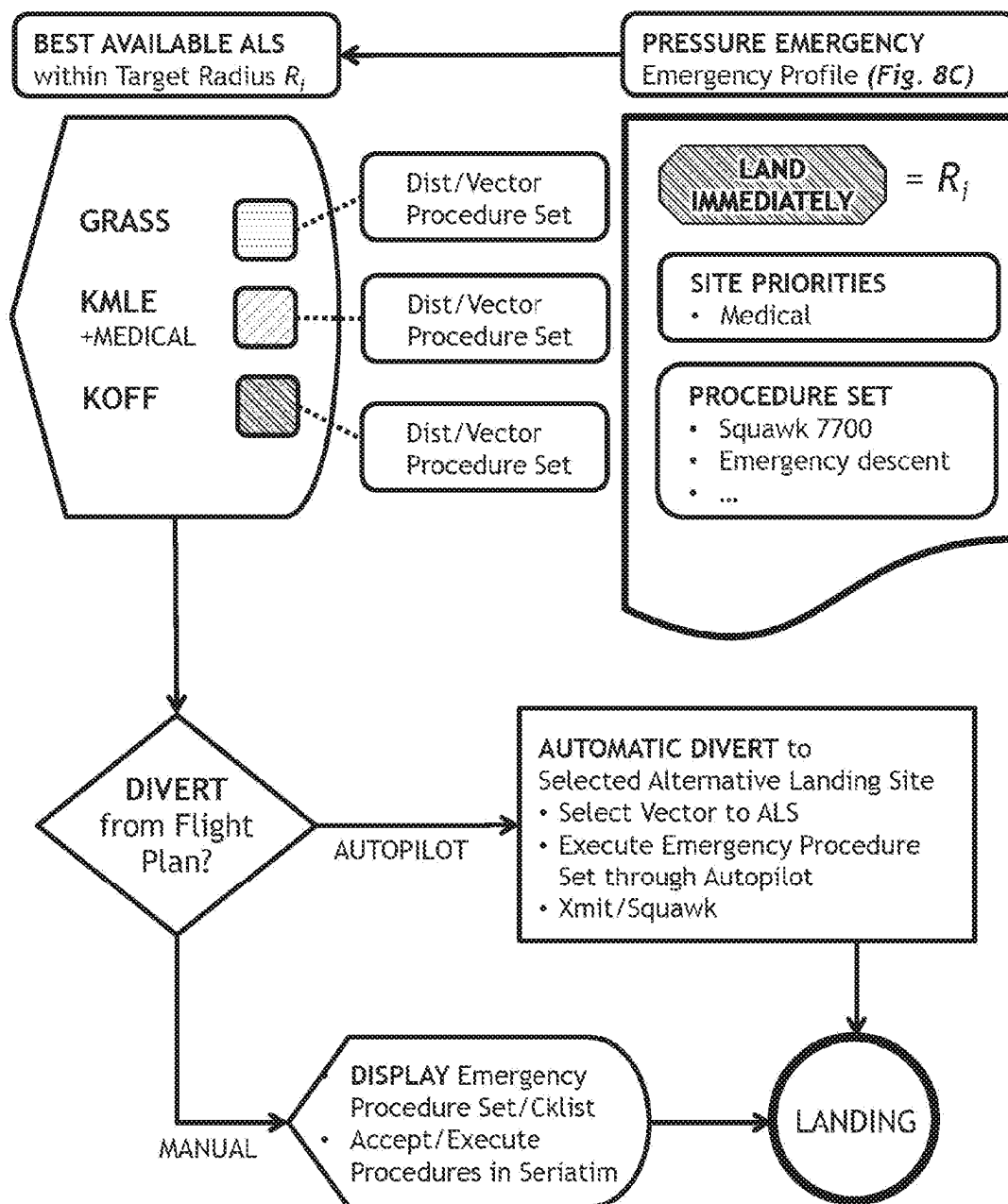


FIG. 8E

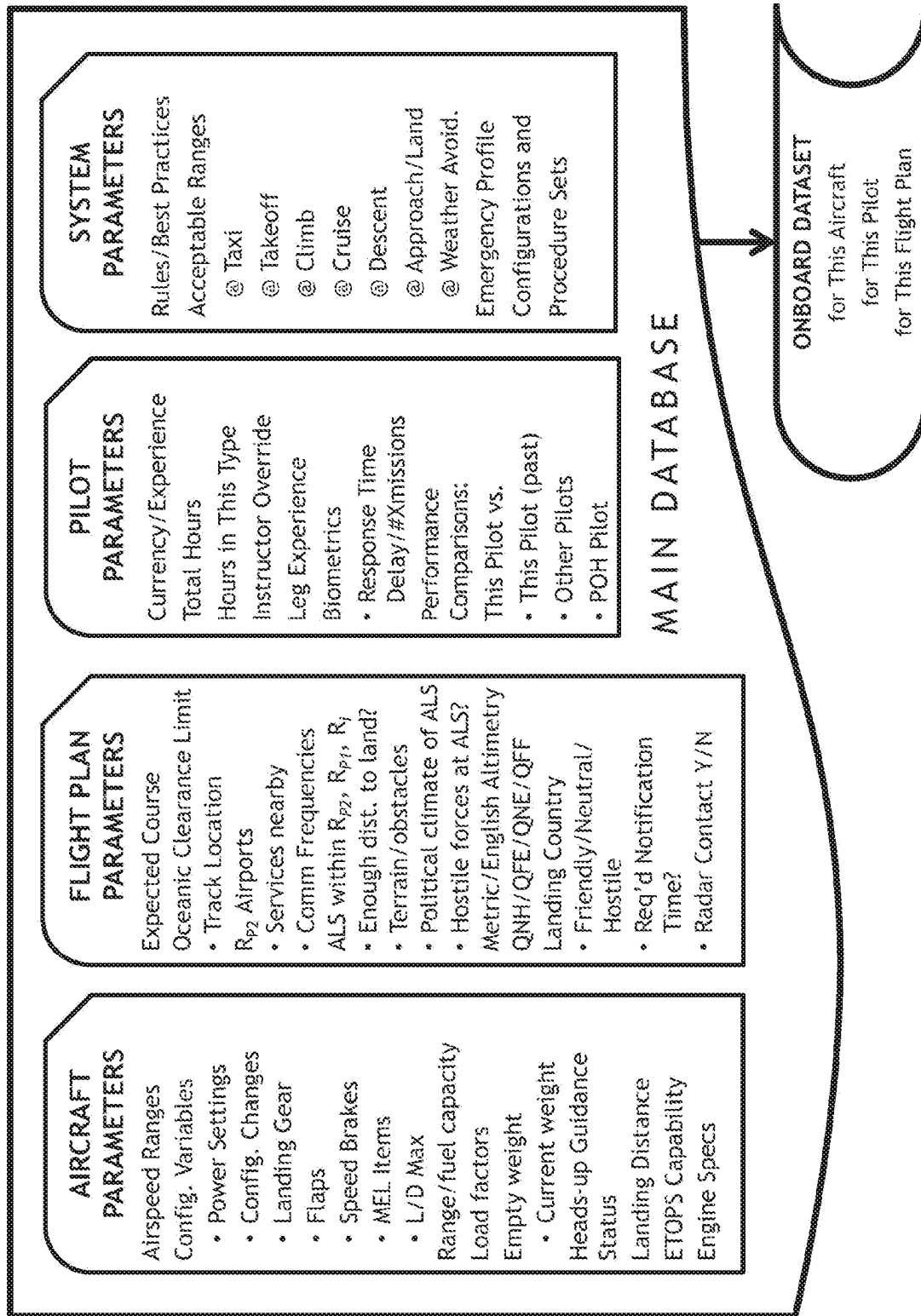


FIG. 9A

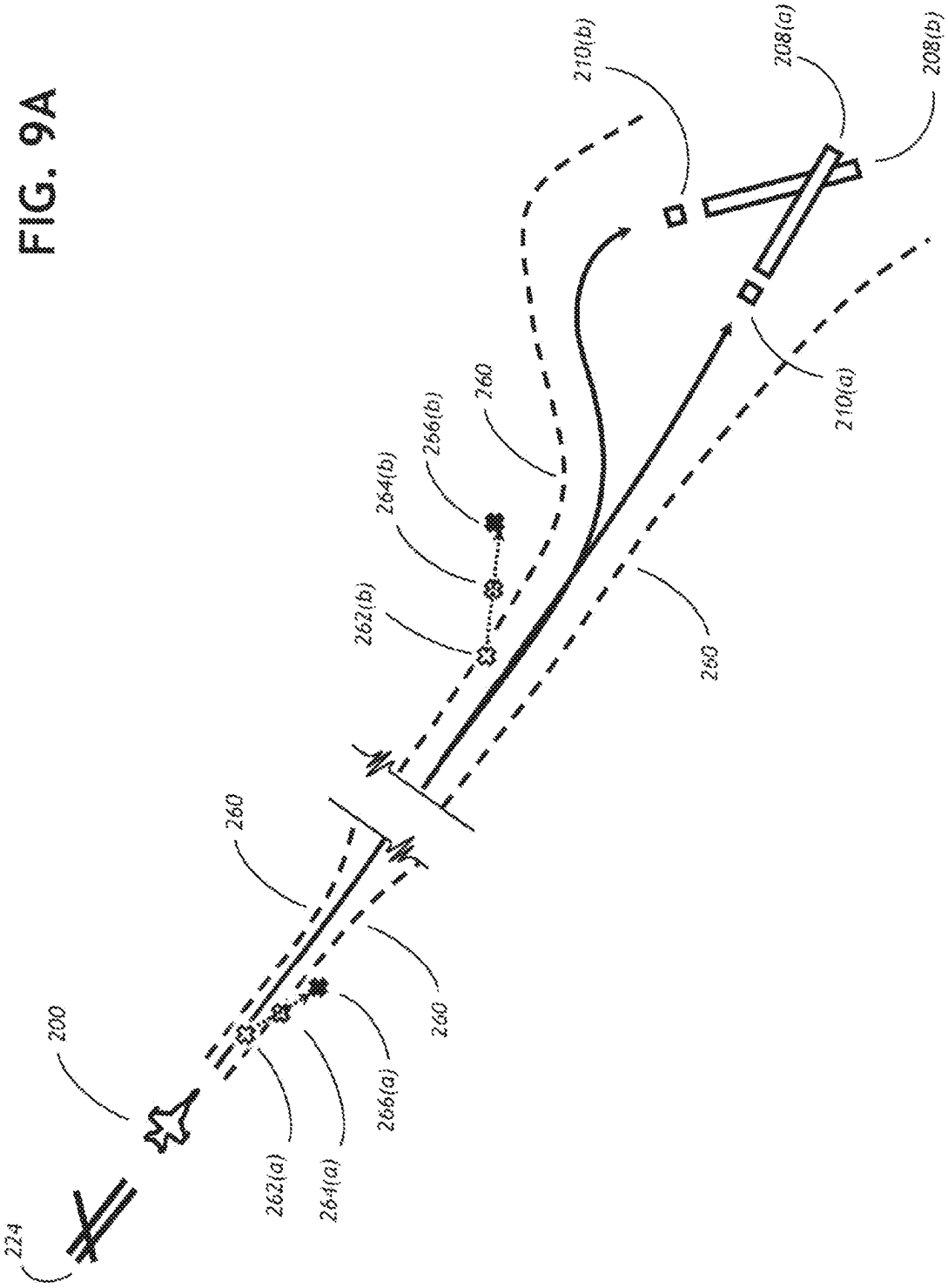
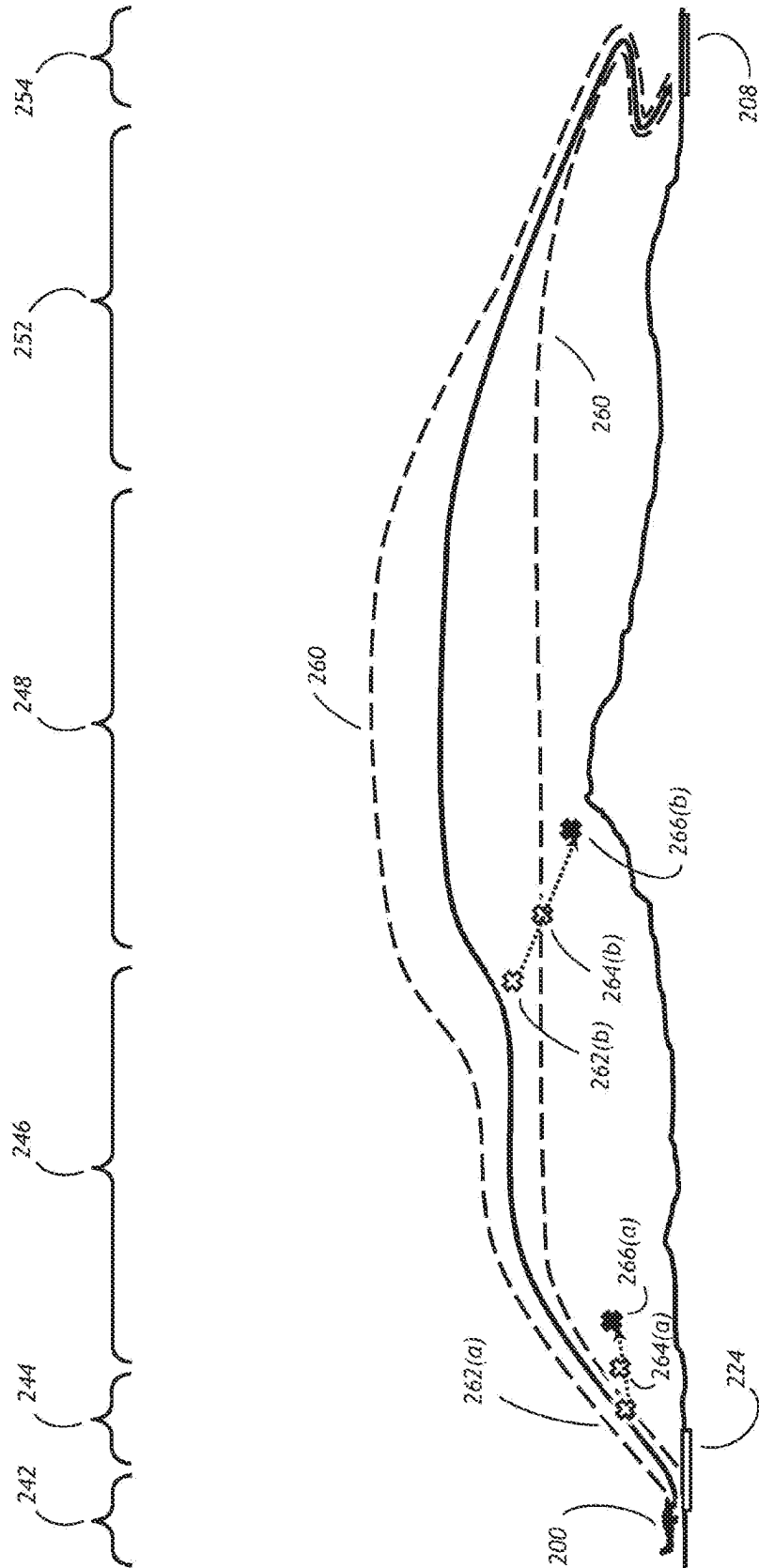


FIG. 9B



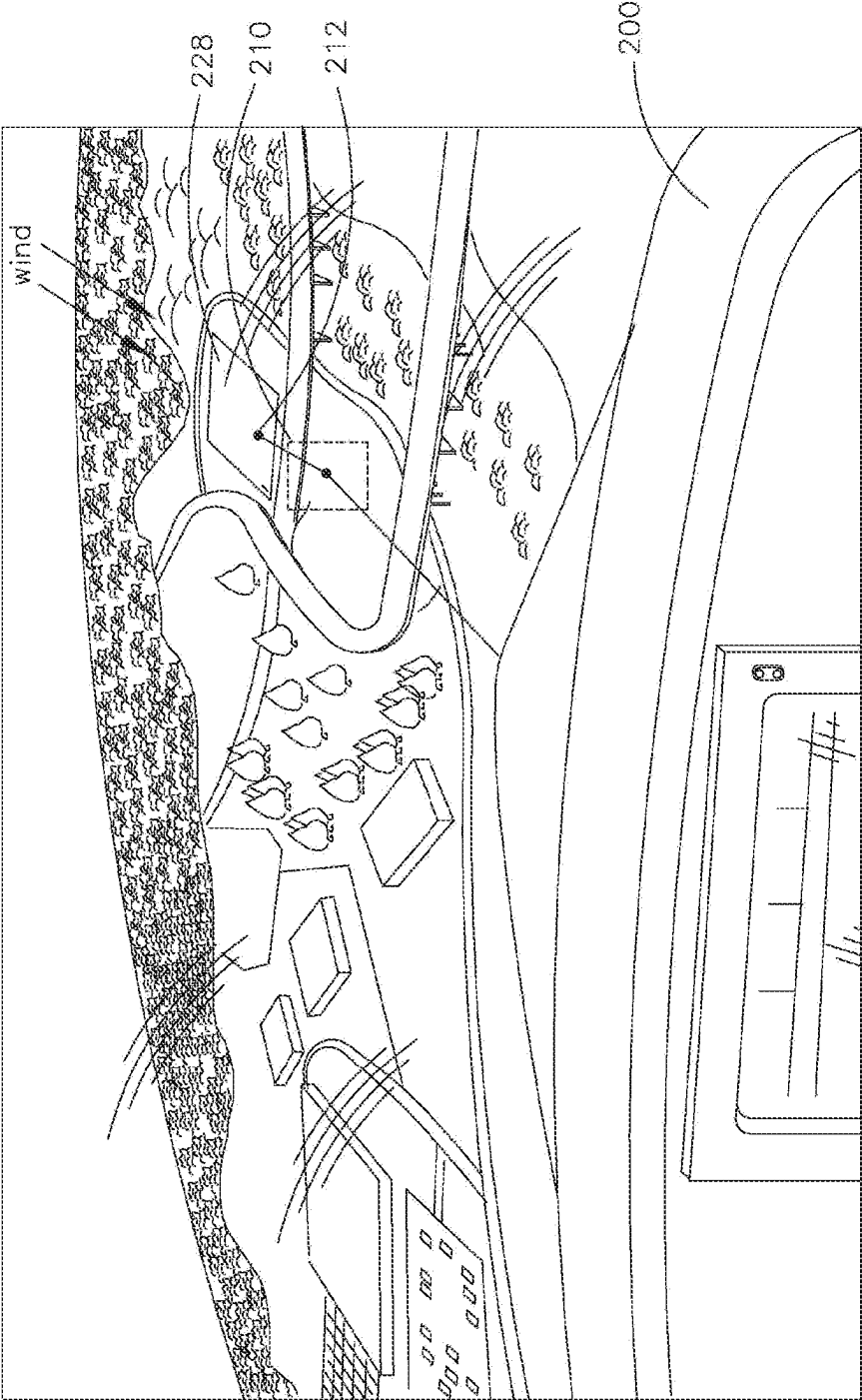


FIG. 10

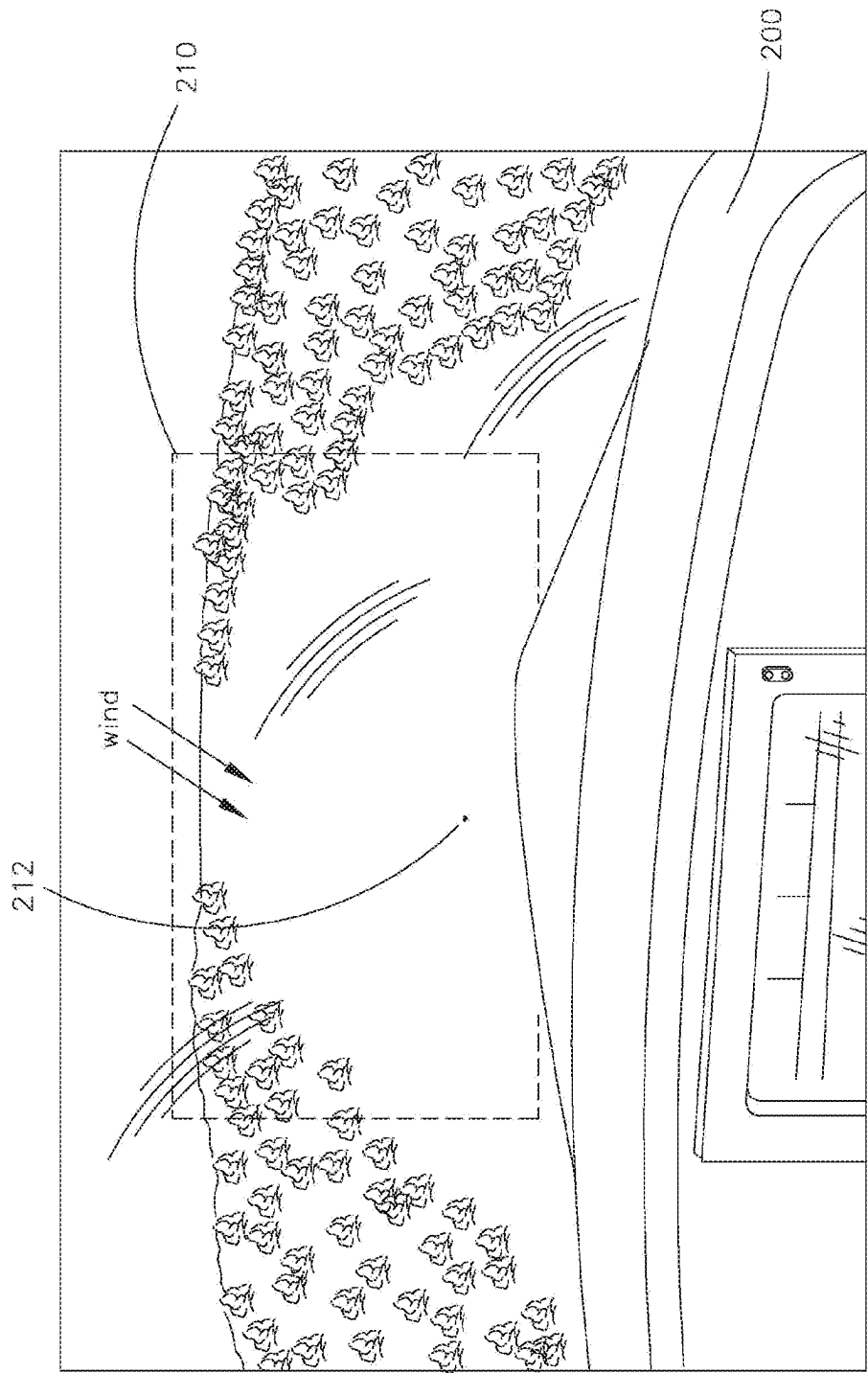


FIG. 11

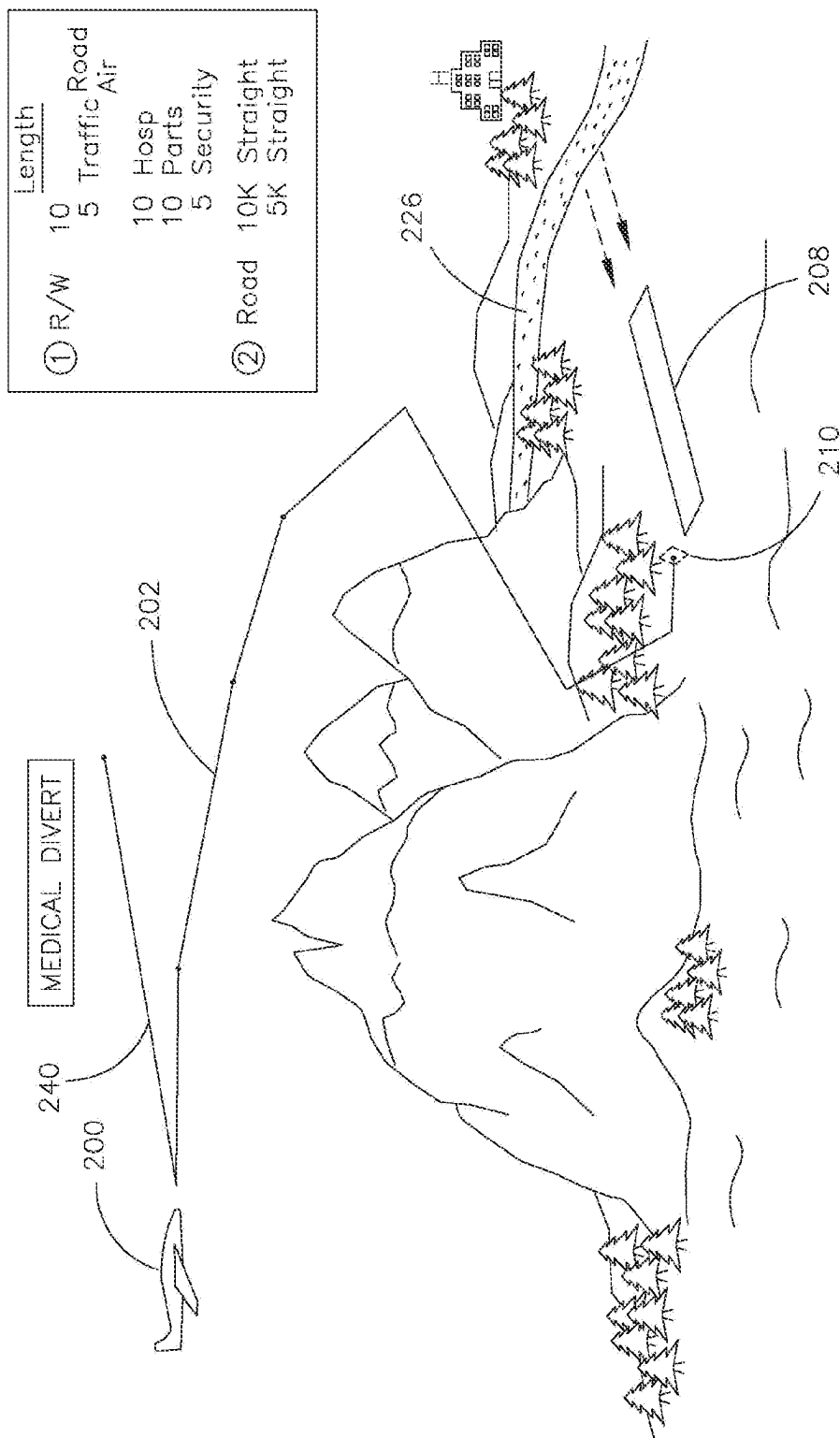
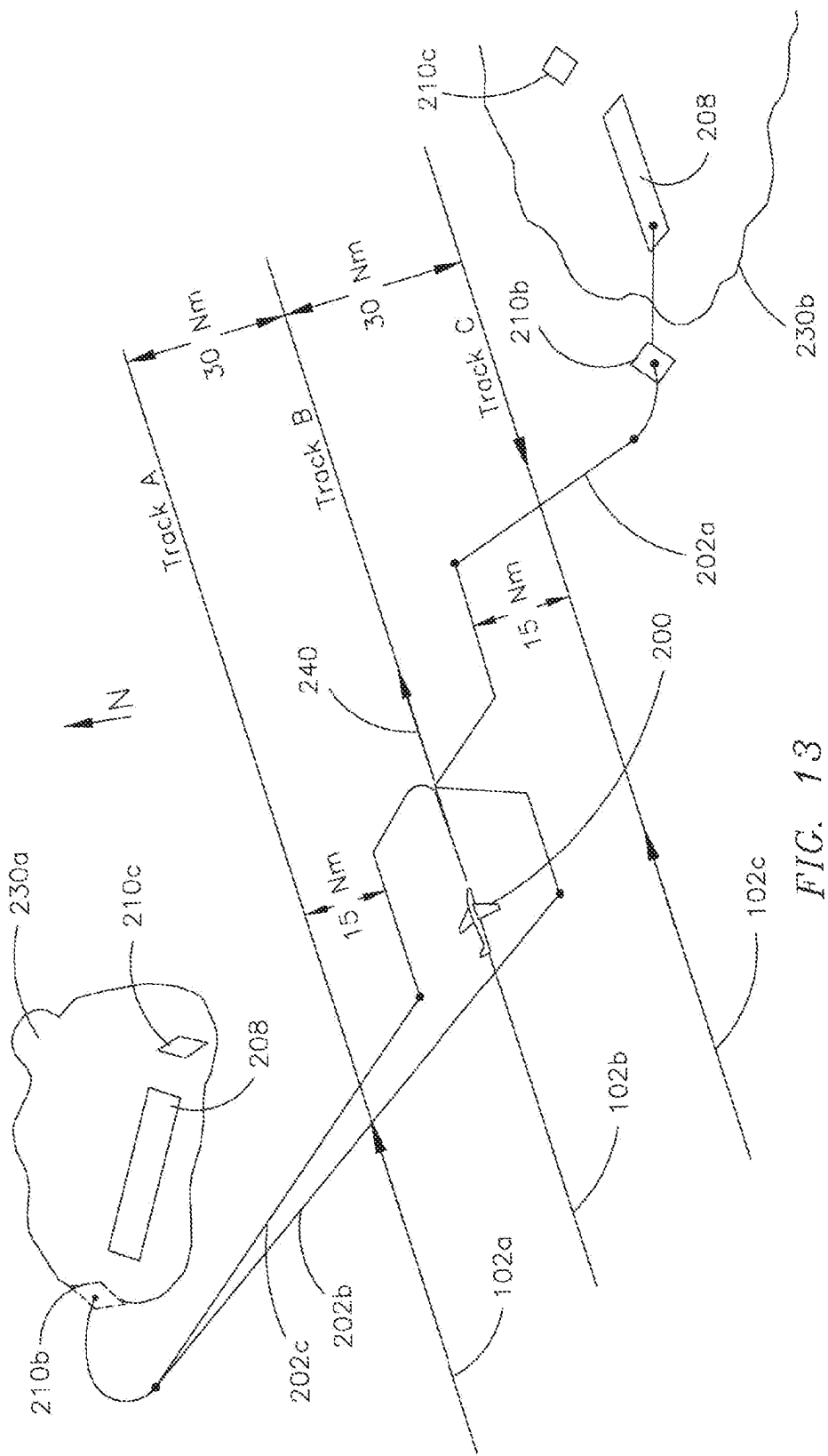


FIG. 12



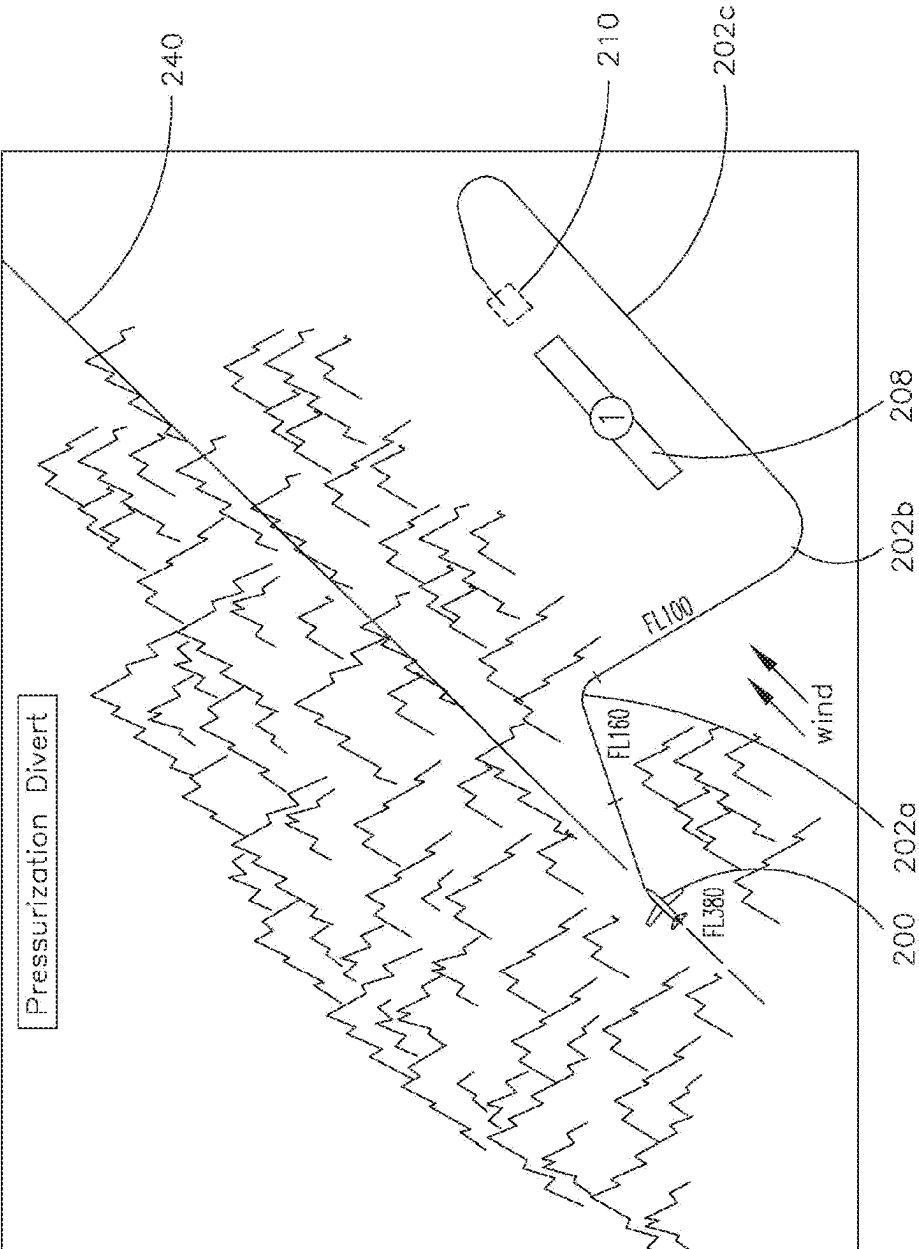


FIG. 14

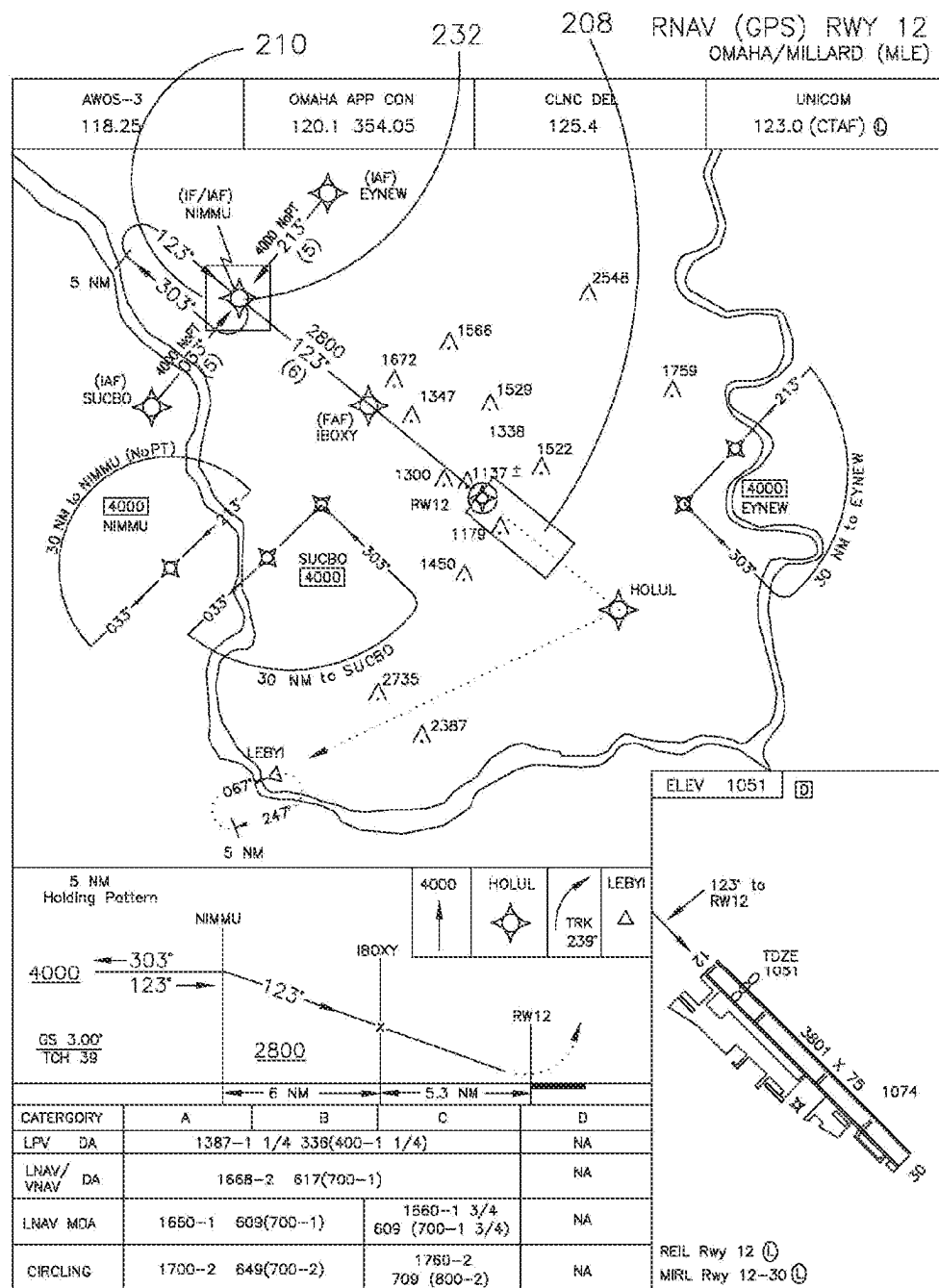


FIG. 15

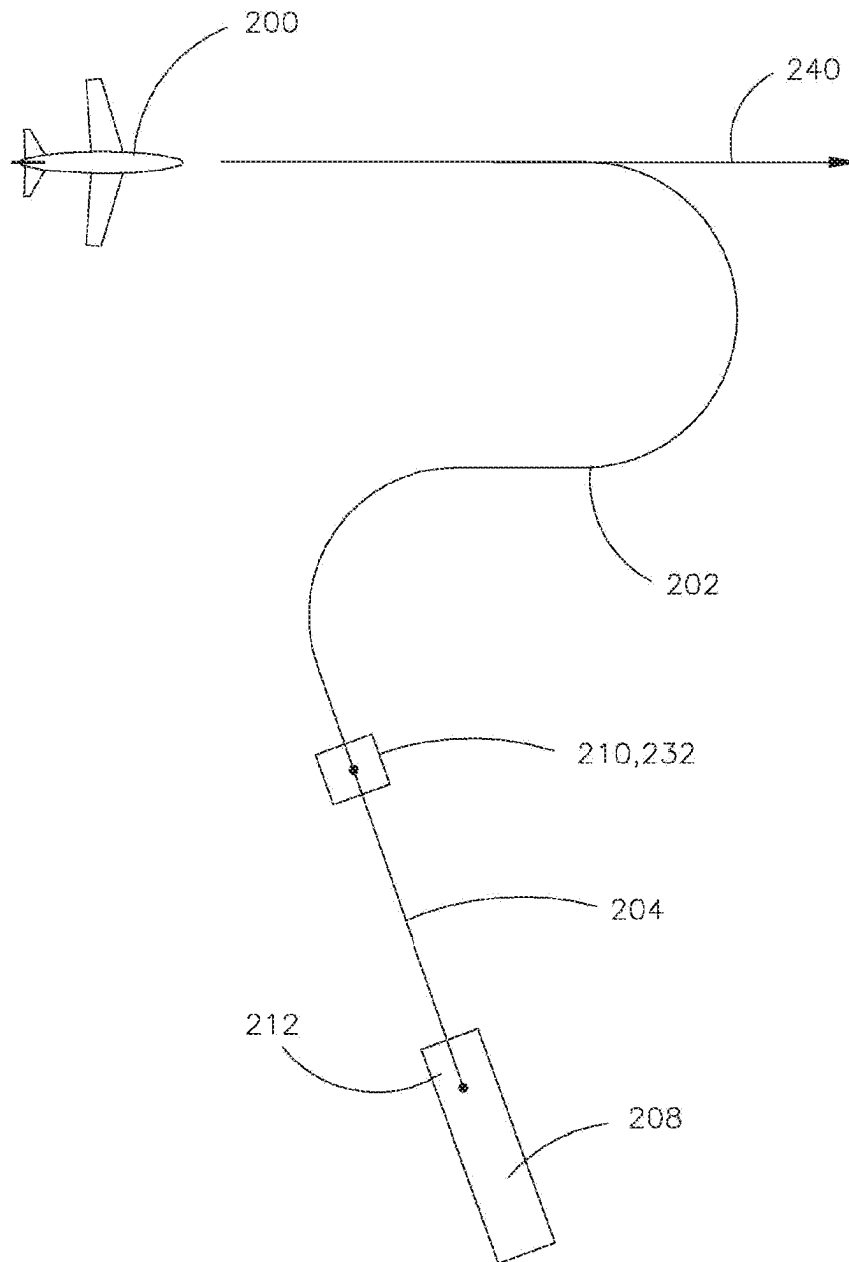


FIG. 16

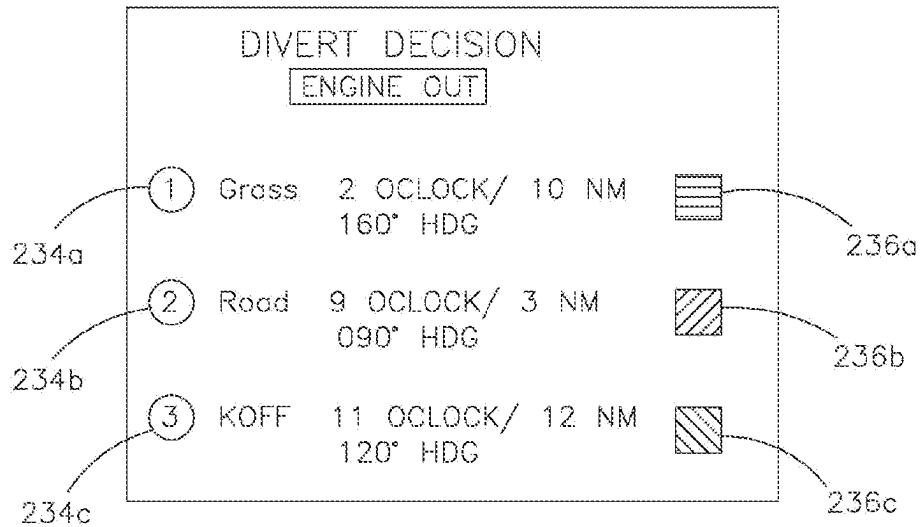


FIG. 17A

Legend:

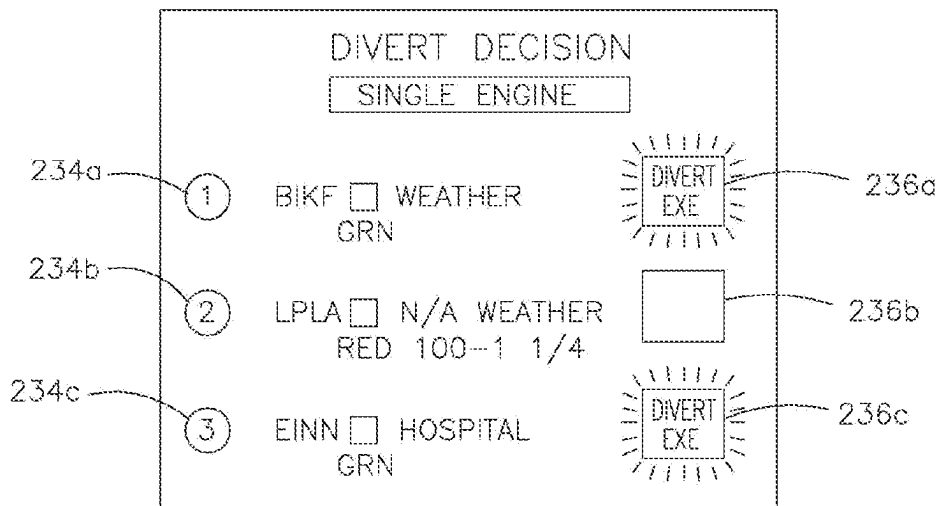


FIG. 17B

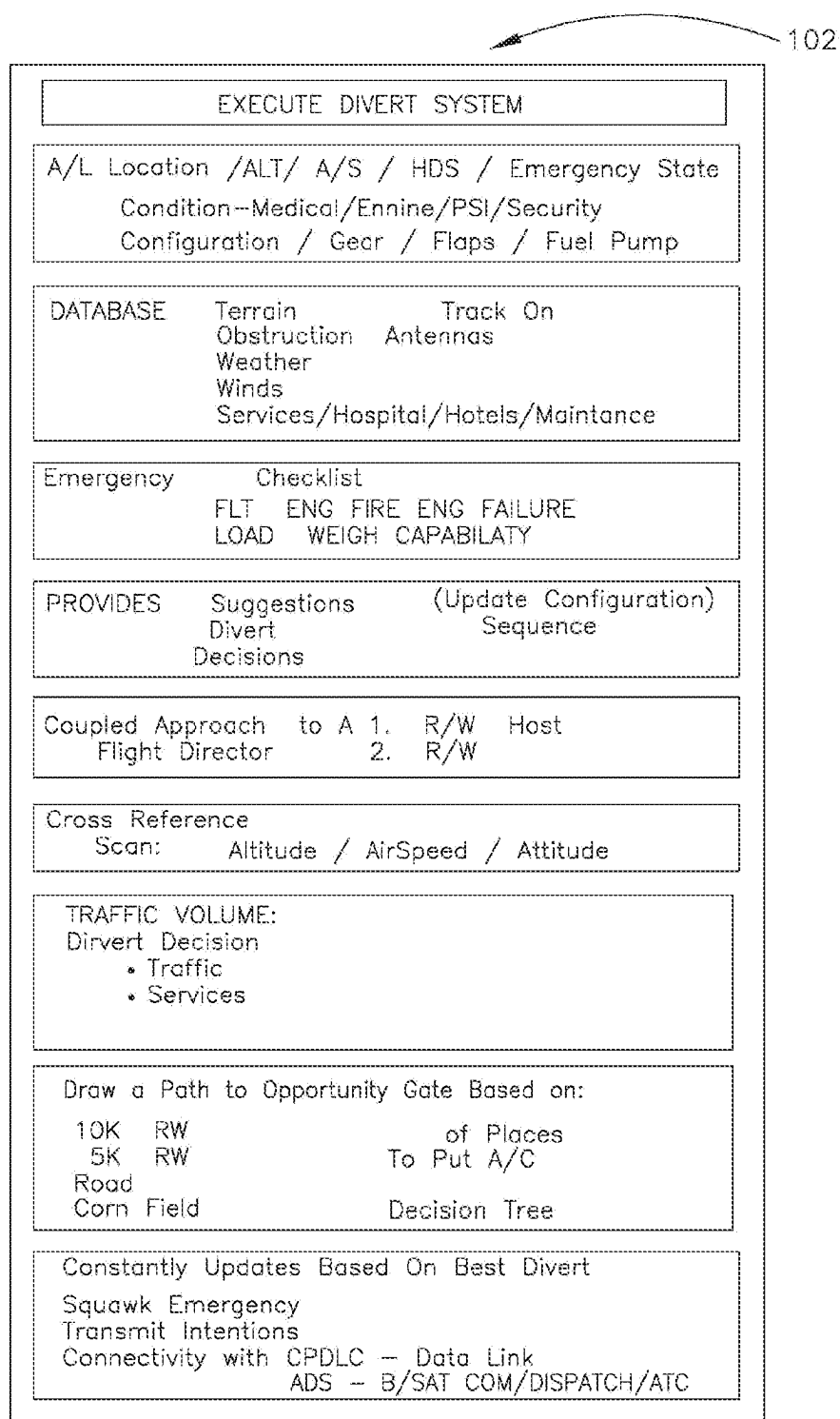


FIG. 18

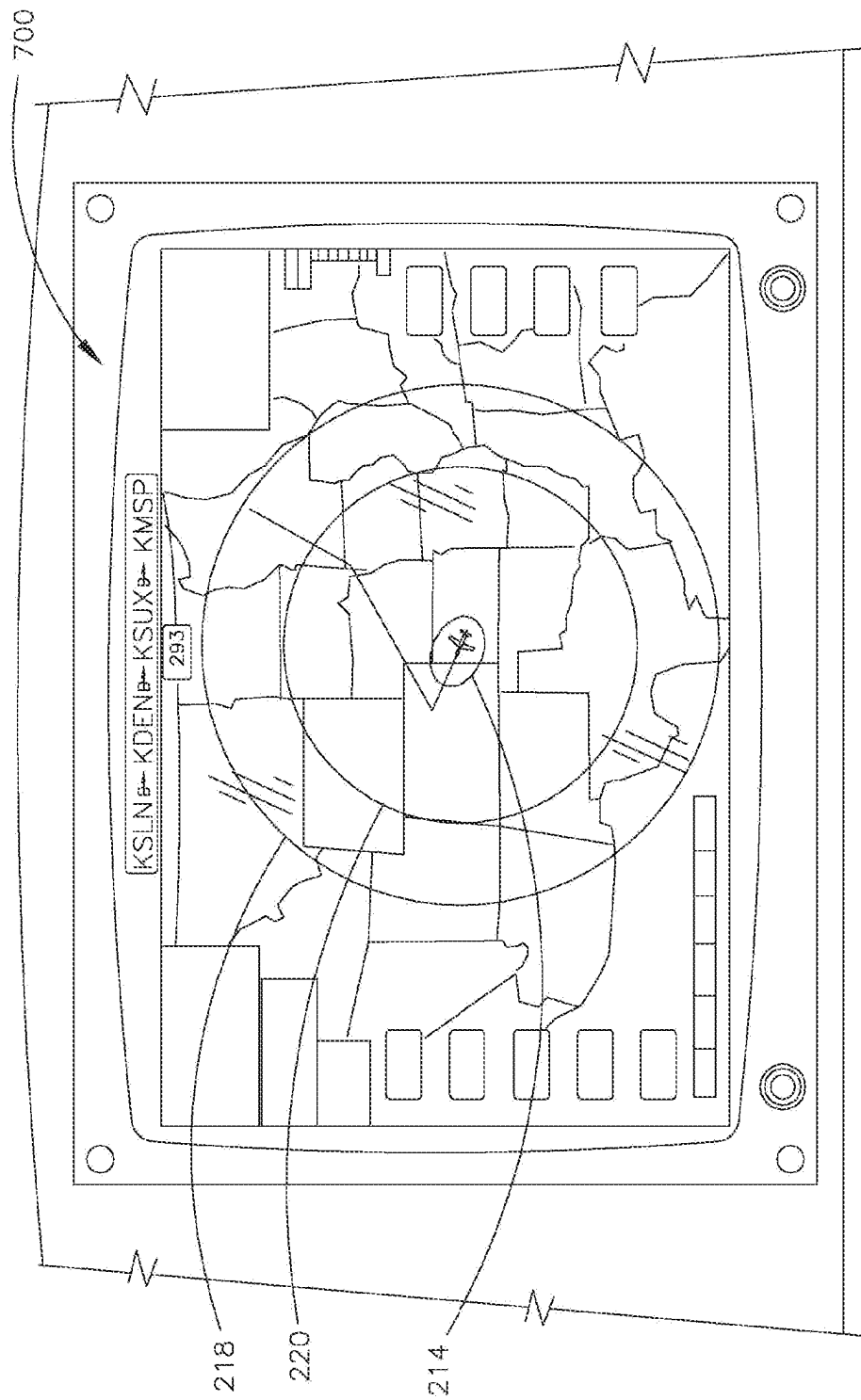


FIG. 19A

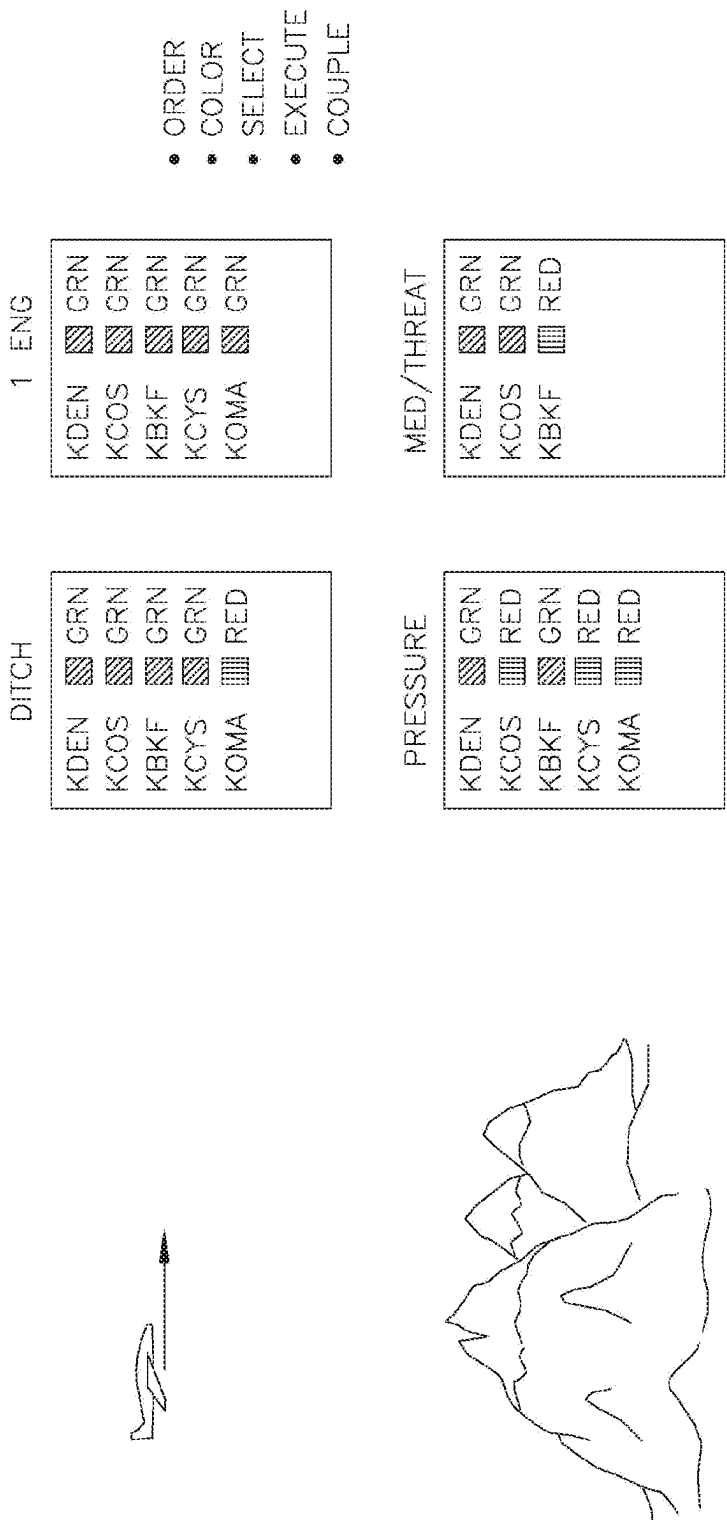
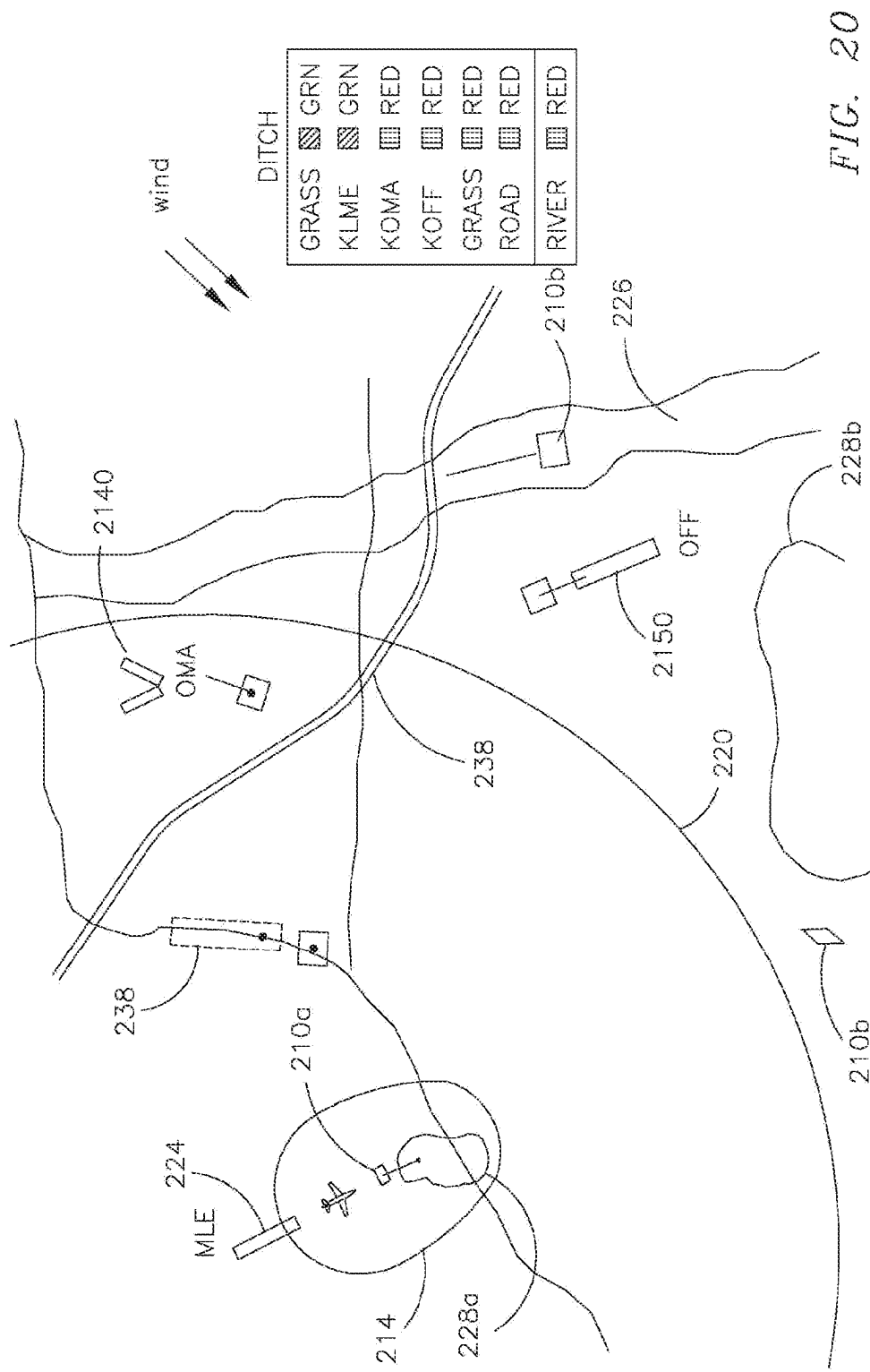
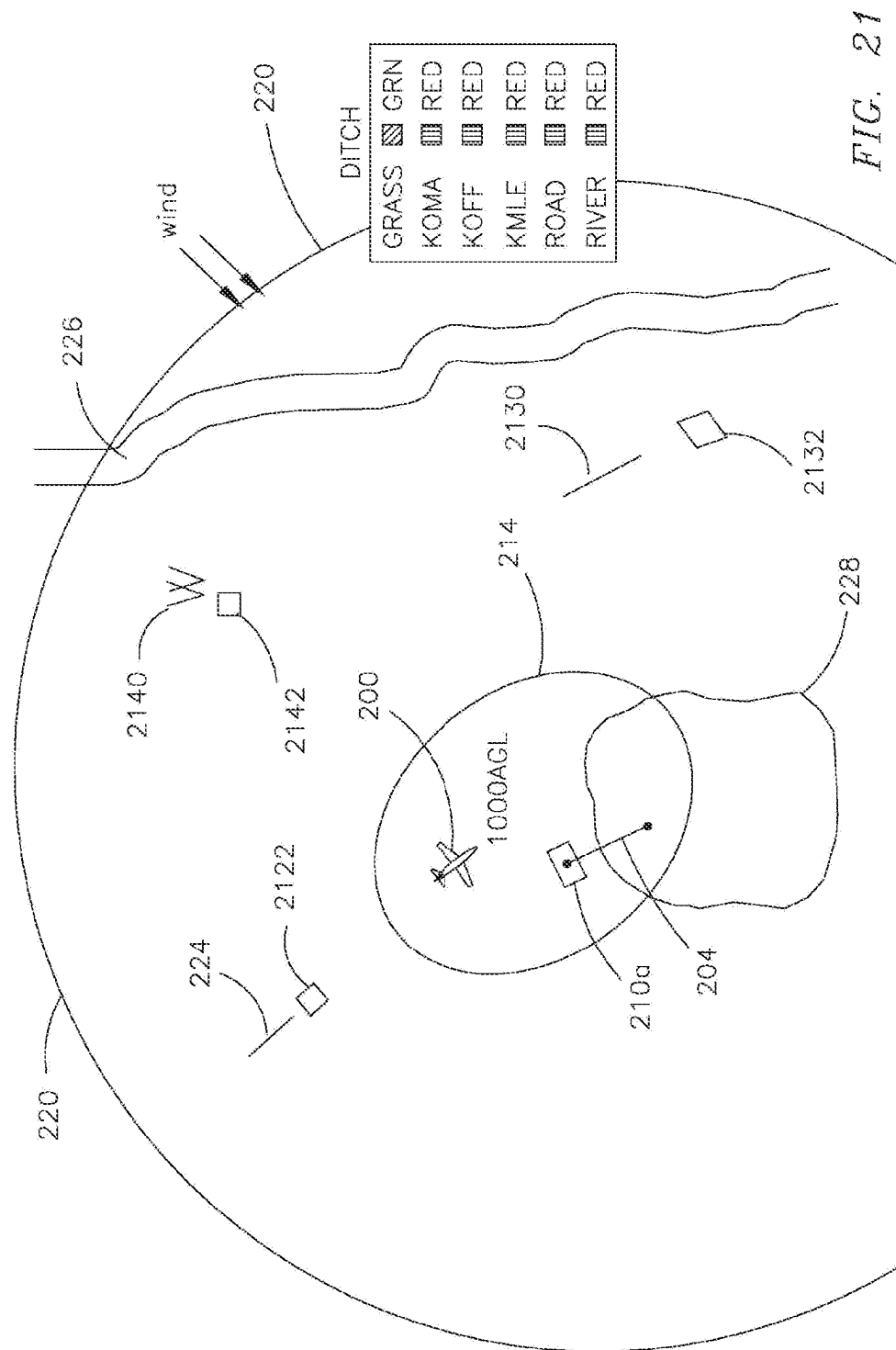
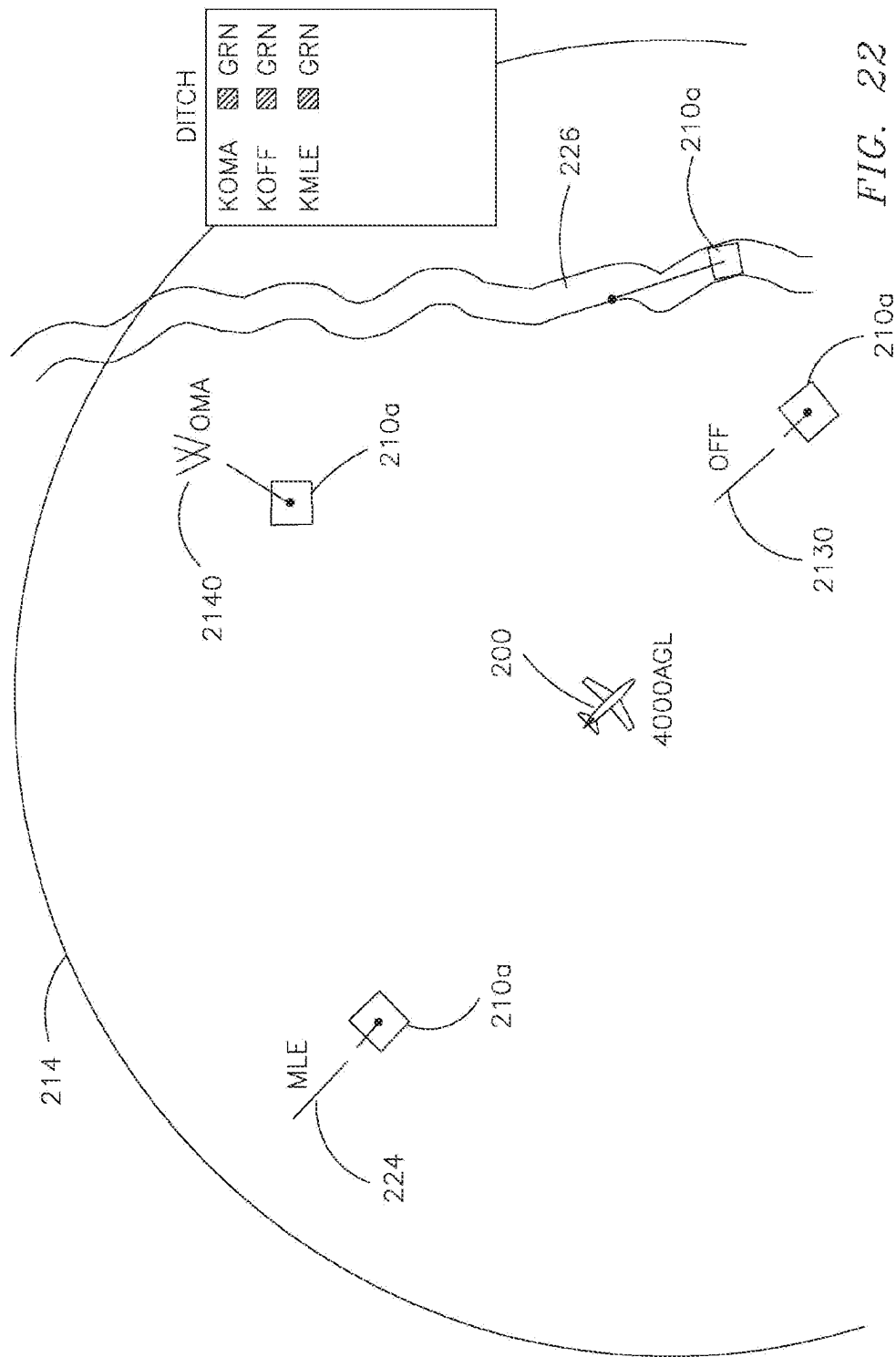


FIG. 19B







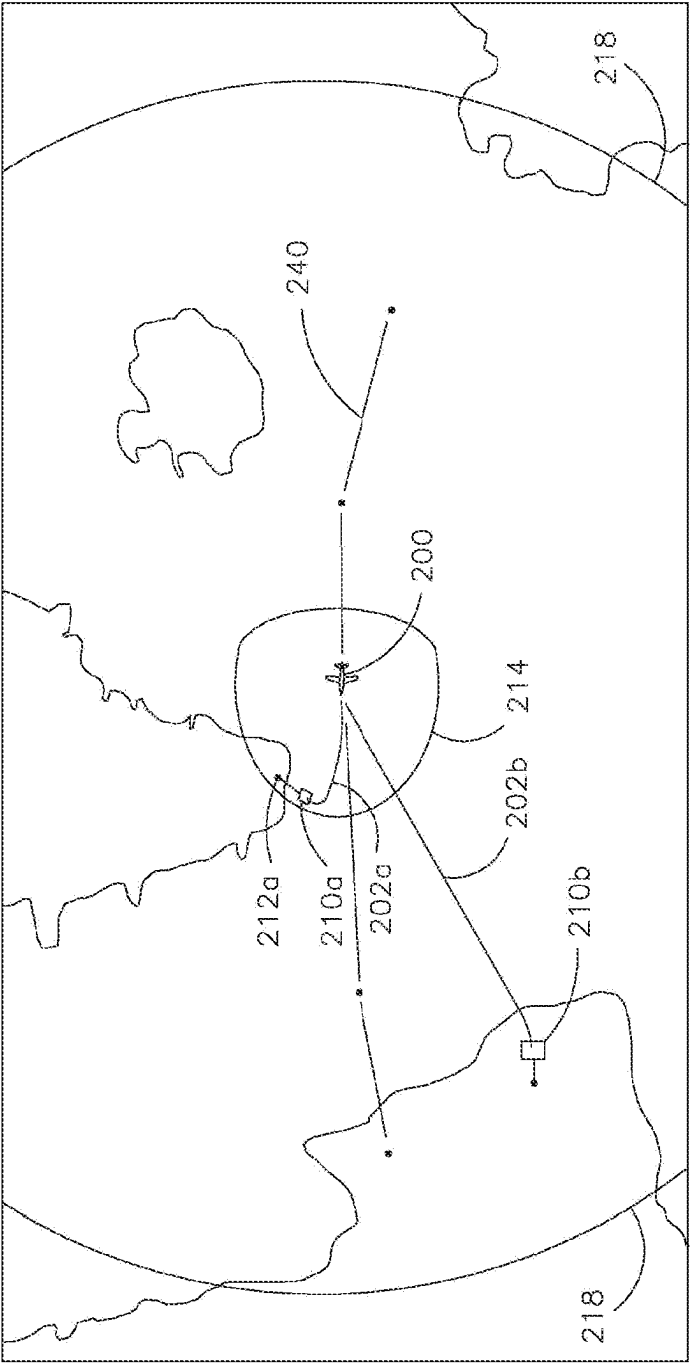


FIG. 23

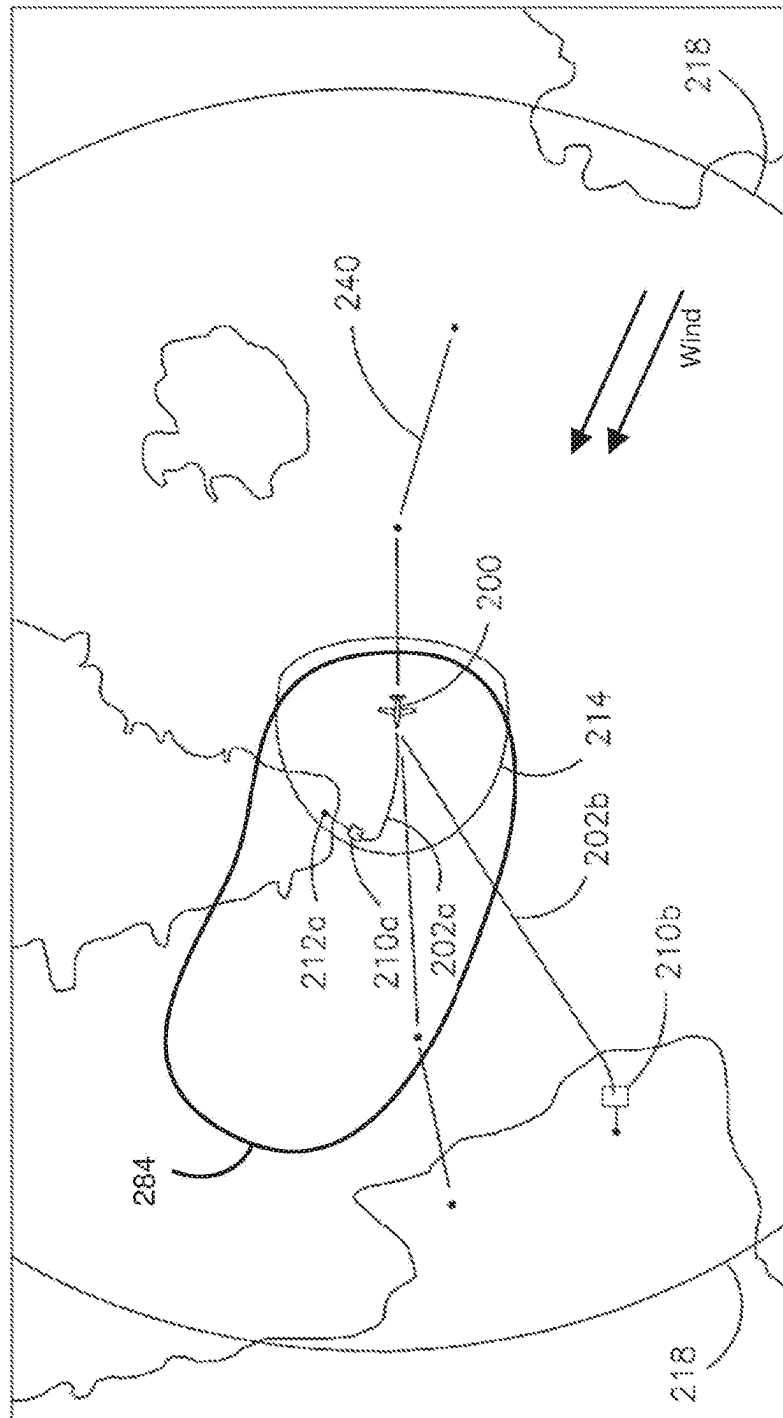
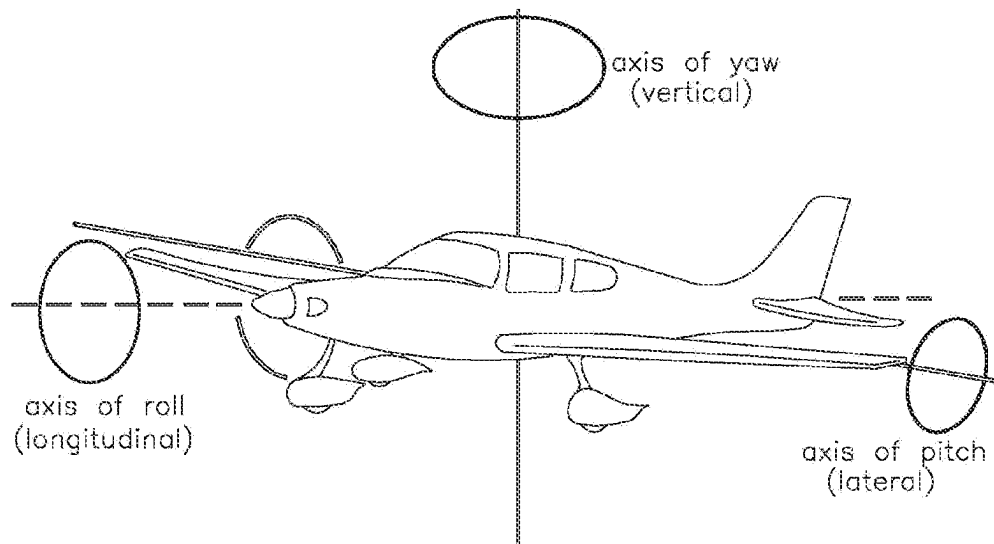
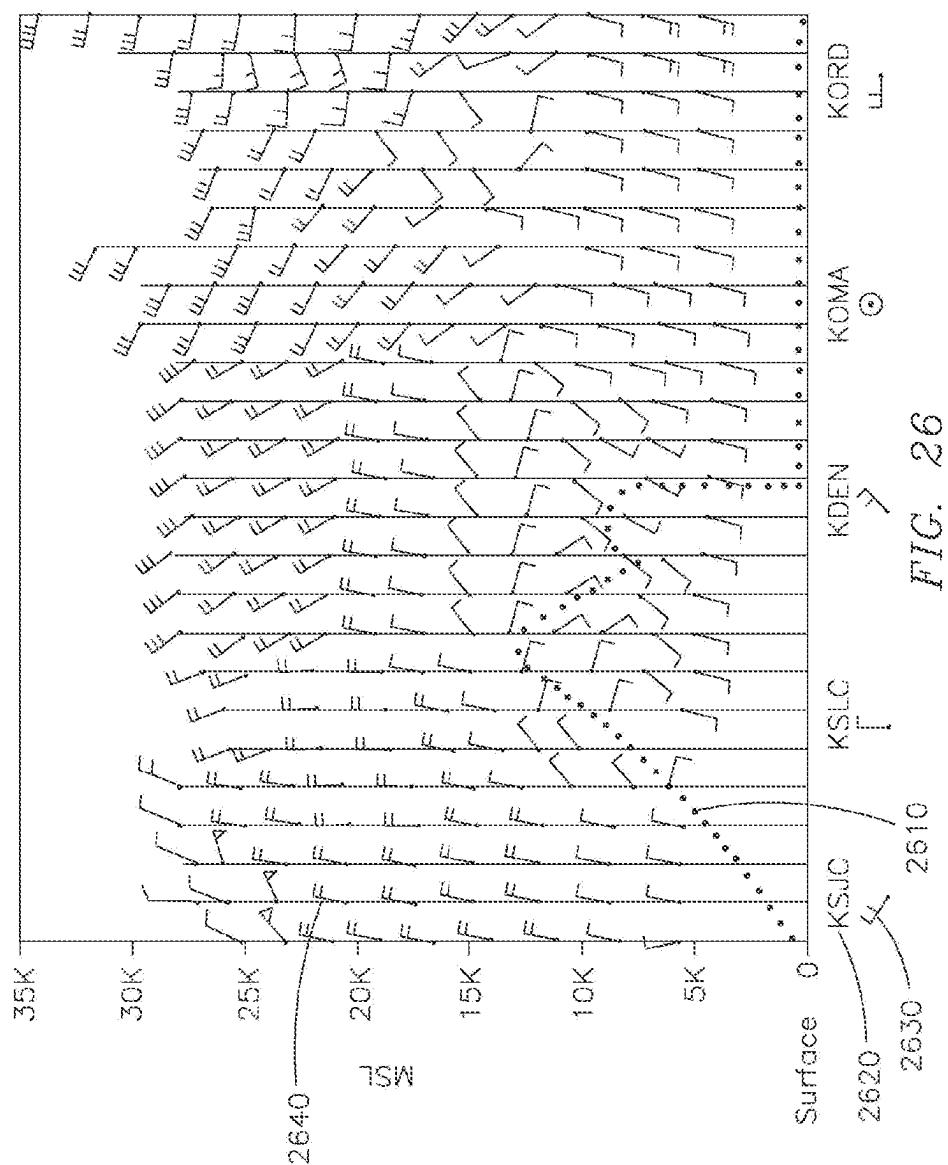


FIG. 24

*FIG. 25*



ID	Name	Details
KANW ANW	Ainsworth Rgnl Anywhere, NE	  Approaches: VOR, RNAV, GPS
KBVN BYN	Ablon Muni Alhorne, NE	 Approaches: RNAV, GPS, NDB
KAIA AIA	Alliance Muni Aplace, NE	    Approaches: ILS, DME, LOC, VOR, RNAV, GPS, NDB
NE32	Burk General Hospital Med-1-Port Aplace, NE	  
H63	Harlan Co Lake Apalco, NE	
4D9	Alpha Muni Alpha, NE	
37V	Arnold Muni Arnold, NE	
ONE0	Merrihaw Ashley, NE	   
ONE1	Runner Landing Area Asphalt, NE	   
53NE	Starns Brothers Apostle, NE	  
8V2	Asman Muni Asman, NE	
K01	Farington Field Apple, NE	
77NE	Fish County Hospital Apple, NE	 
NE64	Memoral Hospital Auroa, NE	 
KAUH AUH	Auroa Muni-Field Auroa, NE	  Approaches: VOR, RNAV, GPS
40NE	Landgren Ranch Barrel, NE	 
6NE7	Lee Field Barley, NE	 
KRBE RBE	Rock County Bubba, NE	 Approaches: RNAV, GPS, NDB
NE31	B.T. Air Better Town, NE	 
KBIE BIE	Better Muni Better, NE	 Approaches: VOR, RNAV, GPS

FIG. 27

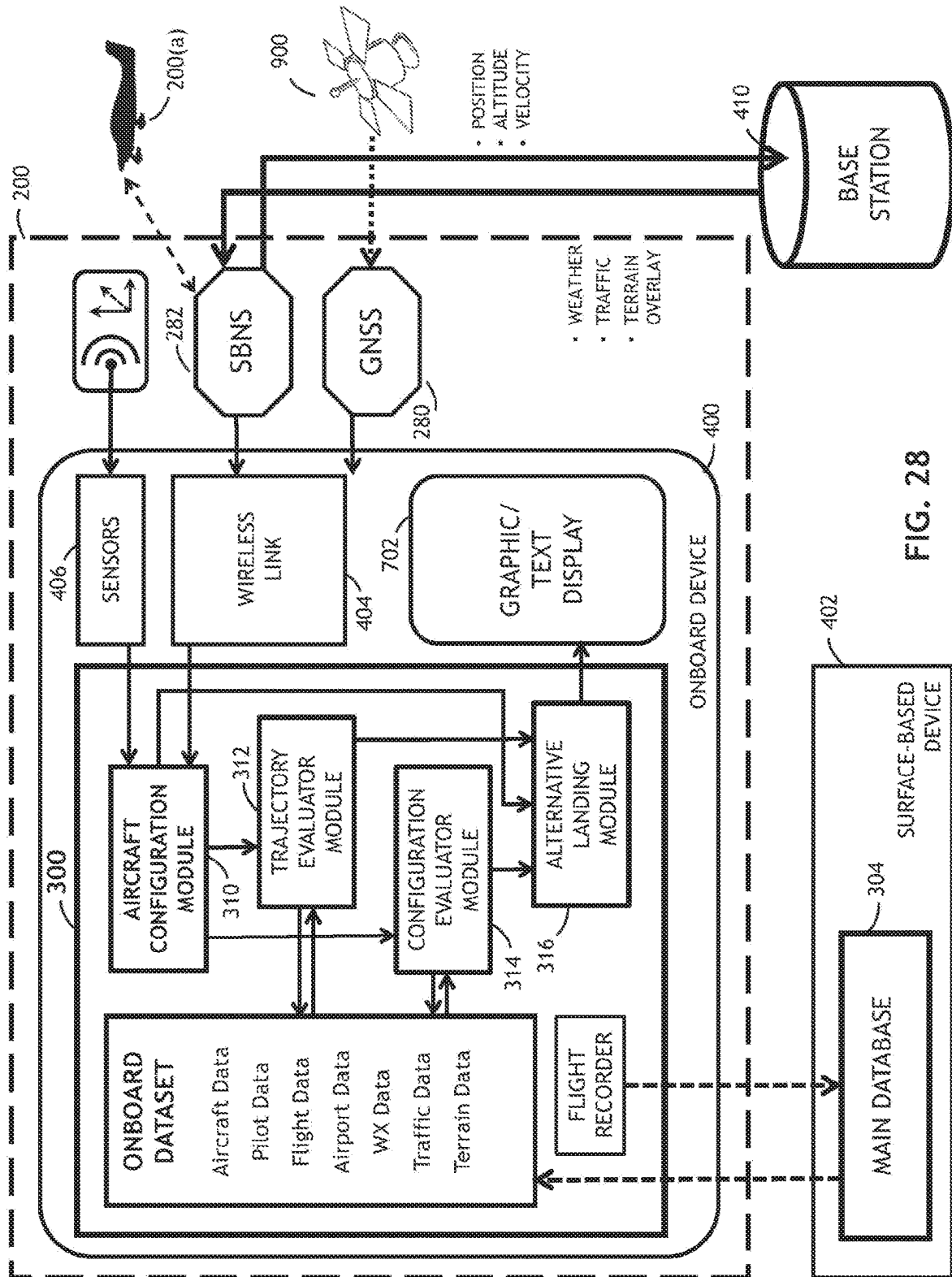


FIG. 28

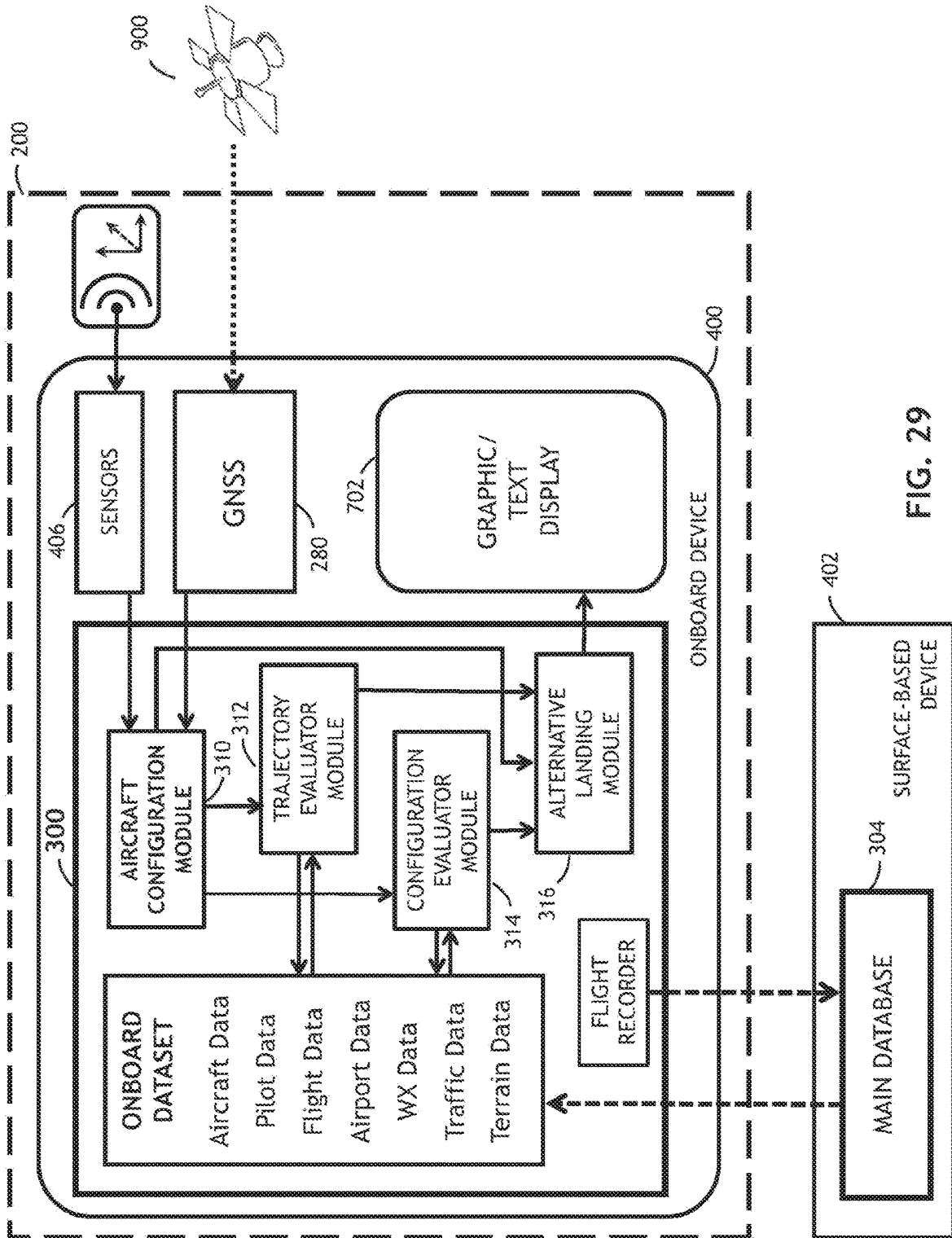


FIG. 29

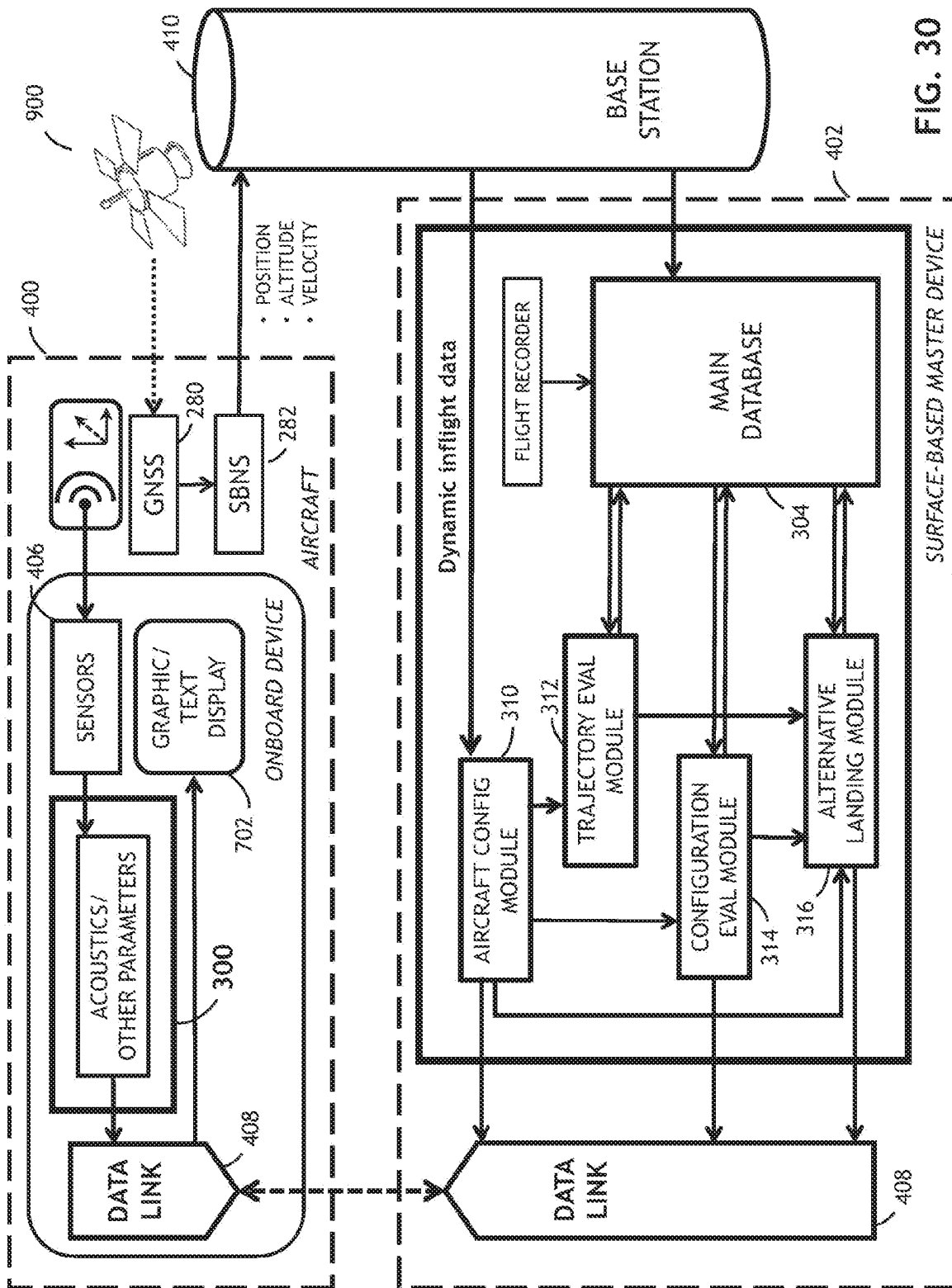


FIG. 30

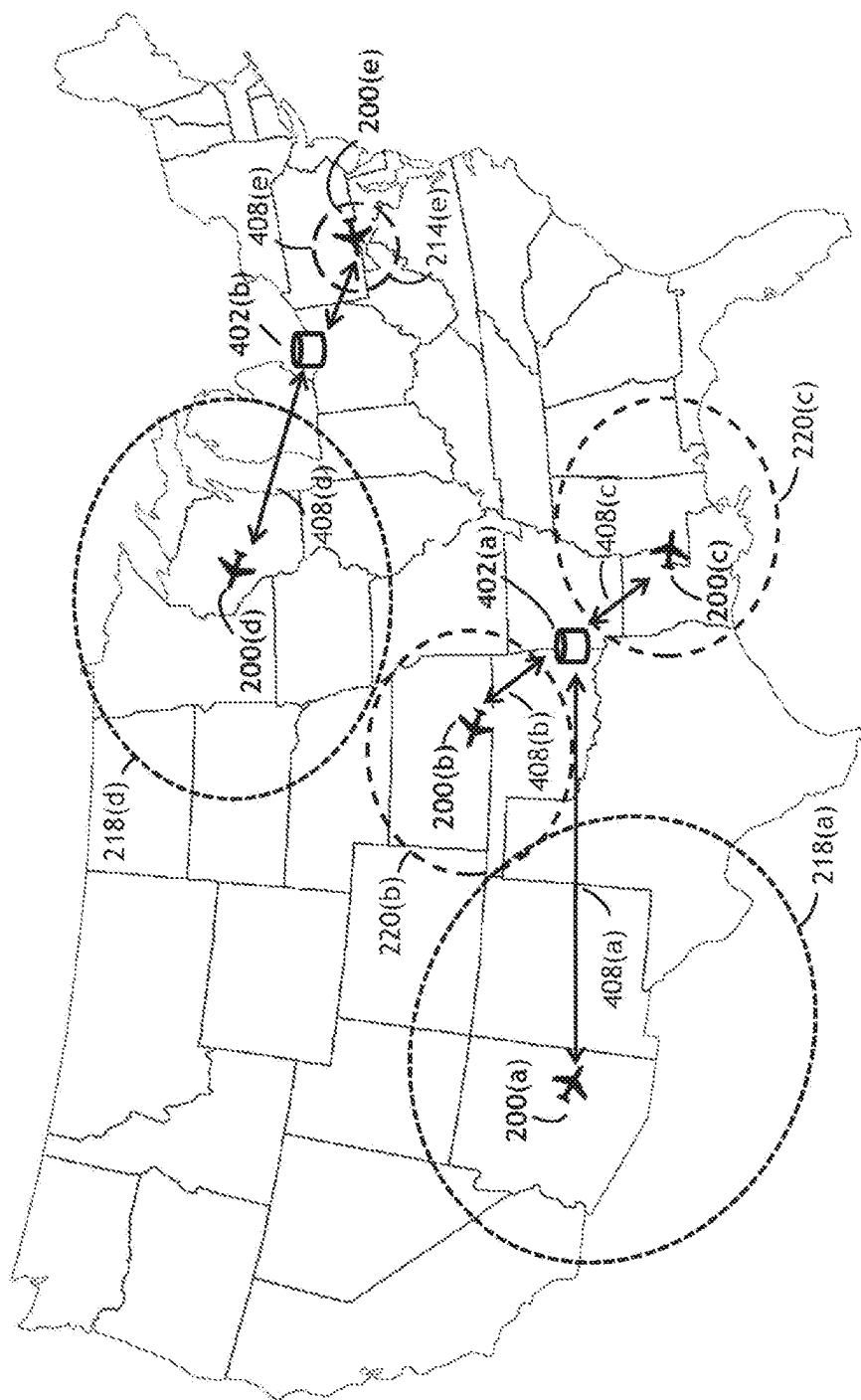


FIG. 31

FIG. 32

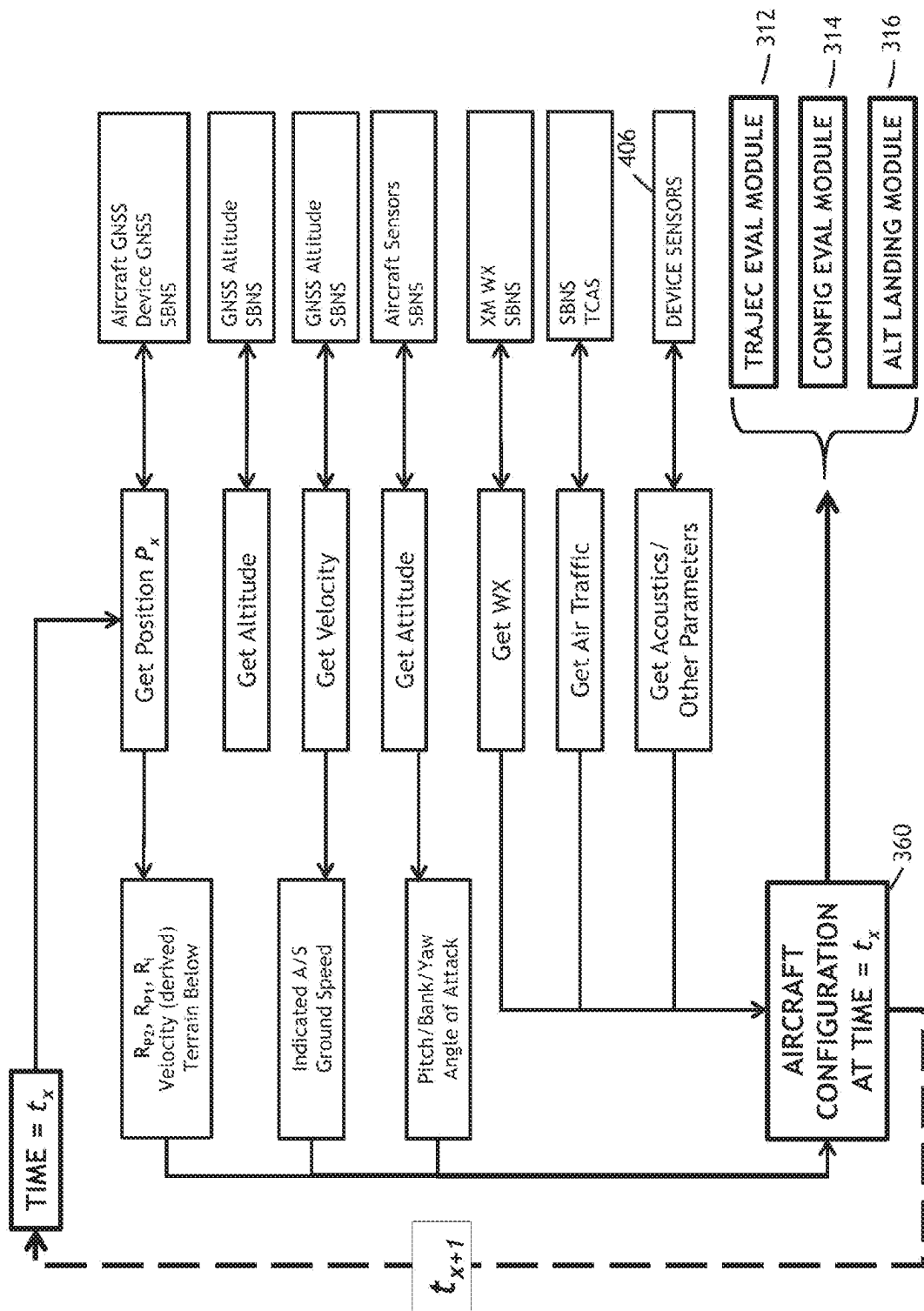


FIG. 33

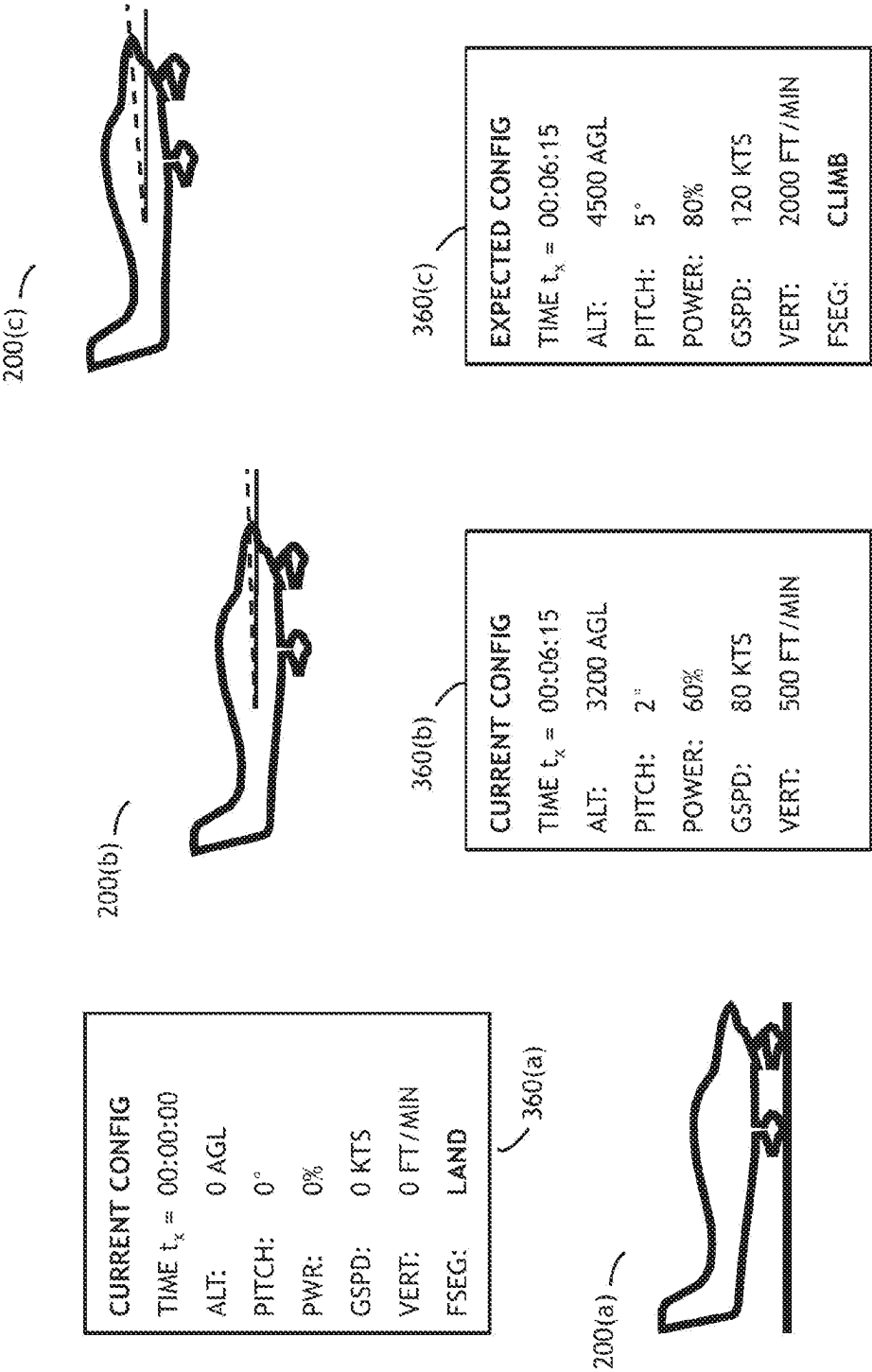


FIG. 34

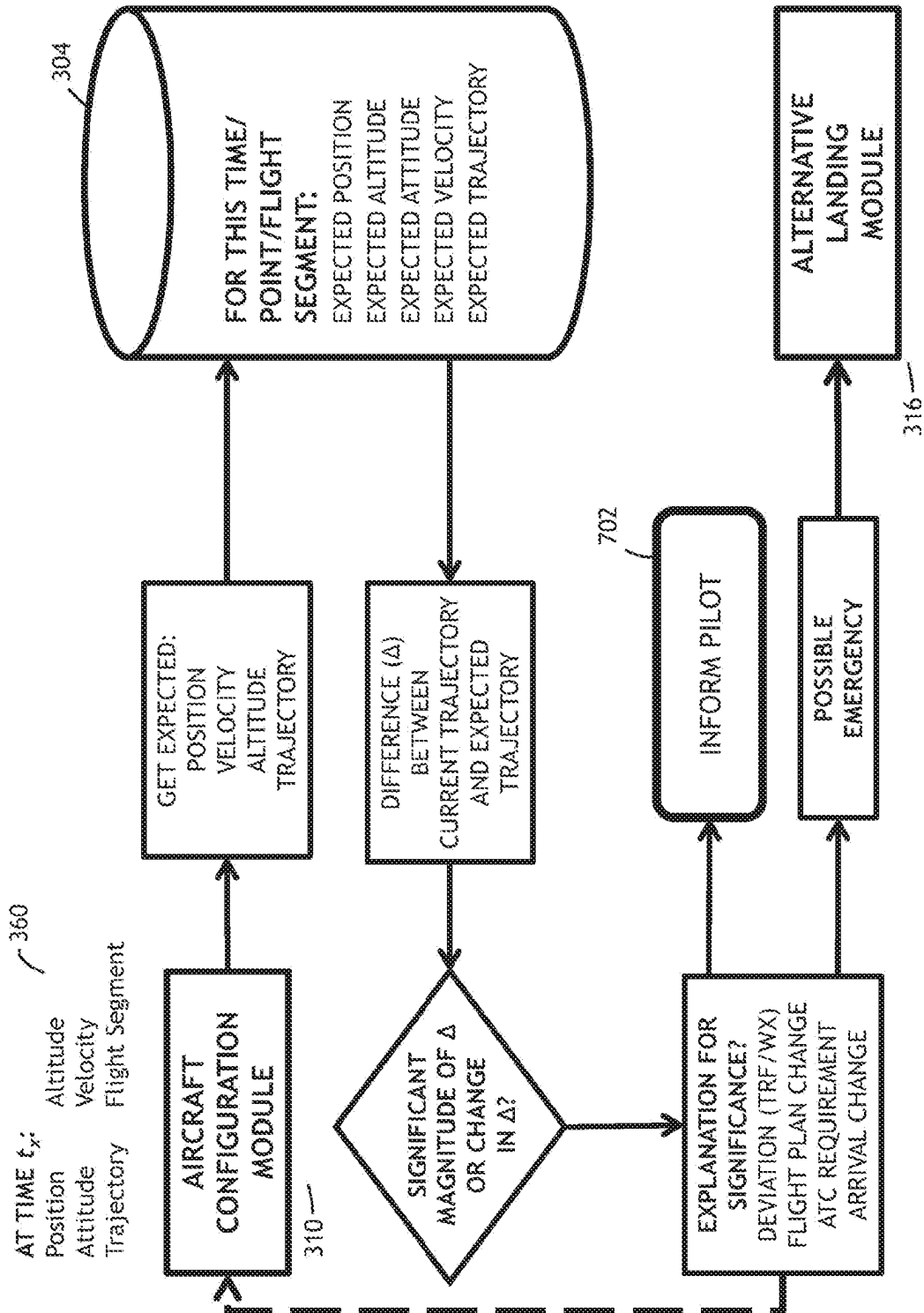


FIG. 35

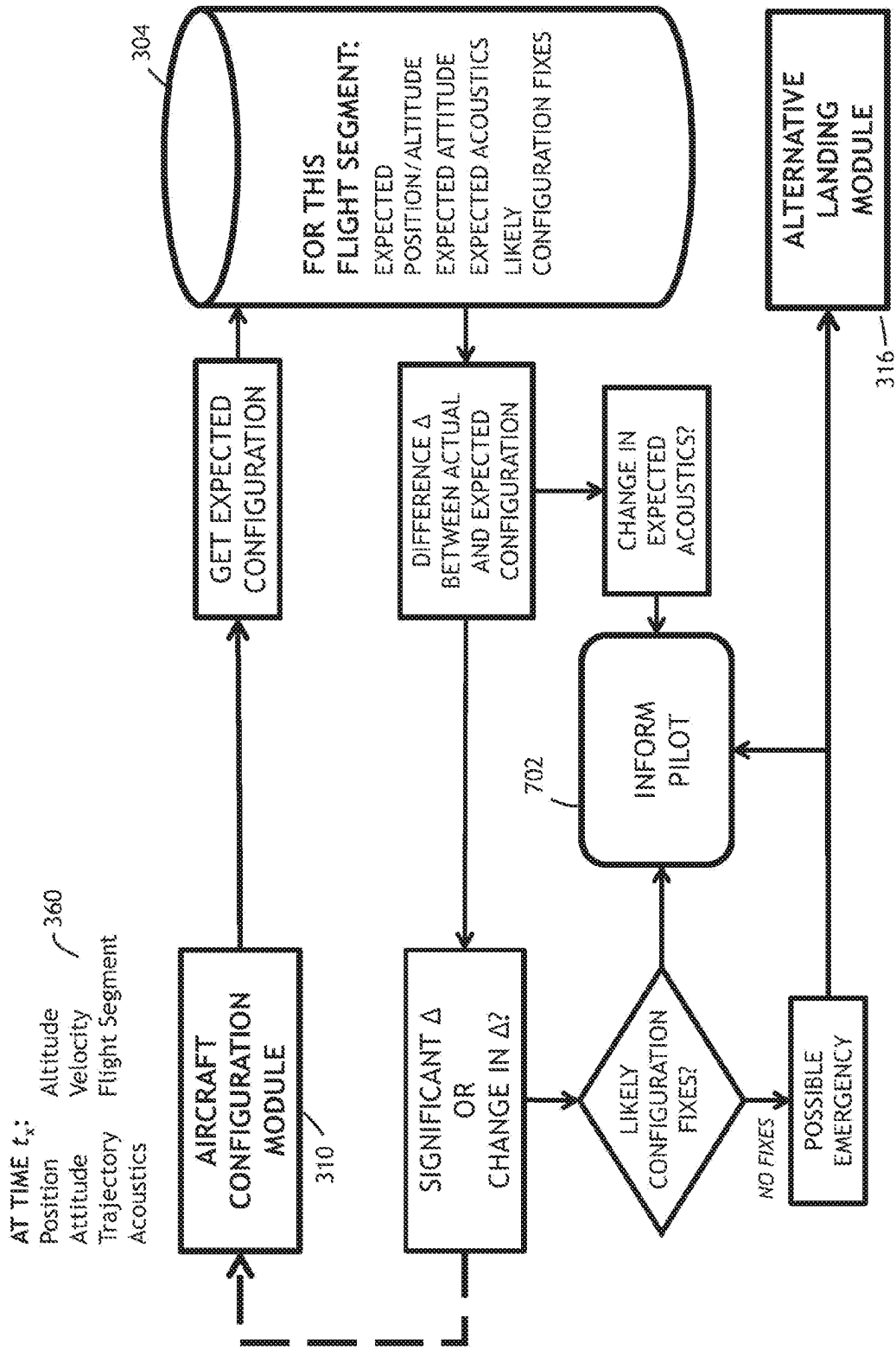


FIG. 36

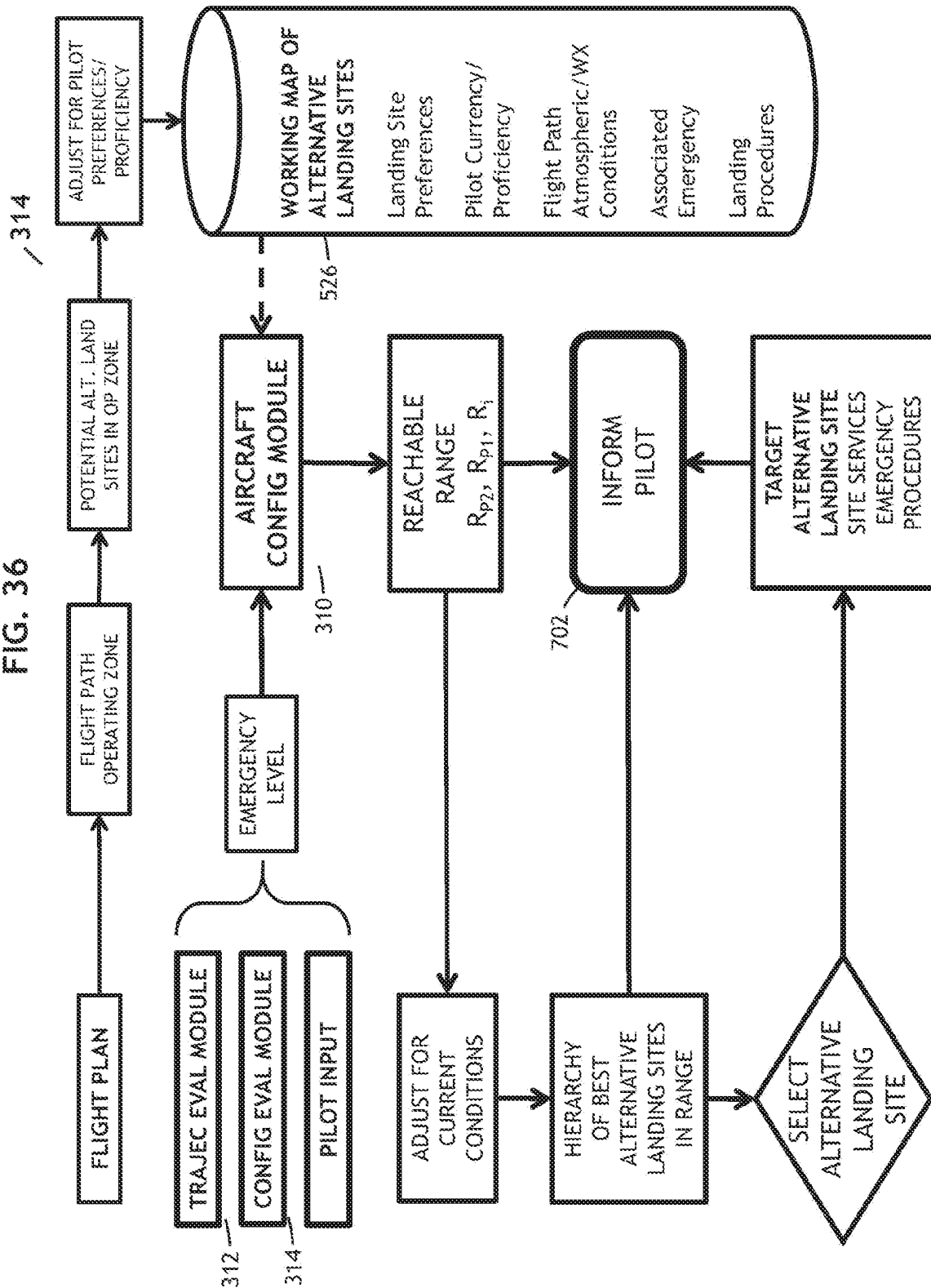


FIG. 37A

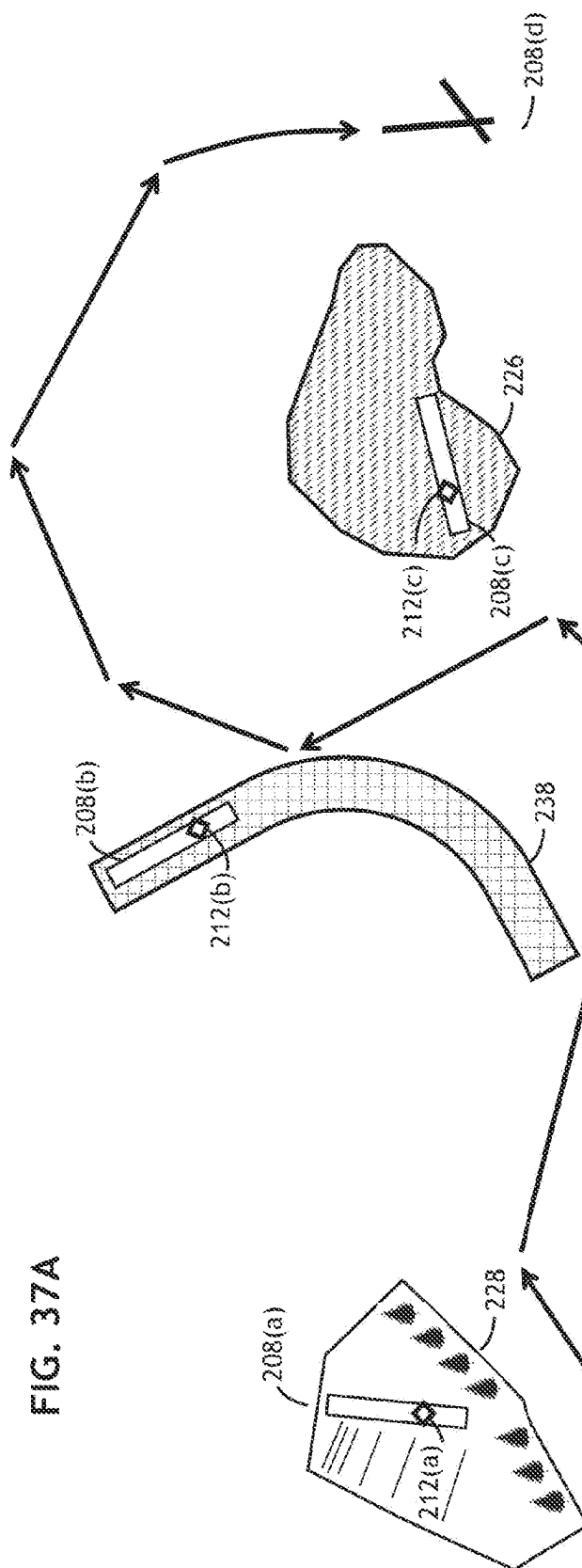
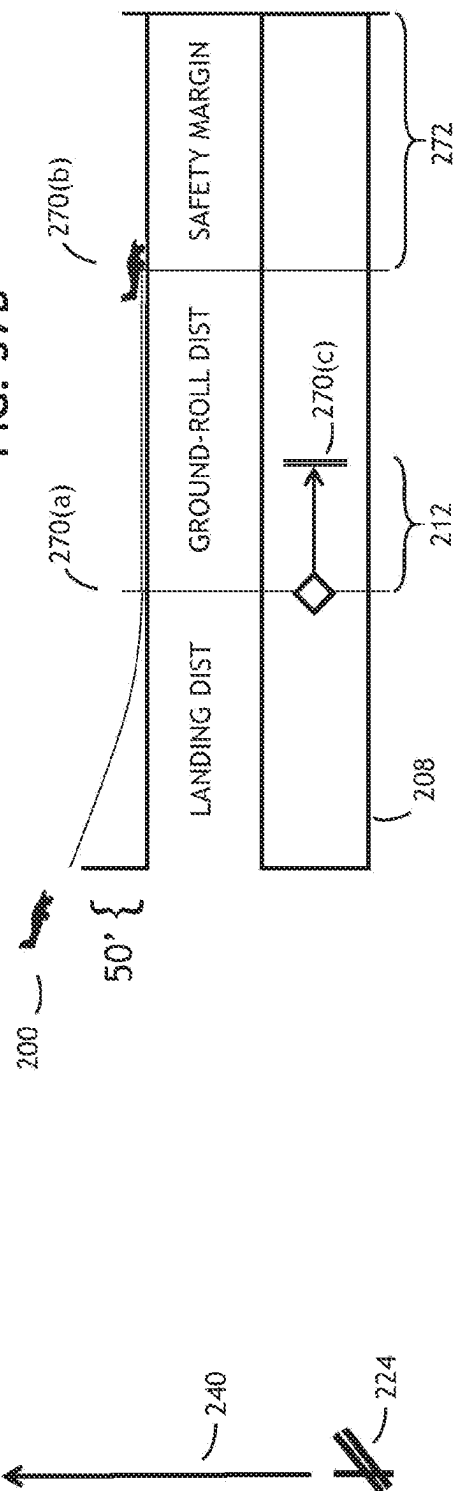


FIG. 37B



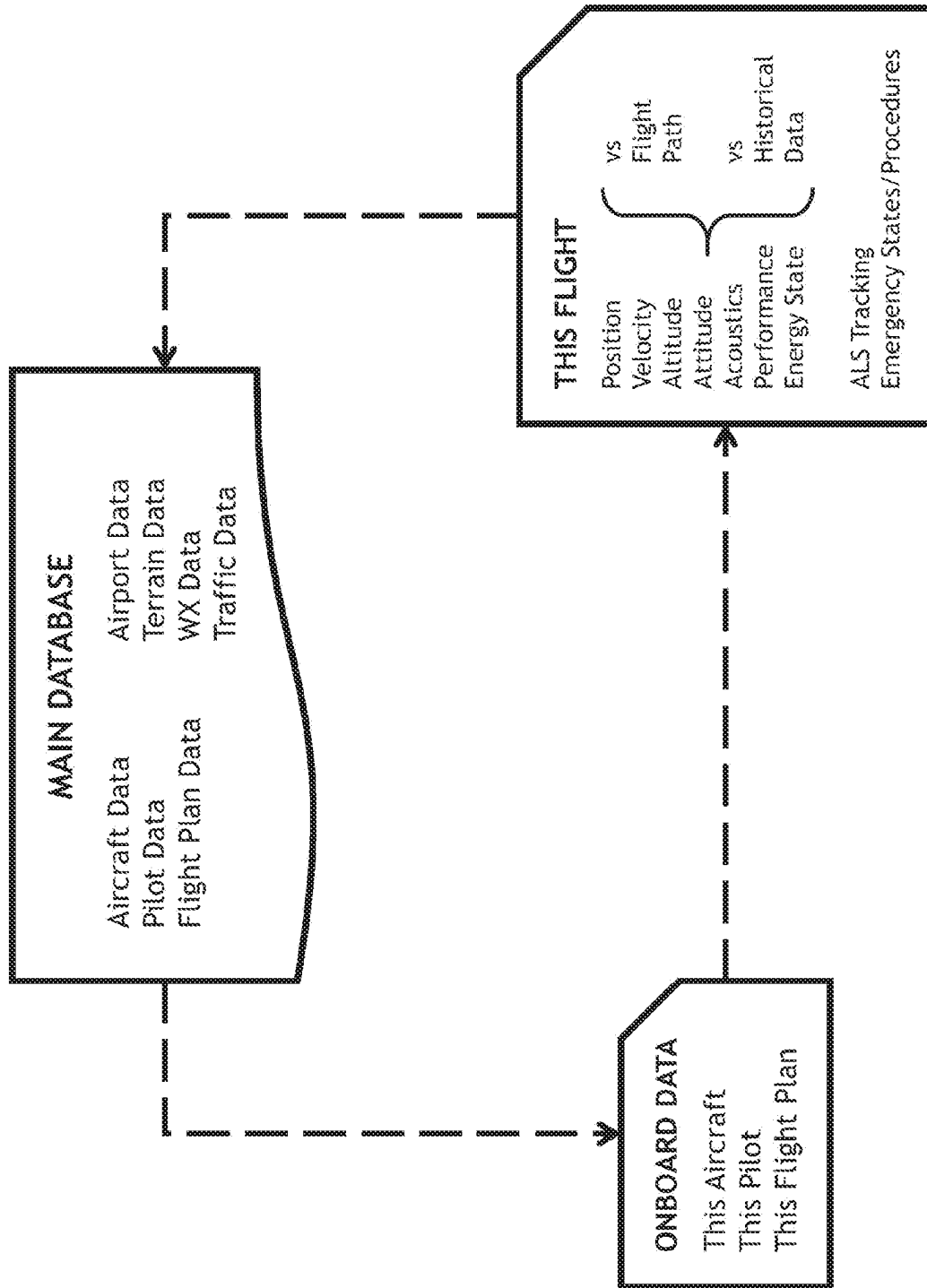


FIG. 38

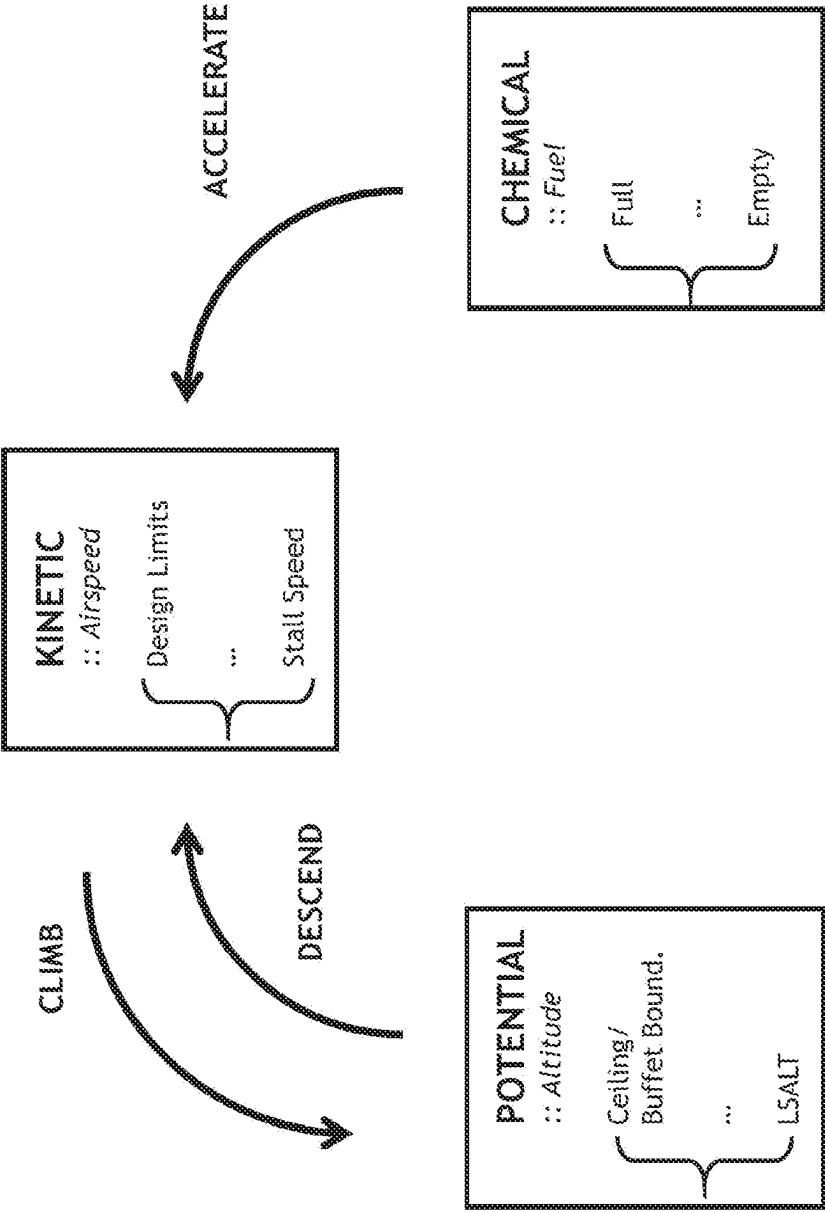
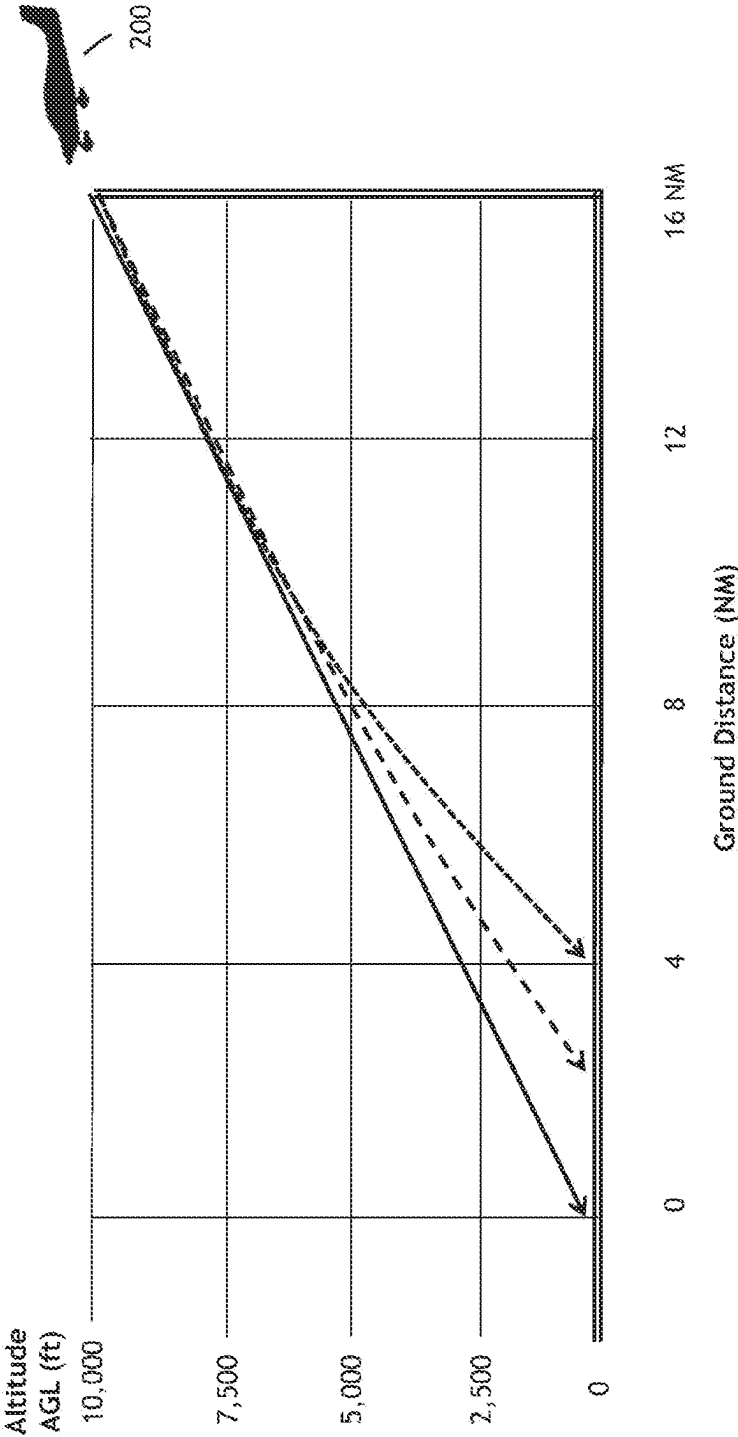


FIG. 39

FIG. 40

OPTIMAL GLIDE RATIO: 9.6/1
OPTIMAL GLIDE DISTANCE/10,000' AGL: 16 NM (96,000')
GROSS WEIGHT OF AIRCRAFT: 3400 LB
BEST GLIDE SPEED V_{bg} : 88 kt IAS

→ OPTIMAL GLIDE ANGLE
→ OPTIMAL GLIDE ANGLE,
LESS PROFICIENT PILOT
→ EFFECT OF HEADWIND



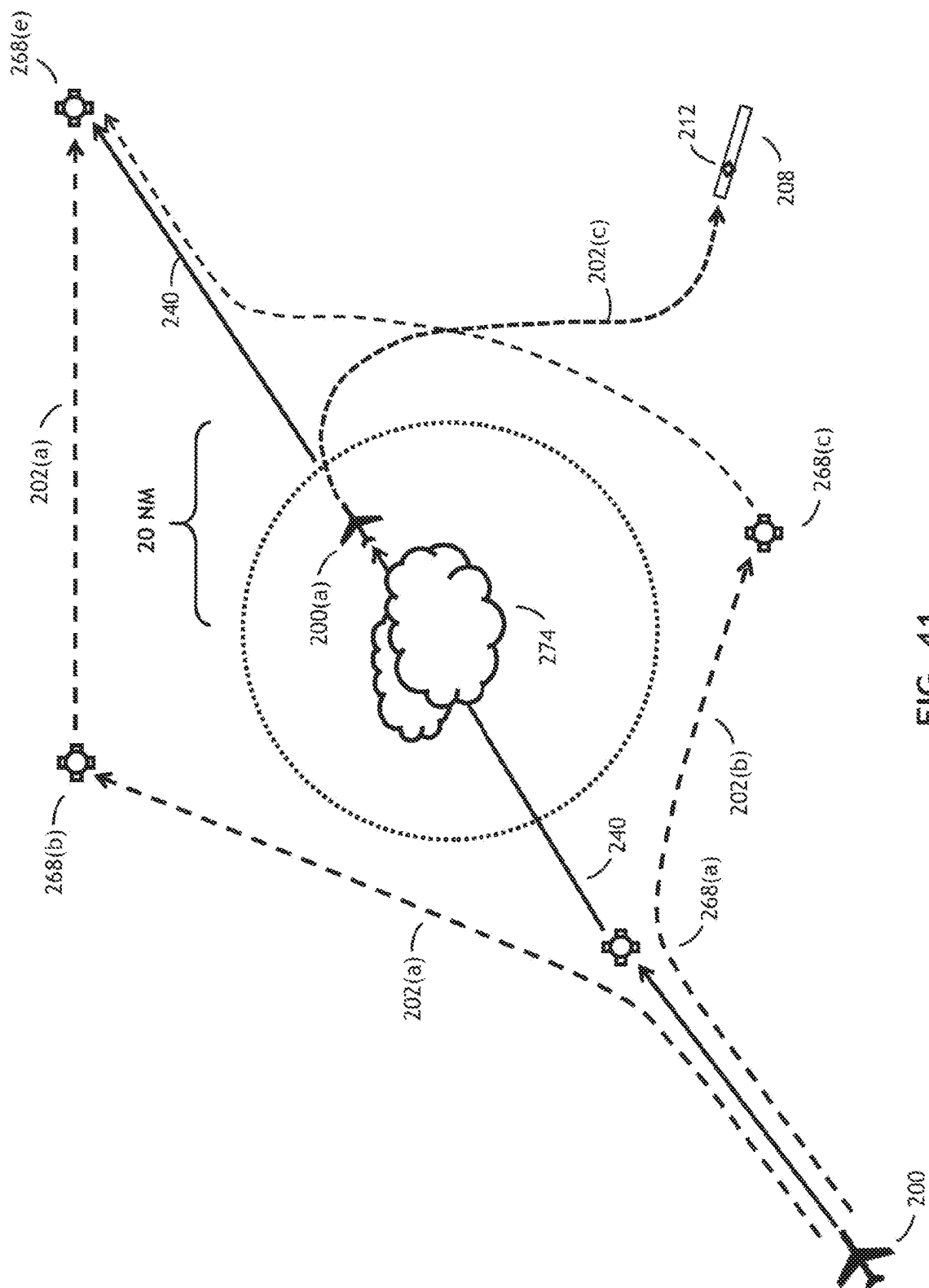
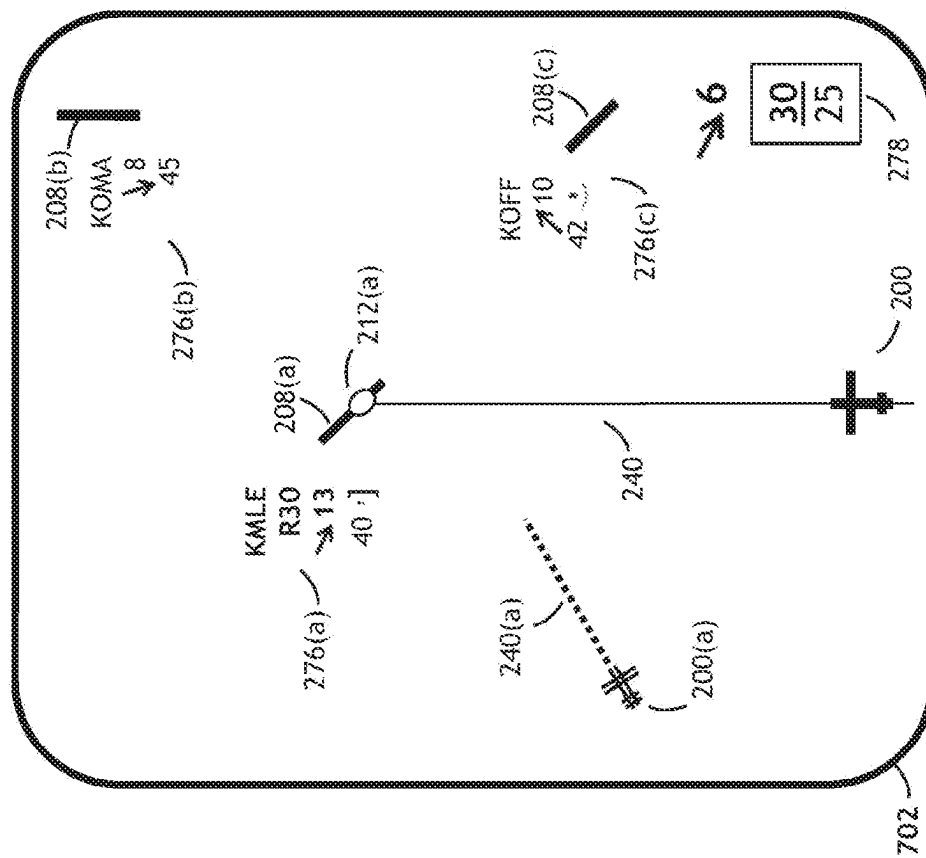


FIG. 41



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G.
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AIRCRAFT FLIGHT OPERATIONS DATA LINK

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application claims priority under 35 U.S.C. 120 to the following U.S. Non-Provisional Applications: U.S. Non-Provisional Ser. No. 17/544,645 filed on 7 Dec. 2021; U.S. Non-Provisional Ser. No. 17/492,448 filed on 1 Oct. 2021 (now U.S. Pat. No. 11,657,721); U.S. Non-Provisional Ser. No. 17/478,821 filed on 17 Sep. 2021; and U.S. Non-Provisional application Ser. No. 17/478,807 filed on 17 Sep. 2021 (now U.S. Pat. No. 11,551,564); which are continuations of U.S. Non-Provisional application Ser. No. 16/673,725 filed on 4 Nov. 2019; said Ser. No. 16/673,725 filed on 4 Nov. 2019 was a continuation of U.S. Non-Provisional application Ser. No. 15/152,437 filed on 11 May 2016; said Ser. No. 15/152,437 filed on 11 May 2016 was a divisional of U.S. Non-Provisional application Ser. No. 13/831,398 filed 14 Mar. 2013 which claims priority under 35 U.S.C. 119(e) to U.S. Non-Provisional Application of U.S. Provisional Application Ser. No. 61/754,522 filed on 18 Jan. 2013; U.S. Provisional Application Ser. No. 61/750,286 filed on 8 Jan. 2013; and U.S. Provisional Application Ser. No. 61/747,051 filed on 28 Dec. 2012.

Additionally, the present application claims priority under 35 U.S.C. 120 to the following U.S. Non-Provisional Applications: U.S. Non-Provisional Ser. No. 17/478,821 filed on 17 Sep. 2021 and U.S. Non-Provisional application Ser. No. 17/478,807 filed on 17 Sep. 2021; which are continuations of U.S. Non-Provisional application Ser. No. 16/673,725 filed on 4 Nov. 2019; said Ser. No. 16/673,725 filed on 4 Nov. 2019 was a continuation of U.S. Non-Provisional application Ser. No. 15/152,437 filed on 11 May 2016; said Ser. No. 15/152,437 filed on 11 May 2016 was a divisional of U.S. Non-Provisional application Ser. No. 14/142,390 filed on 27 Dec. 2013 which claims priority under 35 U.S.C. 119(e) to U.S. Provisional Application Ser. No. 61/900,199 filed on 5 Nov. 2013 and U.S. Provisional Application Ser. No. 61/870,125 filed on 26 Aug. 2013.

All of said patents and applications are hereby incorporated in their entirety by this reference.

TECHNICAL FIELD

The present invention is generally related to aircraft and more specifically to a system and apparatus for monitoring a plurality of flight conditions and parameters, and on a condition selectively suggesting either a new flight profile or assuming flight control and then flying the suggested flight profile.

BACKGROUND

Whether flying a piston-powered personal craft or a multi-engine commercial jet, pilots are taught the same general priorities in emergency situations: aviate, navigate, and communicate—in that order. The pilot's first duty is self-evident: to fly the aircraft. To successfully do so requires the continual processing of a vast amount of data received via any number of different sources. During flight operations a pilot may be confronted with the loss of an engine on takeoff. In such a situation the pilot must immediately decide the safest option for the particular altitude and set of flight conditions, e.g., whether to: (a) turn approximately 180° and make a tail-wind landing; (b) turn at least

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270° and re-land; (c) crash straight ahead; or (d) limp or glide to another nearby airport. Altitude, position, aircraft performance, terrain, atmospheric and weather conditions, and pilot capability dictate the safest option. A pilot's options increase with altitude, performance, and the availability of landing sites (each providing different services). The pilot's options are inversely proportional to the severity of the emergency.

Autopilot, automated navigation and GPS systems have significantly increased the information available to pilots. More information, however, means more potential calculations for the pilot to make, more options to consider, and more information to filter. Other than destination, most of this information is dynamic, for example, position (including attitude), traffic, and weather (including wind speed and direction—both vary by altitude and heading). The pilot must balance the ongoing assessment of this continual stream of data (information) while aviating, navigating, and communicating. Unexpected conditions must be assessed and acted upon decisively and correctly. Depending on criticality, options narrow as time passes. Once a decision is made, the die is substantially cast.

These informational processing factors are complicated when conditions are less than ideal. Available information may not be complete or accurate. For example, a pilot climbing after takeoff over unfamiliar terrain experiencing an emergency is likely 1) aware that the airport runway lies only a few miles behind, and 2) aware of the vague location of additional airfields nearby in possibly deteriorating weather. In this example the pilot may not be aware, however, that an open field (or road or the like) a few miles distant would be a better emergency landing site, in that it would be more likely to be reached with altitude and time to execute a stabilized approach.

An emergency complicates these factors, and the corresponding pressure on the pilot, even further. The means of propulsion or other onboard systems may fail, making a safe landing simultaneously more urgent and more difficult to execute. A structural failure, cabin depressurization, or onboard medical emergency may occur, requiring the pilot to rapidly divert from the initial flight plan and find an alternative landing site (ALS). Emergency conditions add yet another degree of difficulty to the already complex responsibilities of piloting.

Therefore, a need exists for a system and method to aid the pilot of a distressed aircraft, thereby reducing pilot workload, the number of decisions based on inaccurate data, and the potential loss of life and property.

SUMMARY

The present invention relates to a system and apparatus for assisting pilots (flight crews) in determining the best option at successive points in a flight for any category of emergency. Generally, the system categorizes emergencies as (1) land immediately (red), 2) land as soon as possible (yellow), or (3) land as soon as practicable (green). The apparatus of the present invention alerts the pilot (crew) to the available options given a particular category of emergency (and set of flight conditions). In the alternative, an apparatus of the present invention (1) may assist the pilot (crew) in the form of a flight director (or checklist or the like) in executing a proposed landing solution to a given emergency, or 2) may direct the aircraft autopilot in executing the suggested (and selected) landing solution. In either case, the system of the present invention provides the pilot with time to contemplate and consider the emergency and its

resolution (while the aircraft is directed toward and configured for the best landing option given a particular emergency condition).

The present invention may ascertain the configuration of the aircraft from changes in the aircraft's position over time. For example, in a particular planned portion of a planned flight segment the aircraft may be expected to gain altitude over a particular distance at a particular rate. If the aircraft is not climbing at the expected rate it may be an indication that the aircraft is configured incorrectly, for example, improper power setting, or the gear or flaps may remain in takeoff position. Likewise, enroute, a particular aircraft may be expected to perform in a known set of atmospheric conditions within a known range of values. Any deviation from these values indicates a potential problem. Based on the aircraft such deviations may refer to a single likely source or a narrow set of sources. The system of the present invention may alert the aircraft crew to the potential problem and its likely source(s).

In a preferred embodiment of the present invention, even under normal flight conditions (planned or expected conditions) an audio display, a graphic display (HUD or smart glasses or the like) of available landing options given a category of emergency is continuously displayed. This display of information alerts a pilot to available options under various conditions and assists in training pilots in learning aircraft capabilities in various conditions and locations. Likewise, crew and dispatch may alter protocol in an effort to mitigate risks identified through the operation of an aircraft or a fleet of aircraft operating with the present invention.

Preferably, the present invention may continuously update ALS options and make the data available to the pilot upon request. For example, an ALS page on a well-known multifunction display (MFD), or a display screen on a tablet or like portable computing device, may indicate each ALS and graphically indicate the ability of the aircraft to reach each ALS. In addition, a graphical display of range data on the primary flight display or screen may aid the pilot in decision-making. For example, a pilot may opt to select on (or off) a graphical range ring indicating an engine out (zero-thrust) best glide range.

With additional data points made available through existing or added controls and sensors (e.g., auto throttles, acoustic profiles, flight control position indicators, rate of climb/descent, heading, and the like), the accuracy of the configuration data ascertained by an embodiment of the present invention increases. The present invention may ascertain over a series of data collection intervals the presence and scope of an unusual condition (correctable error or emergency), plan an emergency descent profile to the safest (most preferred) available landing site, and suggest troubleshooting options as the emergency profile is accepted and executed. In this manner, the present invention may assist the pilot in discovering and correcting aircraft configuration errors which if left uncorrected may lead to undesirable consequences, e.g., gear-up landings, overstressing aircraft components, flight delays or passenger discomfort.

In a presently preferred embodiment the apparatus comprises: (1) at least one computer processor; (2) a data bus or at least one sensor for collecting flight condition information such as aircraft position, weather, traffic, terrain, aircraft systems status, aircraft flight envelope parameters, pilot (crew) status/condition; (3) a ground-to-air data link; (4) a display system, and (5) a current database of information relating to (a) the aircraft (performance data), (b) pilot (biometric data, flight history, currency, proficiency), (c)

flight plan, and (d) bulk route characteristics (weather, terrain, landing sites, traffic, airspace, navigation). In an embodiment the system continually monitors a plurality of flight parameters, provides the pilot with current information based on those parameters, and upon a given condition prioritizes a course of action. In a preferred embodiment the system may either execute or guide a flight crew in flying a series of control inputs calculated to safely secure the aircraft on the ground. The system of the present invention at least temporarily relieves a pilot (crew) from the task of quickly calculating an emergency plan with its associated set of procedures so they may fly, configure, and troubleshoot (for the situation) while contemplating the acceptability of option(s) suggested by the present invention.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of an aircraft in a condition that necessitates altering or modifying an assigned or planned profile wherein the system of the present invention suggests a new or modified procedure to accommodate a condition, e.g., land immediately, land as soon as possible, and/or land as soon as practicable;

FIGS. 2A and 2B are environmental block diagrams representing embodiments of the present invention;

FIG. 3 is a top plan view of an "on" condition pilot display of an embodiment of the present invention illustrating areas of landing (ditching) opportunities (target radius) available upon a particular condition, wherein the pilot may operatively select to accept and fly a suggested new (emergency) procedure set, or in the alternative to accept each item from said new procedure set in seriatim;

FIG. 4A is a top plan view of a continuously updated pilot display of an embodiment of the present invention illustrating areas of landing (ditching) opportunities (target radius) available upon a particular condition, wherein the pilot may operatively select to accept and fly a suggested new (emergency) procedure set, or in the alternative to accept each item from said new procedure set in seriatim;

FIG. 4B is a diagram explaining generally how wind speed and direction may affect the target radius of available alternative landing sites reachable by a given aircraft under undesirable conditions;

FIG. 5 is a highly diagrammatic perspective view of an emergency condition (engine failure at takeoff) forced landing (straight ahead) procedure overview (display) with an associated queue wherein an embodiment of the present invention suggests a risk profile accessed emergency procedure set providing a return to airport (re-land) procedure;

FIG. 6 is a highly diagrammatic perspective view of an emergency condition (engine failure at takeoff) ditching procedure overview with an associated queue wherein an embodiment of the present invention suggests a risk profile accessed emergency procedure set (the procedure set being dependent on the specific nature of the engine failure) providing a best safe landing (ditching) opportunity procedure;

FIG. 7 is an environmental drawing of alternate landing site (ALS database, and subscription of an embodiment of the flight assistant of the present invention);

FIGS. 8A, 8B, 8C, 8D, and 8E are flow diagrams for presently preferred embodiments of operations of various aspects of the present invention;

FIGS. 9A and 9B are a diagrammatic plan elevation and plan of a flight plan of an embodiment of the present invention from and to an airport illustrating unusual condition detection and reporting;

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FIG. 10 is a perspective cockpit view of an embodiment of the present invention with a HUD (Heads Up Display) illustrating a suggested landing site selected by a system of the present invention (at least partially on aircraft and pilot performance, position, configuration, propulsion, traffic, weather, terrain, cabin environment, ground resources (services)), and preselected risk profile hierarchy;

FIG. 11 is a perspective cockpit view of an embodiment of the present invention with a HUD (Heads Up Display) illustrating a suggested landing site selected by a system of the present invention at least partially on aircraft and pilot performance, position, configuration, propulsion, traffic, weather, terrain, cabin environment, ground resources (services), and a preselected risk profile hierarchy;

FIG. 12 is a highly diagrammatic perspective view of an emergency condition (medical divert) landing procedure overview with an associated queue wherein an embodiment of the present invention suggests a risk profile accessed emergency procedure set providing a landing procedure at an airport with suitable nearby medical facilities;

FIG. 13 is a highly diagrammatic perspective view of a landing procedure overview with an associated queue wherein an embodiment of the present invention suggests a risk profile accessed emergency procedure set providing for safe departure from the North Atlantic track route system and emergency landing at suitable nearby airports;

FIG. 14 is a highly diagrammatic perspective view of an emergency condition (pressurization emergency) forced landing procedure overview with an associated queue wherein an embodiment of the present invention suggests a risk profile accessed emergency procedure set providing for immediate descent to safe altitude and landing at the nearest available airport;

FIG. 15 is a pilot view of an onboard display unit wherein an embodiment of the present invention displays navigational information related to a final approach and landing;

FIG. 16 is a highly diagrammatic top plan view of emergency condition landing procedure overview with an associated queue wherein an embodiment of the present invention suggests a risk profile accessed emergency procedure set providing a landing procedure at a nearby airport;

FIG. 17A is a pilot view of an onboard display unit in an emergency condition (engine out) diversion procedure overview with an associated queue wherein an embodiment of the present invention suggests a hierarchy of risk profile accessed emergency procedure sets providing for best safe landing opportunity procedures at suitable emergency landing sites;

FIG. 17B is a pilot view of an onboard display unit in an emergency condition (single engine) diversion procedure overview with an associated queue wherein an embodiment of the present invention suggests a hierarchy of risk profile accessed emergency procedure sets providing for best safe landing opportunity procedure at suitable emergency landing sites, and the pilot may operatively select to accept and fly a suggested new (emergency) procedure set, or in the alternative to accept each item from said new procedure set in seriatim;

FIG. 18 is a tabular representation of data inputs utilized by an embodiment of the present invention to generate procedure sets, and actions related to the generation of procedure sets and to the outcomes of those procedure sets;

FIG. 19A is a pilot view of an onboard display unit wherein an embodiment of the present invention displays the initial flight plan and illustrates initial areas for landing (ditching) opportunities (target radii) available upon a particular condition;

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FIG. 19B is a highly diagrammatic perspective view of an emergency condition diversion and landing procedure overview with an associated queue wherein an embodiment of the present invention suggests a hierarchy of risk profile accessed emergency procedure sets (the precise procedure set/s, and the hierarchical weight of each set, being dependent on the specific nature of the emergency) providing a best safe landing (ditching) opportunity procedure;

FIG. 20 is a highly diagrammatic top plan view of an emergency condition (engine failure at takeoff) overview with an associated queue wherein an embodiment of the present invention suggests a hierarchy of risk profile accessed emergency procedure sets providing a best safe landing (ditching) opportunity procedure;

FIG. 21 is a highly diagrammatic top plan view of an emergency condition (engine failure at takeoff) overview with an associated queue wherein an embodiment of the present invention suggests a hierarchy of risk profile accessed emergency procedure sets providing a best safe landing (ditching) opportunity procedure;

FIG. 22 is a highly diagrammatic top plan view of an emergency condition (engine failure at initial climb) overview with an associated display queue wherein an embodiment of the present invention suggests a hierarchy of risk profile accessed emergency procedure sets providing a return to airport (re-land) opportunity procedure;

FIG. 23 is a highly diagrammatic top plan view of an embodiment of the present invention illustrating areas of landing (ditching) opportunities (target radius) available upon a particular condition, wherein the pilot may operatively select to accept and fly a suggested new (emergency) procedure set, or in the alternative to accept each item from said new procedure set in seriatim;

FIG. 24 is a highly diagrammatic top plan view of an embodiment of the present invention illustrating areas of landing (ditching) opportunities (target radius) available upon a particular condition;

FIG. 25 is a diagram illustrating angle of attack and rotational axes;

FIG. 26 is a graph illustrating surface and aloft winds at selected altitudes along a flight plan;

FIG. 27 is a table illustrating available services at various airports;

FIG. 28 is an environmental block diagram of a surface- and space-based embodiment of the present invention, configured before or during flight for implementation by a portable computing device aboard an aircraft;

FIG. 29 is an environmental block diagram of a space-based embodiment of the present invention, configured before or during flight for implementation by a portable computing device aboard an aircraft;

FIG. 30 is an environmental block diagram of a surface- or non-space-based embodiment of the present invention, configured before or during flight for implementation by a surface-based computing device in concert with a portable computing device aboard an aircraft;

FIG. 31 is a highly diagrammatic overhead view of an embodiment of the present invention, configured for implementation by at least one surface-based computing device in concert with a plurality of aircraft in flight, whereby each aircraft in flight maintains a two-way stable data link with a surface-based device for the purpose of implementing the system of the present invention;

FIG. 32 is an environmental diagram of the aircraft configuration module component of an embodiment of the present invention;

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FIG. 33 is a diagram illustrating at least one of the individual components of an aircraft configuration determined by an embodiment of the present invention, and a condition whereby the individual data components of an aircraft configuration at any given point in time are compared to an expected configuration of said aircraft at said point in time as determined by the system of the present invention;

FIG. 34 is an environmental diagram of the trajectory evaluator module component of an embodiment of the present invention;

FIG. 35 is an environmental diagram of the configuration evaluator module component of an embodiment of the present invention;

FIG. 36 is an environmental diagram of the alternative landing module component of an embodiment of the present invention;

FIG. 37A is a highly diagrammatic overhead view of a procedure whereby an embodiment of the present invention initially evaluates potential alternative landing sites appropriate for a given aircraft commanded by a given pilot along a given flight path, based on surface terrain characteristics and aircraft landing requirements;

FIG. 37B is a diagram illustrating the process by which an embodiment of the present invention evaluates a potential landing site by determining the landing distance required for a given pilot to safely land a given aircraft;

FIG. 38 is an environmental diagram illustrating the main database component of an embodiment of the present invention, a component dataset of said database exported for use by the system of the present invention for a given flight, pilot, and aircraft, and an additional dataset representing information collected during said flight with which the system of the present invention updates said main database;

FIG. 39 is a diagram representing the relationship between the energy states of an aircraft in flight;

FIG. 40 is a diagram representing the optimal engine-out/zero-thrust glide distance achievable by an aircraft of given glide ratio, and how said optimal distance may be affected by atmospheric conditions or pilot proficiency;

FIG. 41 is a highly diagrammatic top plan view of an emergency condition whereby an embodiment of the present invention suggests at least one alternative flight path based on adverse weather conditions in the original flight path or, in the alternative, suggests a hierarchy of risk profile accessed emergency procedure sets providing a best safe landing opportunity procedure to an aircraft affected by said adverse weather conditions; and

FIG. 42 is a pilot view of the display unit of a portable computing device whereby an embodiment of the present invention displays high priority alternative landing sites along the flight path of an aircraft, as well as current atmospheric conditions at the said alternative landing sites.

DETAILED DESCRIPTION

In the following detailed description, embodiments are described in sufficient detail to enable those skilled in the art to practice the invention. It is understood that other embodiments may be utilized without departing from the scope of the present invention. The following detailed description should therefore not be taken as limiting in any way the scope of the present invention.

Features of the present invention in its various embodiments are exemplified by the following descriptions with reference to the accompanying drawings. These drawings depict only selected embodiments of the invention, and

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should not be considered to limit its scope in any way. The present invention may be described with further detail and specificity through use of these drawings.

The present invention relates to a system and apparatus for monitoring and processing a plurality of flight parameters in order to minimize workload and stress on a pilot due to unexpected (undesired) conditions. The apparatus includes a database of information from which is extracted a dataset that is static relative to any given flight at the point of departure (a "flight" referring to a set whose elements are: an aircraft; one or more pilots; an initial, unexecuted flight plan; and enroute flight path conditions, both dynamic and static. In operation, the system continually processes dataset components in concert with dynamic data relative to a particular point along the flight path, including: the aircraft's position, altitude, heading, and airspeed; its performance relative to benchmark values as determined by the aircraft's flight envelope and flight plan; and current cabin, flight, and engine conditions (including emergency states that might require an unscheduled landing). Additionally, a preferred embodiment determines the availability of airports or other alternative landing sites (including fields, roads, and bodies of water) within the aircraft's range at that moment, as well as current conditions at those landing sites (such as available services, weather, wind, terrain, obstacles, or ground traffic). An embodiment of an apparatus of the present invention may then continually ascertain and display options for any given point and suggest a procedure executable by the pilot or autopilot system providing for an emergency landing at an alternative landing site (including any necessary course changes or aircraft reconfiguration).

In addition, the system may note any significant deviations in the aircraft's performance (relative to its performance envelope and expected performance at a particular point on a flight plan profile), which might result from inappropriate configuration (of landing gear, flaps, or the like). Generally an aircraft may be configured for taxi, takeoff, climb, cruise, descent, approach, landing, or penetrating turbulent air. If an aircraft is inappropriately configured for a particular segment of flight, the present invention may notify the pilot/crew and suggest a configuration correction, such as lowering landing gear or adjusting flaps, and the like such that the aircraft may be correctly configured for the desired flight segment. The suggestion may be generic, e.g., "CHECK CONFIGURATION: _____ AIR SPEED _____ GRADIENT _____ HEADING EXPECTED". Additionally, in further embodiments the system and apparatus may provide a detailed or specific suggestion, e.g., "CHECK GEAR/FLAPS", "CHECK AIR-SPEED", "CHECK PITOT/STATIC", "CHECK PITCH ANGLE" and the like.

FIG. 1 depicts an aircraft 200 in flight; the system has plotted a path and corresponding emergency procedure set 202 to approach and land on a nearby ALS consisting of an airport runway 208, accounting for obstacles (i.e., natural and manmade) in the flight path (vicinity) and wind conditions on descent and at the landing site. In addition to navigating to the best available ALS, procedure set 202 preferably provides (when possible) for aligning aircraft 200's pitch angle for best possible glide speed, configuring aircraft 200 for landing, and the like. Procedure set 202 also provides (when possible) for a landing into wind relative to aircraft 200's approach, thereby reducing the landing speed and required landing distance. FIG. 1 also illustrates operation of an embodiment of the present invention where the apparatus of the present invention has identified an opportunity gate 210 for aircraft 200 final approach 204 and

targeted touchdown point **212** on runway preferred touchdown zone nearest aircraft **200** (system effort to maximize safety).

FIGS. 2A & 2B are diagrammatic representations of the environment of an embodiment of the present invention, including both hardware components of the apparatus and data components used by the system. For a given aircraft, the dataset includes information about its specifications, e.g., size, empty weight, engine(s) operational parameters, fuel capacity and consumption characteristics, flight configuration performance characteristics, and general performance envelope **506** plus derivative data (e.g., the runway length required for a normal landing and the suitability of various landing surfaces). In a currently preferred embodiment, all or substantially all of the information in the Pilot's Operating Handbook (POH, -1, or Flight Manual) may be included (e.g., all performance data and emergency procedures).

For a given pilot (aircraft, leg), the dataset may include general information such as pilot experience level (such as the pilot's flight time for any given aircraft) and past performance **534**, as well as specific and derivative flight performance data. In operation a preferred embodiment of the present invention may continuously ascertain current position relative to an expected position (per the flight plan or previous leg/segment position) and determine whether a reportable problem exists. An apparatus of the present invention may be selectively programmed with a range of operational values (via a menu or the like to a user/operator set of parameters) for each operational segment. Ideally, takeoff and landing segments may have tighter (tighter/narrower more periodically detected values) so a pilot/crew may more quickly be alerted to deviations.

In a preferred embodiment the invention may, for example, monitor previous traffic and arrival/departure information to improve accuracy (of operation of the present invention). For example (selectably per the user/operator) on a flight plan to KSTL/St. Louis the system may detect that landing traffic (from the North) is now landing on runway 11 via the AARCH ONE arrival (rather than on runway 30 via the QBALL EIGHT arrival) due to a wind shift or the like. Within the constraints of this changed condition, the system may alert the crew to a heading deviation consistent with the new AARCH path. Additionally, satellite (or the like) data over a period of intervals, or road traffic applications, may be utilized by an embodiment of the present invention to ascertain the relative risk associated with a potential ALS (field, e.g., growing crop, plowed, row direction; or road, e.g., slope, width, and current or predicted level of traffic).

For a given flight plan **502**, the dataset may include the initial flight path and related information (including information derived by the present invention). This may include, for example, terrain **520**, including both topographic features and manmade obstructions) along the path; nearby airports along the path suitable for landing **524** and their navigation/communication frequencies; and any services near those airports (parts/repair, fuel, hospitals, currently available accommodations and/or transportation). Derivative data may include risk ranked ALS along or near the initial path, along with surface (typical/predicted), gradient, elevation, obstacles, and other information relevant to an attempted approach and landing **512**. Alternative landing sites may include open fields of sufficient dimension for the aircraft, paved surfaces such as highways or parking lots, or bodies of water.

In-flight, the system may monitor dynamic values for any given point along the flight path, including: the aircraft's current position **510**; its airspeed, heading, and altitude

above sea level; atmospheric conditions such as air pressure/temperature and wind speed/direction **522**; cabin conditions **508**; flight controls and settings **514**; and propulsion system conditions **518**. The system may also monitor information available via data link, including: local air and ground traffic for a given position **516**; current and forecast weather along the initial path **504**; and conditions at any nearby ALS where available. Some embodiments may also monitor biometric information about the pilot and/or crew **530**, including: brain activity; breathing and heart rate; reaction times (and changes thereto); signs of nervousness or drowsiness; or other vital signs. In such an embodiment of the present invention, collected biometric information may be utilized to ascertain acceptable (obtainable) ALS.

The system bus **532** connects the various components responsible for the collection of these diverse data points to the processor **800** and comparator **540** for data processing. A display unit **700** with user interface **550** displays pertinent information and processing output to the pilot, while at the same time allowing for pilot input to reflect a change in conditions (for example, declaring an emergency state [pan, pan; mayday, mayday]) that in turn would change system parameters. An apparatus of a preferred embodiment may include a user selectable switch for selectively activating or selecting an emergency protocol (for a particular situation/condition within a segment [R_f , R_{P1} , or R_{P2}]). R_f for land immediately (or e.g., eject, activate airframe parachute); R_{P1} for land as soon as possible; and R_{P2} for land as soon as practicable.

In a presently preferred embodiment, the system may continually evaluate both dataset components and dynamic values to determine the best available ALS, or a weighted hierarchy of alternatives (if more than one exists). The system may initially select sites from those nearby landing sites provided by the dataset. The system may consider additional sites suitable for a given aircraft (but more distant from the initial flight plan) if those sites fall within a predetermined range of the aircraft's current position (i.e., the target radius) or an emergency state is declared.

For any given set of more than one ALS, the system may assign a weighted value to each individual ALS, corresponding to that site's suitability for landing based on available current (anticipated at arrival time) conditions. This assignment may account for a variety of factors including: (1) the site's distance from current position; (2) atmospheric conditions at current position and at the ALS (if available or derivable); (3) ground terrain at the ALS, including surface composition and the presence of nearby trees, brush, vegetation, or other obstacles; (4) the presence of hospital, security, repair, or other facilities near the ALS; and (5) the difficulty of navigating to and landing at the ALS for a pilot of given skill (current performance) and experience level.

This assignment, and the resulting best ALS or hierarchy thereof, may continually refresh as conditions and contributing factors change. The results of this assignment may be available for display to the pilot, refreshing as system results update.

The system may plot an optimal path to each identified ALS. This path may be represented by a set of points in three dimensions comprising a navigable path from the aircraft's current position to a ground-level touchdown point at the ALS. In plotting these paths, the system may incorporate aviation rules and best practices (e.g., landing into headwinds where possible to minimize landing speed, maintaining safe distances from neighboring air (ground) traffic, setting a touchdown point that maximizes the available landing surface). The system may continually revise paths as

the hierarchy of potential landing sites (as well as the aircraft's precise position along its flight path) changes.

FIG. 3 depicts aircraft 200 in an engine-out state, in need of immediate landing. Without allowing for wind, the system uses target radius 214 (reflecting an immediate need to land) in order to determine the best available ALS at runway 208a, here the site nearest aircraft 200's position. The system then plots emergency course 202a to opportunity gate 110a and landing on runway 208a. As a substantial wind may affect aircraft 200's glide capability in an engine-out state, it may also affect the system's choice of best available ALS. Here the presence of wind over aircraft 200's port side (100 kts) flattens the circular target radius into an ellipse 216, and runway 208b is selected as the best available ALS despite its greater physical distance from aircraft 200's position. The system then plots paths along 202b, 202c to opportunity gates 110b or 110c for landing at runway 208b.

FIG. 4A depicts aircraft 200 and potential alternative landing sites 208(P2), 208(P1), and 208i. If an emergency state is declared, the system parameters for determining the best available ALS may change. Target radius 218 represents a need to land as soon as practicable (considering any available services within a broad radius), while radius 220 reflects a more pressing need if aircraft 200 is in single-engine state (i.e., one of aircraft 200's two engines has failed) to land as soon as possible (e.g., first safe opportunity depending on VFR). While one landing site 208(P2) may be an airport runway, if aircraft 200 is in a single-engine state it may lose priority as an ALS to closer landing sites 208(P1) and 208i. The presence of hospital 222 adjacent to 208(P1) means that if aircraft 200 is in a medical-emergency state, 208(P1) may gain priority as an ALS due to nearby facilities. Finally, if aircraft 200 is in an engine-out state (all engines have failed), the system's land-immediately target radius 214 is narrower still. Landing site 208i, situated inside target radius 214, may maintain priority as an ALS.

FIG. 4B explains generally how wind speed and direction may affect the range of an aircraft experiencing engine failure, and consequently the shape of the target range used to select best available landing sites. A slight wind over aircraft 200's starboard side elongates aircraft 200's target radii 214, 220, 218 slightly on that side.

Condition Indicator Possible Scenario Radius

Land immediately R_I Dead Stick 214

Land as soon as possible R_{P1} Single Engine 220

Land as soon as practicable R_{P2} Medical Emergency 218

The present invention may display landing opportunities within ranges (R_P , R_{P1} , R_{P2}) as circles (ellipses) on an aircraft's Multi-Function Display (MFD) or the like. Additionally, ALS suitability may be represented and displayed by color-coded icons (green, yellow, orange, red, or the like). The display range at which this and other information is presented may be user or system selected. In operation, an MFD (HUD or the like) serves to display information when the selected (user or system) range makes various classes (types) of information relevant. For example, in an effort to reduce clutter on moving map displays and the like, detailed surrounding terrain is displayed depending on altitude, airspeed, glide range, and distance (e.g., 20 NM). To reduce clutter traffic information may nominally be displayed at ranges between five and ten NM. Weather (and the like) is generally displayed at ranges of 200 NM and less.

Approach courses, and the corresponding target windows projected by the HUD, may vary depending on the specific emergency state. FIG. 5 depicts aircraft 200 experiencing engine failure after takeoff from runway 208. The system has selected a nearby open field as the best available ALS.

Aircraft 200's glide slope and speed may vary depending on whether its emergency state is single-engine (210b), aux-engine (210c), or engine-out (210a), and the optimal approach course (and corresponding opportunity gates) may vary accordingly. If aircraft 200 is in an engine-out state, for example, its glide slope may be steeper and its corresponding opportunity gate (210a) higher relative to the final touchdown point.

FIG. 6 depicts aircraft 200 experiencing engine failure immediately after takeoff from originating airport 224. The system is directing aircraft 200 along emergency course 202, and preparing to ditch in a nearby river 226a (section 226b, being unreachable by aircraft 200, is unsuitable for ditching). Again, the aircraft's specific engine emergency (engine-out/210a, single-engine/210b, aux-engine/210c) may determine its glide slope and speed, and its precise approach and corresponding opportunity gate (210a, 210b, 210c) toward touchdown point 212 may vary accordingly.

FIG. 7 depicts an onboard display unit 700, which shows an aircraft in-flight from KSLN/Salina to KDEN/Denver to KSUX/Sioux City to KMSP/Minneapolis-Saint Paul per its initial flight plan (not over KMCK). Onboard display unit 700 also displays initial target radii calculated by the system for immediate landing 222, landing as soon as possible 220, and landing as soon as practicable 218, as well as a no-wind opportunity radius 214.

The system may record all data generated in-flight in memory storage. Flight data is streamed (or batch loaded) for incorporation into a database of the present invention for auxiliary purposes (e.g., comparison of a current flight to previous flights in order to predict and/or detect unusual conditions).

At least two primary embodiments of the present invention may be delineated in operation by the means utilized to determine the existence of an unusual condition. Where data is available from existing aircraft systems, such as position, airspeeds, flight control positions, attitude and angle of attack, propulsion and cabin condition, that data may be utilized by the present invention via the bus 532 (FIG. 2) to assist an apparatus of the present invention in ascertaining status and/or an unusual condition. For example, an ARINC 429, MIL-STD-1553, UNILINK, or like avionics data bus protocol may be utilized by the present invention. Likewise, an embodiment of the present invention may sufficiently derive the necessary information from limited available data. For example, an aircraft equipped with a GPS or other navigation system (inertial guidance, VOR, RNAV, LORAN, and/or ADF or the like) may be sufficient. Likewise, an embodiment of the present invention may include a GPS. In operation such an embodiment receiving continually refreshed GPS data related to the aircraft's position inflight may derive from those data further information related to the aircraft's airspeed, heading, attitude, rate of climb or descent, configuration, propulsion, or cabin conditions, and thereby assist an apparatus of the present invention in ascertaining status and/or an unusual condition.

For example, the present invention may interpret the aircraft 200's GPS coordinates as placing the aircraft (for any unique time T_x) at a point P_x of coordinates (X_x , Y_x , Z_x), where X_x and Y_x correspond to that point's latitude and longitude, and Z_x to its altitude above mean sea level (MSL) and ground level (AGL). The present invention may then interpret the point of takeoff as (X_0 , Y_0 , Z_0) at time T_0 , where X_0 and Y_0 represent the latitude and longitude of the current flight's origin point and Z_0 (relative to the ground at the point of takeoff) is zero. The present invention may then interpret subsequent GPS data as representing a series of

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points $(P_0, V_{X_0}, Y_0, Z_0), \dots, P_n(X_n, Y_n, Z_n), P_{n+1}(X_{n+1}, Y_{n+1}, Z_{n+1}), \dots, P_L(X_L, Y_L, Z_L)$ along the aircraft's flight path, from liftoff at P_0 to touchdown at P_L . For any two such points $P_a(X_a, Y_a, Z_a)$ and $P_b(X_b, Y_b, Z_b)$, the present invention may easily derive the total distance d traveled relative to the ground (over the time interval T_a to T_b) as

$$\cos^{-1}(\sin x_b \sin x_a + \cos x_b \cos x_a \cos(x_b - x_a)),$$

or

$$2 \sin^{-1} \sqrt{\left(\sin \frac{x_b - x_a}{2}\right)^2 + \cos x_a \cos x_b \left(\sin \frac{y_b - y_a}{2}\right)^2},$$

the initial course from P_a to P_b (of distance d) as

$$\begin{cases} \sin(y_b - y_a) < 0: \cos^{-1} \frac{(\sin x_b - \sin x_a) \cos d}{\sin d \cos x_a} \\ \text{else: } 2\pi - \cos^{-1} \frac{(\sin x_b - \sin x_a) \cos d}{\sin d \cos x_a} \end{cases},$$

and the rate of climb or descent over the time interval as:

$$\frac{(Z_b - Z_a)}{(T_b - T_a)}$$

The present invention may also derive information related to the aircraft's airspeed, allowing for variances in wind speed and atmospheric pressure. For the aircraft's takeoff and initial climb, beginning at liftoff

$\{P_0(X_0, Y_0, Z_0), \text{time } T_0\}$,

and concluding when the aircraft reaches

$\{P_c(X_c, Y_c, Z_c), \text{time } T_c\}$,

the aircraft reaches cruising altitude Z_c , climbing at an average rate of

$$\frac{(Z_c - Z_0)}{(T_c - T_0)}$$

and traveling a distance (relative to the ground) of

$\cos^{-1}(\sin x_c \sin x_0 + \cos x_c \cos x_0 \cos(x_c - x_0))$ while climbing. The present invention may also, for example, derive the aircraft's angle of climb as:

$$\tan^{-1} \frac{Z_c - Z_0}{\cos^{-1}(\sin x_c \sin x_0 + \cos x_c \cos x_0 \cos(x_c - x_0))}$$

A preferred embodiment of the present invention may ascertain the existence of an unusual condition by comparing available data related to the aircraft's performance in-flight (e.g., its position, altitude, airspeed, attitude, heading, rate of climb/descent) to performance norms (ideals) stored in an onboard dataset. Data sources from which these performance norms may be derived include the pilot's past performance history while flying the current route or under similar flight conditions, the aircraft's expected performance along a given flight plan or under similar flight conditions, or optimal performance conditions for a given aircraft at any point within a given flight plan (and/or flight segment).

Likewise, an embodiment of the present invention may note as an unusual condition any deviation of a particular performance factor, or set of factors, from performance

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norms and respond to a detected (ascertained) unusual condition according to one or more user selectable protocols (depending on the nature and severity of the condition). However, the precise course of action recommended by the present invention in response to an unusual condition may vary depending on the specific phase of flight in which the condition occurs (including taxi, takeoff, initial climb, cruise, descent, approach landing, and weather avoidance). Similarly, depending on the specific phase of flight in which a deviation from performance norms occur, the present invention may account for a broader or narrower deviation from performance norms in determining whether a deviation represents a routine event (associated with a configuration fix or procedure set that can be communicated to the pilot or autopilot system) or an unusual condition (including a potential emergency requiring diversion from the initial flight plan). For example, a deviation of two percent from expected cruising altitude may not be interpreted as an unusual condition (requiring only continued observation at that time, with possible action taken if the deviation persists or increases) while a similar deviation in altitude during the initial climb phase (approach or landing) may be interpreted as an unusual condition potentially requiring correction (and brought to the pilot's attention). Similarly, a preferred embodiment of the present invention may ascertain whether an unusual condition is a reroute, minor deviation, a configuration error, or a more serious problem (a potential emergency). The range of acceptable deviations from, for example, an idealized, expected norm, may be user/operator selectable and may vary by flight segment (and/or airspeed and altitude). Generally, in a preferred embodiment, tighter ranges (of acceptable values) are utilized the closer the aircraft is to the ground, other aircraft, or weather and the like.

For example, an embodiment of the present invention may identify a significant loss of airspeed in flight that may in turn indicate a partial or total failure of the propulsion system. If this loss of airspeed occurs at cruise, the present invention may suggest reconfiguration of the aircraft as a remedy, e.g., correction of improper use of flaps, power setting, and/or angle of attack. If the loss of airspeed is not remedied by reconfiguration, the present invention may then suggest other courses of action. In the alternative, immediately after takeoff the present invention may interpret a significant loss of airspeed as an emergency or a potential emergency. Based on a variety of factors (including but not limited to the aircraft's altitude, the availability of alternative landing sites, and wind conditions), the present invention may then suggest an emergency landing, advising the pilot as to possible emergency procedure sets (turning in excess of 180° to land at the originating airport, gliding forward to an alternative airport, or touching down at some nearby alternative site suitable for landing) and the relative risk of each course of action.

In addition, embodiments of the present invention may track, collect and transmit data according to an established set of requirements. Such requirements may include a Flight Operations Quality Assurance (FOQA) program and the like. Such requirements may track operational data over time and transmit data to a central operational facility for follow on analysis. Future training or future simulator scenarios may be based on such analysis. Further, pilot specific data may be recorded for future pilot specific training. For example, should a specific pilot maintain a consistent set of errors over time, the systems of the present invention may create a training scenario for the specific pilot based on the consistent set of errors.

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FIG. 8A depicts the underlying process by which, under routine flight conditions, the system of a preferred embodiment of the present invention may continually collect position data at selectable time intervals. Based on this information, the system may develop a continually refreshing hierarchy of the best available landing sites. FIG. 8B depicts the subroutine by which the system creates the ALS hierarchy, augmenting via ground-to-air data link the information previously downloaded from the onboard dataset (which may include terrain, traffic, and service information about airports and other alternative landing sites along the flight path). When this information is current, the system may then evaluate and rank available landing sites within a given radius, storing the results and displaying them to the pilot via display unit 700.

While this subroutine continually runs, the comparator 540 may also continuously assess incoming and derived flight data (which may include information about the aircraft's position, altitude, airspeed, attitude, etc.) in comparison to data patterns in the onboard dataset. These data patterns represent performance norms and may include expected data points relative to the history of a particular flight plan or leg, and the pilot's past performance on the current or similar routes. If a deviation from performance norms is detected or ascertained, the system may assess whether the deviation is sufficient to constitute an unusual condition. If an unusual condition exists, the system may notify the pilot via display unit 700, and may then further assess whether the unusual condition is associated with a configuration error (change) or, in the alternative, an emergency profile. If there is a configuration change associated with the deviation, the system will suggest the appropriate correction to the pilot via the display unit 700, or communicate the necessary changes to the autopilot system if it is currently active.

If there is no appropriate configuration correction (trouble solution or fix) to address the current deviation, the system may notify the pilot via the display unit 700, either recommending the activation of R_{P2} land-when-practicable status or activating that status through the autopilot system. The system may then compare current flight data with emergency profiles stored in the onboard dataset in order to determine if the current deviation from performance norms is indicative of an emergency or potential emergency. FIG. 8C depicts data components included in an emergency profile by a preferred embodiment of the present invention. Certain conditions, e.g., a rapid descent or significant loss of speed at climb or descent, may be associated with a particular emergency profile such as an engine failure. The magnitude of the deviation (e.g., a 25 percent vs. 50 percent loss of speed at initial climb) may inform the system of the severity of the emergency, and the system may set the corresponding target radius accordingly (R_{P2} , R_{P1} , or R_I) depending on urgency. The system may then adjust landing site priorities depending on the specific emergency, prioritizing medical, security, or other services in addition to the size of the target radius and feeding these new priorities to the ALS search routine (FIG. 8B). Finally, the system may load checklists and emergency procedure sets associated with the particular emergency, displaying them to the pilot via the display unit 700 for execution or sending them to the autopilot system for execution in the event of a diversion. The pilot may also activate an emergency state manually through the system interface 550.

When an emergency state is active, the pilot may be presented with current ALS information pertinent to the current emergency, displayed via the display unit 700. The

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pilot may then divert to an ALS. FIG. 8D depicts the information that may be displayed to the pilot via the display unit 700 when an emergency state is active and a diversion is imminent. For every viable ALS identified by the system, the system may associate with that ALS a calculated path in three dimensions to that ALS, as well as a set of emergency procedures necessary to effect a landing there. The pilot may choose to divert automatically, in which case the autopilot system will execute the associated emergency procedure set. In the alternative, the pilot may choose to execute a manual diversion to a particular ALS. In this case, the display unit 700 will display the associated emergency procedure sets and checklists for the pilot to execute in seriatim.

FIG. 8E depicts the data components of the main database and associated onboard dataset utilized by a preferred embodiment of the present invention. The database may contain information specific to pilots, aircraft and aircraft types, flight plans and legs, and selectable parameters for the system of the present invention. Prior to flight, the pilot may download from the main database information related to the pilot's history and past performance, the specifications and expected performance of his/her aircraft and aircraft type, the expected flight plan (including a history of expected performance associated with that particular flight plan or leg), and system parameters that may be selectable by the pilot or determined by a commercial carrier (including emergency procedure sets, system sensitivity settings and associated flight stages, rules and policies, and best practices). Parameters used by the system of the present invention may be scalable depending on aircraft categorization.

Aircraft are most typically categorized by weight and mission. For the purposes of an embodiment of the present invention aircraft may be categorized as: (1) general aviation (small <12,500 lbs and large >12,500 lbs); 2) commercial transport aircraft; or (3) military aircraft. Small general aviation aircraft tend to be low flying (non-pressurized) and have little excessive reserve performance. Commonly they are single-engine piston powered with only nominal performance reserve during all but taxi, descent, approach and landing flight segments. For this reason a reduction in or loss of propulsion is always an emergency (R_I). Larger general aviation aircraft tend to be pressurized and may have multiple turbine engines. Thus, larger general aviation aircraft are operated at significantly higher altitudes. Upon a loss of or reduction in propulsion, larger general aviation aircraft have an increased gliding distance and generally some propulsion. Thus, the loss of an engine generally requires descent and landing (R_{P1}). Commercial transport aircraft are certified under different standards and have required performance criteria making the continuation of a flight after the loss of an engine safer and less time critical (R_{P2}).

Military aircraft are generally designed to operate in extreme conditions at the boundaries of a broad flight envelope. In a hostile operating theater a damaged or failing aircraft may have few readily discernable options. In an operation of a military embodiment of the present invention, the apparatus may analyze aircraft and pilot performance in a threat theater and offer ALS risk analysis based upon identified options. For example, a wounded crew member, less than optimally performing pilot, a damaged aircraft, in an environment containing multiple threats (ground and/or air) will be greatly assisted by an embodiment of the present invention. As an embodiment of the present invention is notified of threat location and movement, aircraft and crew performance, mission plan, mission capabilities (changed or deteriorating), and position information, it may continuously display or point to a risk assessed option or set of options

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(e.g., mission abort, divert, egress direction). In highly critical situations an embodiment of the present invention may selectively execute a mission abort or selectively execute a return to base (RTB) where the crew is unresponsive. Additionally, such an embodiment of the present invention may be configured to transfer control of the aircraft to itself or ground (or wing) based control.

FIG. 10 depicts aircraft 200 making an emergency landing in a suburban area. The system has selected field 228 (a relatively open area whose location allows the pilot to land into the wind) as the best available ALS, and the pilot has executed to divert. The projected window 210 represents the pilot's opportunity gate, assisting in targeting the near end of the ALS in order to maximize the available emergency landing space.

FIG. 11 depicts aircraft 200 continuing to land in the suburban area. The system has selected an ALS (consisting of an open area, relatively free of obstacles, with favorable winds) and identified an emergency touchdown point; the pilot has diverted to that ALS. Projected window 210 serves to assist the pilot in touching down in such a way as to maximize the available treeless area for landing.

FIG. 12 depicts aircraft 200 in a medical-emergency state. After executing a divert from flight plan 240 to emergency course 202, aircraft 200 navigates to opportunity gate 210 to begin its final approach and land on runway 208, the best available ALS. In addition to runway 208, the system has evaluated nearby river 226 as a potential ALS. However, due to the fact that the medical emergency may require an R_{P2} profile, river 226 may display on a lower level of the hierarchy. Note the presence of hospital facilities in close proximity to runway 208, and that aircraft 200's course enables landing into a headwind.

FIG. 13 depicts aircraft 200 inflight eastbound over the North Atlantic Ocean, along flight plan 240. In order to ensure aircraft separation in an area with high traffic and sporadic radar coverage, air traffic is directed along well-known parallel tracks 102a, 102b, and 102c, here depicted as 30 NM apart). Should aircraft 200 divert north to an ALS on island 230a, or south to an ALS on island 230b, the system may plot a course 202b, 202c) that first directs aircraft 200 parallel to track routes, maintaining a maximum distance of 15 NM from either adjacent track. Then, the system will allow aircraft 200 to exit the track system at an altitude safely below other aircraft.

FIG. 14 depicts aircraft 200 in a pressure-emergency state, at cruising altitude (FL380="flight level 380"=pressure altitude 38,000 ft) over mountainous terrain. In the event of a pressurization failure, aircraft 200 executes a diversion from initial flight plan 240. Aircraft 200's emergency path takes it through a mountain pass at FL160 (flight level 160=16,000 ft, 202a). Once clear of mountainous terrain, aircraft 200 descends along path 202b to FL100 (flight level 100=10,000 ft, 202b), at which altitude lack of pressure is no longer an immediate danger. Aircraft 200's emergency course then proceeds downwind 202c) to opportunity gate 210 for final approach and emergency landing on ALS 208, an airport runway.

FIG. 15 depicts RNAV navigational display information to aircraft 200 on approach to KMLE/Millard Airport, including navigational beacons and waypoints and nearby obstacles (and their elevations). Waypoint NIMMU serves as the initial approach fix 232 and opportunity gate 210 for final approach to land at KMLE Runway 12 208. Should aircraft 200 fly toward initial approach fix 232 in an incorrect configuration, the system may alert the pilot via display unit 700 of a suggested configuration change.

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FIG. 16 depicts aircraft 200 diverting from its initial flight plan 240 to emergency path 202 (and executing the associated emergency procedure set). Emergency path 202 includes opportunity gate 210, which may also represent an initial approach fix 232 for final approach 204 to landing on runway 208; emergency path 202 provides for an obtainable (safe and desirable) touchdown point 212 on the portion of runway 208 nearest the position of aircraft 200 (in order to maximize available landing space).

In some embodiments of the present invention the pilot may, under emergency conditions, "divert" to a particular ALS by selecting the colored indicator displayed next to that ALS. Diverting to an ALS has several consequences. First, ground control may be immediately alerted of the diversion and of the pilot's intentions. Second, the pilot (or autopilot system, if active) may be directed to the selected ALS along the emergency course plotted by the system. Third, when the aircraft approaches the landing site, the heads-up display may project a virtual "window". This window may provide the pilot with a quick visual reference to use in approaching what may be an unfamiliar or unmarked site, and in targeting a touchdown point selected by the system to maximize the chance of a safe and normal landing. In the alternative, the system may suggest emergency procedures to the pilot, who may then accept and execute them in seriatim.

FIG. 17A and FIG. 17B depict embodiments of the present invention displaying onscreen divert options available to a pilot who has declared an inflight emergency state. The aircraft in FIG. 17A has declared an engine-out state over eastern Nebraska. Three potential alternative landing sites are indicated along with their distance, heading, and suitability status: a grass field, 234a (rated "green", 236a); a nearby road, 234b (rated "yellow", 236b); and KOFF/Offutt AFB, 234c (rated "red", 236c). The aircraft in FIG. 17B, inflight over the North Atlantic Ocean, has declared a single-engine state. Divert options are available to nearby emergency landing strips, along with their suitability color codes. The system has rated LPLA/Lajes Field 234b) "red" (236b) on account of adverse weather. However, divert options to either BIKF/Keflavik AFB 234a or EINN/Shannon Airport 234c-near which hospital facilities are found—are rated "green" (236a and 236c respectively) and available to the pilot by selecting the "DIVERT" indicator. In the alternative, the pilot may accept each component of the emergency divert procedure set in seriatim, manually changing heading (queuing configuration changes), contacting ground control, and so forth.

A display of the present invention may preferably indicate possible options to the pilot. The peace of mind of knowing one has "green" runway options available may offer the pilot valuable choices. As an aircraft leaves a runway at takeoff, all ALS options are red. As the aircraft climbs, ALS options turn green as they become viable R_i options. With all options red, the pilot has limited options: eject or activate the airframe chute. As landing options turn green, the pilot has options from which to choose to safely land.

Color Indicator/s	Options
ALL RED	EJECT/CHUTE/DITCH
1 GREEN	LAND
2+ GREEN	CHOICE & LAND

An aircraft may encounter emergency conditions inflight that require diversion from the initial flight path. Conditions may require a precautionary landing (if further flight is

possible but inadvisable), a forced landing (if further flight is not possible), or an emergency landing on water (generally referred to as a “ditching”). Emergencies may also dramatically reduce the time frame within which such a landing must occur. Emergency conditions may include: the failure of one or more engines (single-engine, aux-engine, or engine-out states); the failure of pitot/static, electrical, hydraulic, communications, or other onboard systems; a rapid decompression or other pressurization emergency; a medical emergency; an onboard fire; or an attempted hijacking (or similar security threat).

System input in the event of an emergency may be simplified to minimize demands on the pilot’s attention and time. Should an emergency occur, the pilot may select from a menu of emergency states and “declare” the relevant emergency by selecting that state (R_{P2} , R_{P1} , R_f). Declaring an emergency state results in two immediate consequences. First, the system parameters for selecting an ALS may change depending on the specific emergency. Second, the pilot may be given the option to divert from the initial flight path to an ALS. The “divert” option represents an emergency procedure set ascertained/suggested by the system; in the alternative, the pilot may maintain manual control and accept each item of the emergency procedure set in seriatim (from a list, a flight director (with a queue key (scroll switch) or the like)).

When an emergency state has been declared and the “divert” option is available to the pilot, the hierarchical list of potential alternative landing sites (weighted according to their suitability as an ALS) may be displayed using a green/yellow/red color scheme. The most suitable landing sites (those closest to current position (for example), or with favorable surface and/or wind conditions, easily navigated headings, or nearby services) may be marked “green”. “Yellow” sites may be acceptable for an emergency landing, but conditions there may be less than ideal (e.g., ground traffic, uneven landing surface, crosswinds, obstacles). A site rated “red” is contraindicated as an ALS. Pertinent information about each potential “divert” destination (e.g., airport designation if any, surface conditions, other information relating to the site assessment) may be displayed along with its distance, heading, and color indication. In addition, should an aircraft be on one of the selected profiles (R_{P2} , R_{P1} , R_f) the system may continue to update possible ALS data if conditions change. For example, an aircraft is flying an R_{P2} profile and all engines fail. In this condition, the pilot may select and/or execute the R_f profile, allowing for safe forced landing at the selected ALS.

FIG. 18 represents the datasets and components that inform the system’s selection of best available alternative landing sites and corresponding emergency procedure sets: the aircraft’s airspeed, location, condition and configuration; current data on terrain and obstacles (natural and manmade), weather systems, winds, and ground services; available emergency states; available decision trees and courses of action (automatic diverts vs. manual decisions in sequence); coupled approaches to identified runways and landing sites; cross-referencing for airspeed/altitude/attitude; current data about local air traffic; glide potential; obtainable descent profiles; and factors influencing the selection of emergency courses and opportunity gates such as the length of runway, road or field required for landing; and other actions to be taken in the event of a divert such as transmitting intentions, squawking emergency, and maintaining contact with ground control (dispatch/ATC) and other authorities.

FIG. 19B depicts an aircraft in-flight over the Colorado Front Range; nearby fields suitable for landing are KDEN/

Denver, KCOS/Colorado Springs, KBKF/Buckley AFB (Aurora, CO), KCYS/Cheyenne, and KOMA/Eppley (Omaha). If an emergency state is declared, available divert options may vary depending on the specific emergency. If R_1 , a land-immediately or “ditch” state, is declared KOMA is rated “red” due to its extreme distance. If the aircraft is in R_{P1} , a single-engine state (“1 ENG”) KOMA may not be ruled out if it would be feasible to reach that destination safely and without incident. If a medical emergency is declared, however, landing is a more imminent priority and the system’s target radius therefore narrows considerably. Only KDEN and KBKF are rated “green” due to their proximity and services. Finally, if a medical or threat emergency is declared both KOMA and KCYS are disregarded due to distance, KDEN and KCOS are both rated “green” as a suitable ALS, while KBKF is assessed and rated “red” despite its proximity due to lack of appropriate facilities.

FIG. 20 depicts aircraft 200 in R_1 , a land-immediately or “ditch” state, having experienced engine failure immediately after takeoff from originating airport KMLE/Millard, NE 224. Several alternative landing sites have been evaluated, but most have been rated “red” due to their distance: KOMA/Eppley 2140, KOFF/Offutt AFB 2130, Interstate 680 238 (indicated by the system display as ROAD), a large field south-southwest of Offutt AFB 228b (indicated as GRASS), and the Missouri River 226 (indicated as RIVER). Within R_f range 214, however, are two options rated “green”: a field 228a directly ahead (opportunity gate 210a) and originating airport 224. Field 228a, however, is given higher priority than airport 224 as an ALS.

While returning aircraft 200 to the originating airport might appear to be the obvious emergency landing option in the event of engine failure, circumstances often indicate otherwise. Depending on aircraft 200’s airspeed and altitude, as well as the experience and reaction time of its pilot (among many other considerations), executing a turn in excess of 1800 back to originating airport 224 while in glide descent (the excess being necessary to realign the aircraft with the runway) may not be the safest available option. In FIG. 20, a field directly ahead of aircraft 200’s position serves as a suitable, and safely reachable, ALS. If otherwise unfamiliar with the local terrain, aircraft 200’s pilot may not have considered a landing in the field, instead attempting to return to originating airport 224 at considerable risk.

Similar to FIG. 20, FIG. 21 depicts aircraft 200 in a land-immediately (“ditch”) state at 1,000 feet above ground level (AGL). Multiple landing sites fall within aircraft 200’s R_{P1} radius 220, some of them airport runways: originating airport KMLE/Millard 224, opportunity gate 2122; KOMA/Eppley 2140, opportunity gate 2142; KOFF/Offutt 2130, opportunity gate 2132; and the Missouri River 226. Only one site, however, lies within R_f radius 214: open field 228 (indicated as “GRASS”) at a roughly 1 o’clock heading relative to aircraft 200. Therefore field 228 has been rated “green” as an ALS (opportunity gate 210a, final approach 204). Because hard-surface runways are available, the system may not display river 226 as an option.

FIG. 22 depicts aircraft 200 in a land-immediately (“ditch”) state, but at 4,000 feet AGL. At this higher altitude, an ALS that might not have been within 200’s glide range at 1,000 feet may now be safely reachable. Therefore aircraft 200’s land-immediately radius 222 now includes originating airport KMLE/Millard 224 (opportunity gate 210a), KOFF/Offutt 2130, KOMA/Eppley 2140, and the Missouri River 226. All three airports are rated “green” as an ALS.

FIG. 23 depicts aircraft 200 in-flight westbound over the North Atlantic Ocean. In-flight under normal transatlantic

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conditions via course **240**, the system may search broadly for an ALS within target radius **218**, reflecting a need to land only when practicable. Within target radius **218**, the system has identified an ALS in Greenland along path/procedure set **202a** to opportunity gate **210a** and touchdown point **212a**, and an ALS in Labrador along path/procedure set **202b** to opportunity gate **210b**. Should aircraft **200** declare an engine-out state, however, the search radius narrows to R_t , or target radius **222**, reflecting an immediate need to land. As the Greenland ALS lies within the engine-out radius, it may be the only option available in an engine-out situation.

FIG. **25** diagrammatically depicts the various axes of rotation of an aircraft, which together produce a particular angle of attack into the relative wind. Flight control surface position selected via pilot or autopilot inputs largely dictate angle of attack. Aircraft airspeed is at least partially selectable by angle of attack and aircraft configuration. Power settings in combination with pitch and configuration control altitude and airspeed. Thus, in operation of a preferred embodiment of the present invention, a particular aircraft configuration is desired for each of the various segments of flight, generally, taxi, takeoff, climb, cruise, descent, approach, landing, and the like. Each model (type) of aircraft has known flight performance characteristics in a particular configuration within a particular flight (operational) segment. The present invention utilizes a lookup table (register or the like) extracted from the dataset, containing these aircraft performance characteristics for comparison with current or realized characteristics against expected (most likely desired) flight segment characteristics (given weather, traffic, routing, and the like).

FIG. **26** depicts how wind speeds and headings can vary dramatically along a flight plan, and at a single geographic location, depending upon the altitude. The dotted line **2610** depicts a flight from KSJC/San Jose **2620** to KSLC/Salt Lake City to KDEN/Denver. A weather report might describe the wind as 10 knots from the northwest (300°) at KJSC **2630**, 5 knots from the north (360°) at KJSC, and 10 knots from the northeast (60°) at KDEN. At any given altitude **2640** over any of these points, however, the reported wind direction and speed may not be accurately represented or forecast. The apparatus of the present invention may be preferably programmed/set for offsetting the variability of forecasts and/or moderating the display based upon most likely (worst case) conditions.

FIG. **27** depicts a hypothetical dataset display utilized/presented by the system to catalog available services (hospital, lodging, security, maintenance, and the like) and approach procedures for various airports, ALS's, and other pertinent locations. For example, at KAIA/Alliance Municipal Airport, hospital services and police are nearby, and a broad variety of instrument approach procedures are available, including RNAV, VOR, ILS, and LOC/DME.

In the context of embodiments of the present invention configuration and configured mean (1) the position of the aircraft relative to an expected position, (2) the attitude of the aircraft relative to an expected attitude, and (3) the position of controllable members and settings (e.g., gear, flaps, elevator, rudder, ailerons, spoilers, throttle(s), selection of navigation/communication frequencies, and the like) relative to expected settings. A flight plan may be described as a series of scalars describing the vector of an aircraft from one location to another (gate-to-gate, hanger-to-ramp, runway to runway, and the like). The vector describing this path will be altered in operation by, for example: (1) ATC (altitude changes, course changes, airspeed restrictions, arrival and departures, traffic, and holds or the like), (2)

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weather (deviations around, ground speeds, turbulence, and the like), and (3) pilot and aircraft performance. In an embodiment, system experience with a particular pilot or leg may be stored, compared, and made part of an analysis in determining what constitutes a departure from an expected vector (path). Deviation from what is expected may be tolerance dependent. For example, on takeoff, climb out, approach, and landing, system sensitivity to a deviation may be higher. Deviations resulting from ATC or weather may be ascertained, for example, by ATC communication patterns (i.e., a change in heading, altitude, and/or airspeed precedes an ATC/pilot communication) or by a change in weather condition or forecast received by an embodiment of the present invention not preceded by a change in heading, altitude, and/or airspeed (or the like). Thus, where a deviation is found unlikely (improbable) by the system of an embodiment of the present invention to be associated with ATC and/or weather, depending of flight phase/segment and the magnitude of the deviation, the system may warn the pilot of a likely configuration error and under certain conditions it may suggest a configuration change. However, if altitude, airspeed, weather, or traffic indicate few safe options (e.g., loss of power on takeoff) an embodiment may immediately suggest an ALS with an associated procedure set (insufficient ALS options given the total energy TE available to aircraft **200**).

In the context of embodiments of the present invention unusual condition (**262**, **264**, **266**) means a deviation having a magnitude outside of a preselected range of acceptable values for a particular flight segment/phase. In a preferred embodiment a pilot, user, dispatcher, owner, or other entity may preselect what constitutes an unusual condition for each segment/phase of flight. Conversely, a system of a preferred embodiment of the present invention may operationally determine a range of acceptable values for a particular pilot, aircraft, segment, leg, or the like from past flight data.

In the context of embodiments of the present invention flight segment, flight phase, segment, phase, or segment/phase means a portion of a flight having a particular aircraft configuration or desired aircraft configuration. More particularly, in the context of an embodiment of the present invention an aircraft in a certain configuration will produce a corresponding airspeed, rate of ascent/descent, course change, or the like. Aircraft being operated on a flight plan with an embodiment of the present invention and its associated database(s) (FIG. **2** and the like) in a particular segment of flight should be progressing along the desired vector (path) at an expected rate (relative to the ground and destination) within an expected tolerance. Deviations from expected tolerances may be user (pilot and the like) selectable and are presented to the pilot.

In addition, during each phase/segment of flight an aircraft possess a finite energy state (kinetic+potential energy=total energy ($KE+PE=TE$)). Aircraft energy state (total energy) directly effects range. For example, an aircraft at FL380 (38000 MSL) has more energy than one at 8000 MSL. Similarly an aircraft at 500 knots and 500 MSL in a bombing run with full stores has more energy than one at 200 knots and 500 MSL. Energy equals options. An embodiment of the present invention monitors total energy and utilizes known total energy to ascertain available options by, for example, criticality and flight segment.

FIG. **9A** diagrammatically illustrates an aircraft **200** departing from an airport **224** for a destination airport having runways **208(a)** and **(b)**. Depending on wind conditions the aircraft may be landing via and approach **210(a)** and **(b)**. In operation an aircraft may be expected to operate

between an area of expected operation **260** (dashed lines) while on a flight plan (solid line) during flight operations and associated phases of flight (**242-256**). An aircraft on takeoff and climb, in a preferred embodiment, will be considered in an unusual condition **262(a)** with even a slight deviation from the expected flight path. Corrective actions may be expected or prompted by the system when the unusual condition **262(a)** is detected. If the aircraft **200** proceeds, to for example, a likely unusual condition **264(a)** the system of an embodiment of the present invention may be more insistent (perhaps requiring a pilot acknowledgement or the like) in order to neutralize further warnings. Should the aircraft **200** appear to the system to be proceeding to a position **266(a)** (anticipated position) dangerous to the aircraft, the system of an embodiment of the invention may become still more insistent (or the like), requiring some aircraft reconfiguration, flight plan cancellation or alteration, corrective action (or the like). An aircraft **200** enroute (or transiting another less critical flight phase) may deviate from expected position substantially more before an embodiment of the present invention detects an unusual condition **262(b)**. An embodiment of the present invention may not draw the pilot's (crew's) attention to the deviation until the aircraft **200** has exited the expected operation area **260** and is in a likely unusual condition **264(b)**. The system may become more insistent if it predicts the aircraft **200** is proceeding to a position **266(b)** (anticipated position) dangerous to the aircraft.

FIG. 9B is a diagrammatic elevation of a flight plan (expected path/course) of an aircraft **200** from an originating airport **224** to a second airport **208**. The expected area of operation of the aircraft **200** is schematically defined by dashed lines (**260**). The expected area of operation is diagrammatically illustrated as substantially narrower during take off **242**, climb **244**, descent approach **252**, and landing **254**. A variation in position is tolerated by an embodiment of the present position most preferably by altitude (AGL), airspeed (GS), and aircraft configuration.

An aircraft on takeoff and climb (FIG. 9B), in a preferred embodiment, will be considered in an unusual condition **262(a)** with even a slight deviation from the expected flight path. Corrective actions may be expected or prompted by the system when the unusual condition **262(a)** is detected. If the aircraft **200** proceeds to, for example, a likely unusual condition **264(a)** the system of an embodiment of the present invention may be more insistent (perhaps requiring a pilot acknowledgement or the like) in order to neutralize further warnings. Should the aircraft **200** appear to the system to be proceeding to a position **266(a)** (anticipated position) dangerous to the aircraft, the system of an embodiment of the invention may become still more insistent (or the like), requiring some aircraft reconfiguration, flight plan cancellation or alteration, corrective action (or the like). An aircraft **200** enroute (or transiting another less critical flight phase) may deviate from expected position substantially more before an embodiment of the present invention detects an unusual condition **262(b)**. An embodiment of the present invention may not draw the pilot/crew's attention to the deviation until the aircraft **200** has exited the expected operation area **260** and is in a likely unusual condition **264(b)**. The system may become more insistent if it predicts the aircraft **200** is proceeding to a position **266(b)** (anticipated position) dangerous to the aircraft.

Thus, in various preferred embodiments of the present invention, the invention may provide at least one of emergency guidance (Safety Hierarchical Emergency Pilot Helper Engageable Runway Diverter: "SHEPHERD") and

configuration error identification and configuration suggestions (Safety Interface Mission Operations Navigation: "SIMON").

In operation, a database of potential alternative landing sites (ALS's) may be created and maintained utilizing airport directory information, satellite imagery, survey data, surface temperature data (variations over time), traffic data, current and historic Landsat imagery, remote sensing (road and field), LDCM (Landsat Data Continuity Mission), TIRS (thermal infrared sensor), and the like. Airport directories such as AeroNav (www.aeronav.faa.gov), AOPA (www.aopa.org/members/airports), AirNav (www.airnav.com/airports), and world airport directories such as www.airport-directory.com and airport.airlines-inform.com may be utilized by embodiments of the present invention. The present invention may utilize satellite imagery such as Landsat, LDCM, TIRS and with terrain data from USGS (www.usgs.gov), WeoGeo, and TopoQuest, Google Maps and the like to determine the acceptability of potential off-airport landing sites. Likewise, road and traffic information may be analyzed for additional potential off-airport landing sites and incorporated into the ALS database **526** via the flight assistant **100** and through a subscription **600** of an embodiment of the present invention. Generally, traffic data may be obtained via the onboard database associated with the network of GPS satellites (for general traffic patterns), US Department of Transportation traffic sensors, reflected data from GPS-enabled vehicles and mobile devices, or from aftermarket data providers and data aggregators such as Google Maps, Inrix, Radio Data Service, Sirius/XM, MSN, and the like.

An embodiment of the present invention may utilize data from the Automatic Dependent Surveillance-Broadcast (ADS-B) as well as the full compliment of the Next Generation Air Transportation System (NextGen). In operation an embodiment of the present invention may receive traffic, weather, terrain, and flight information from ADS-B as an exclusive source (or enhancing cumulative or partially cumulative source) for processing by an apparatus of the present invention for detecting unusual conditions (positions) and configuration errors (and the like) and selectively suggesting either a new flight profile or flying a suggested flight profile.

FIG. 28 is a diagrammatic representation of an environment of a surface-based embodiment of the present invention, including both hardware components of the apparatus and data components used by the system. In this configuration the system **300** is implemented by a portable computing device **400**, which can be carried aboard aircraft **200**, and a surface-based computing device **402**. Device **402** includes a main database **304**. For a given aircraft, database **304** includes information about the aircraft's specifications (e.g., empty and gross weight, engine operational parameters, fuel capacity and consumption, flight configuration characteristics and performance envelope, optimal glide ratio and best glide speed, emergency procedures and checklists, safe landing and ground-roll distance, and how landing is affected by a variety of surface characteristics such as slope, moisture, temperature and pressure at ground level, etc. In a preferred embodiment, all or substantially all information located in the Pilot's Operating Handbook or aircraft flight manual (DASH-1) may be included in database **304**, along with any additional data derivable therefrom.

For a given pilot, database **304** may include information about pilot currency and experience level, such as total flight time for a given aircraft. Database **304** may also include a

general assessment of the proficiency of a given pilot, either generally or with respect to specific flight segments or procedures (e.g., engine-out landings). Once a flight is completed, database 304 may be updated with additional information specific to the completed flight.

For a given flight, database 304 may include an initial flight path in addition to terrain data for the flight path and a surrounding operating zone of predetermined radius. Terrain data may include: topographic information about any obstacles, natural or manmade, along the flight path; any airports or runways along the flight path suitable for landing; information about approaches to these airports, including navigational/communication frequencies and traffic patterns; and information about services offered at or near these airports, including fuel, maintenance, transport, medical, or emergency services. Terrain data may also include information about potential alternative landing sites in the operating zone and their suitability for landing. Database 304 may indicate as a potential alternative landing site any open area (field, clearing, etc.) of sufficient size to accommodate a given landing distance, any sufficiently large paved or level surface (such as a parking lot, highway, or road), or any sufficiently large body of water. For any highway or road so identified, database 304 may contain information about ground traffic, including current traffic data or traffic projections based on historical patterns. Database 304 may also include weather information for the operating zone, such as current forecasts, temperatures and dew points, historical weather patterns, ground level and aloft wind characteristics, and visibility. Data recorded inflight may help calibrate flight data to reflect performance along a certain route by various pilots, aircraft, or types of aircraft.

In a preferred embodiment of the present invention, prior to departure system 300 may export from database 304 a dataset specific to the aircraft, the pilot, and the flight plan. This dataset may be downloaded to portable device 400 for use by system 300, and may include a preflight assessment of likely weather conditions, obstacles, and alternative landing sites for a particular flight. This preflight assessment may take into consideration the pilot's proficiency in determining the feasibility of a given potential alternative landing site. A less proficient pilot may require a greater safety margin in determining the distance necessary to safely land the aircraft and bring it to a complete stop on the ground. Given available information about adverse weather and atmospheric conditions along the flight path, device 402 may identify likely points along the flight path where aircraft 200 may encounter said adverse conditions, and prioritize the selection of alternative landing sites safely reachable from those points.

Once flight has commenced, aircraft configuration module 310 may continually assess, on a selectable periodic basis, the comprehensive configuration of aircraft 200, including its current position, current altitude, current airspeed and groundspeed, current attitude, angle or rate of climb or descent, power levels and energy state, component configuration, flight segment, acoustic profile, etc. In a preferred embodiment of the present invention, aircraft configuration module 310 may continually receive updated positional information via wireless link 404, communicatively connected to onboard GNSS receiver 280 and surface-based navigational service (SBNS) receiver 282 of aircraft 200. SBNS may include any surface-based navigational or surveillance system broadcasting or transmitting ground-to air, air-to-air, or via space-based satellites (e.g. ADS-B In and ADS-B Out). SBNS receiver 282 may provide system 300 with continually updated data about nearby air traffic,

terrain overlay, and weather conditions (e.g., METARs, TAFs, AIRMET/SIGMETs, TFRs). System 300 may then display relevant portions of this data to the pilot via display unit 702. GNSS receiver 280 may continually receive updated position data from satellite constellation 900, said constellation including GPS, GLONASS, Galileo, Compass, IRNSS, or other navigational satellites. SBNS receiver 282 may continually receive updated traffic and weather information from base station 410, as well as traffic data broadcast air-to-air by aircraft 200(a). Aircraft configuration module 310 may also continually receive data from at least one sensor 406. In a preferred embodiment, sensor 406 may include a microphone incorporated into portable device 400, configured to analyze audio levels and frequencies onboard aircraft 200. Sensor 406 may additionally include a barometer, magnetometer, or accelerometer incorporated into portable device 400, or otherwise installed onboard aircraft 200 and communicatively connected to portable device 400.

In a preferred embodiment, output data from aircraft configuration module 310 may be processed on a continual and selectively periodic basis by trajectory evaluator module 312 and configuration evaluator module 314. As aircraft configuration module 310 produces a continually updated "snapshot" of the configuration of aircraft 200 at any given moment, trajectory evaluator module 312 may analyze each successive aircraft configuration in comparison to the expected configuration of aircraft 200. The expected configuration of aircraft 200 may be based on the flight plan of aircraft 200 and historical performance by the pilot, by the aircraft, or along the flight path; if the current aircraft configuration represents where aircraft 200 is and what it is doing, the expected aircraft configuration represents where aircraft 200 should be, and what it should be doing. Based on sensitivity parameters selectable by the pilot of aircraft 200, or defined by appropriate business policy or military protocol, trajectory evaluator module 312 may determine that aircraft 200 has significantly deviated from its expected trajectory, suggesting corrective action or indicating a possible emergency.

Similarly, configuration evaluator module 314 may analyze each successive aircraft configuration in order to ascertain possible solutions to configuration errors at a given point in time. A configuration error may include improper positioning of landing gear, improper extension or retraction of flaps, improper flight control position, improper rotational angle or angle of attack, insufficient or excessive speed or climb/descent rate, or any other potentially correctable aspect of the performance of aircraft 200. Configuration evaluator module 314 may then compare the current configuration of aircraft 200 to its expected configuration, depending on the current flight segment or other conditions. Configuration evaluator module 314 may then determine (to a selectable level of confidence) that aircraft 200 is improperly configured, recommending corrective action relevant to the specific configuration error (e.g., extending or retracting flaps) or indicating a possible emergency.

In a preferred embodiment of the present invention, alternative landing module 316 may continually assess the best potential alternative landing sites for aircraft 200, based on landing site data downloaded from database 304 as well as the current configuration of aircraft 200 and external conditions such as weather and local traffic. Alternative landing site 314 may determine an operative range defined by the area reachable by aircraft 200 under a set of conditions. The effective range of aircraft 200 from a given position or altitude may be determined in part by the performance capabilities of aircraft 200 (e.g. glide ratio,

configuration), the proficiency of its pilot, and dynamic inflight parameters subject to constant change (e.g., weather patterns, atmospheric conditions, air and ground traffic). In the alternative, the pilot may select the effective range: alternative landing module 316 may use as its default effective range the area reachable by aircraft 200 when it must land whenever practicable (R_{P2}).

For a given effective range, alternative landing module 316 may review the list of alternative landing sites in the operating zone defined by database 304, determining which sites fall within the effective range. Alternative landing module 316 may then generate a hierarchy of sites within the effective range, evaluated according to a selectable set of criteria allowing for current conditions. Given a hierarchy of alternative landing sites, alternative landing module 316 may then display candidate alternative landing sites, along with any available information about current conditions (atmospherics, precipitation, traffic) via display unit 702 of portable device 400. In the event that the pilot of aircraft 200 elects to divert to a particular alternative landing site, alternative landing module 316 may then display associated emergency procedures to the pilot in order to position and configure aircraft 200 for a safe landing at the chosen site.

FIG. 29 is a diagrammatic representation of the environment of an embodiment of the present invention as implemented aboard an aircraft not equipped with SBNS. In this embodiment a GNSS receiver 280 may be directly incorporated into portable device 400, directly supplying continual position data to aircraft configuration module 310. In scenarios where data connectivity to the ground or to other aircraft may be limited, system 300 may compensate by defining a larger operating zone around its flight path, and therefore downloading a larger dataset of alternative landing site information to portable device 400.

FIG. 30 is a diagrammatic representation of the environment of a surface-based or non-space-based embodiment of the present invention. Stable data link 408 connects portable device 400 aboard aircraft 200 with surface-based master device 402. Stable data link 408 may be a ground-to-air broadband link such as Gogo, a Ka-band, Ku-band, or other satellite- or space-based data link, or a high frequency (UHF, VHF) surface-based data link. Stable data link 408 may transmit data between device 400 and device 402 in real time or in batches. The functionality of system 300 may be affected by the robustness of stable data link 408; however, as there is no need to download large amounts of data to portable device 400, portable device 400 may include a smartphone.

In a surface-based or non-space-based embodiment of the present invention, the processing functions of system 300 may be performed in significant part by surface-based master device 402. Position data relevant to aircraft 200 (collected by GNSS receiver 280) may be transmitted to base station 410 by SBNS receiver 282. Aircraft configuration module 310 may receive continual data updates from base station 410. In addition, at least one sensor 406 on board aircraft 200 may collect additional inflight data (e.g., aircraft acoustics) and transmit those data to aircraft configuration module 310. Trajectory evaluator module 312, configuration evaluator module 314, and alternative landing module 316 may receive data from aircraft configuration module 310, accessing main database 304 directly in order to make continual assessments of the configuration and trajectory of aircraft 200 and continually transmit those assessments to portable device 400 via stable data link 408.

In a surface-based or non-space-based embodiment of the present invention, a single computing device including a

single main database may provide configuration evaluation, trajectory evaluation, and alternative landing site processing for multiple aircraft at once. FIG. 31 is a highly diagrammatic view of a condition whereby a surface-based or non-space-based embodiment of the present invention may be implemented through one or more surface-based master devices 402(a) and 402(b). Master device 402(a) may continually assess the configuration of aircraft 200(a), 200(b), and 200(c), receiving position data from the said aircraft and transmitting real-time trajectory and configuration information back to the said aircraft via stable data links 408(a), 408(b), and 408(c). Similarly, master device 402(b) may continually receive position data from aircraft 200(d) and 200(e) and transmit configuration and trajectory information back to the said aircraft via stable data links 408(d) and 408(e). Aircraft 200(a) and 218(d), for example, may be operating under normal or default conditions, as indicated by their broader effective ranges 218(a) and 218(d). Aircraft 200(b) and 200(c), however, have declared a need to land where possible, as indicated by their narrower effective ranges 220(b) and 220(c). Finally, aircraft 200(e) has declared an emergency and must land immediately; the effective range 214(e) of aircraft 200(e) is narrower still, and when aircraft 200(e) selects an alternative landing site, master device 402(b) may transmit the associated emergency procedure set in order to assist aircraft 200(e) in configuring the aircraft and positioning for approach and safe landing at the selected site.

FIG. 32 is a diagrammatic representation of the environment of aircraft configuration module 310 of the present invention. Aircraft configuration module 310 will attempt to determine the configuration of aircraft 200 to a selectable degree of sensitivity at a selectable time interval. For every time t_i corresponding to a point along the flight path of aircraft 200, aircraft configuration module 310 collects the current position and altitude of aircraft 200 (via GNSS, VOR, VORTAC, ADF, LORAN, ADS-B, inertial navigation, radar, pilot input); current velocity, including airspeed or groundspeed (derived from position over time); current attitude (via sensors); weather and atmospheric conditions at the current position; nearby air traffic (via ADS-B or TCAS); and any other available parameters such as aircraft acoustics. The set of these elements represents a "snapshot" of the aircraft in time; the data may be recorded and sent to the other modules of system 300 for processing.

FIG. 33 represents a diagrammatic representation of current expected configurations for a given aircraft on a given flight path. Flight plan data downloaded from database 304 may include a prior history of performance by a given aircraft or a given pilot along a given flight path or route. Multiple data points from multiple flights may define a set of expected configurations for that flight plan. At any point in time along a flight path an expected configuration may describe, according to past performance and a history of safe and completed flights, where an aircraft should be, how fast it should be traveling (horizontally or vertically), how it should be configured according to the current flight segment (FSEG), how the aircraft acoustics should sound, etc. An aircraft configuration may be defined by as many or as few variables as is selected. Aircraft 200(a) is at time zero (t_0), on a runway preparing to accelerate for takeoff, and the configuration 360(a) of aircraft 200(a) reflects this: zero altitude AGL, zero forward or climbing speed, zero power usage, horizontal (zero-degree) pitch angle, etc. Aircraft 200(b) has taken off, and therefore its configuration 360(b) at time t_x (6 minutes, 15 seconds elapsed since time zero) reflects an altitude of 3,200 feet AGL, pitch 2° above

horizontal, ground speed 80 kts, a climb rate of 500 ft/min, and 60% power output. Aircraft **200(c)**, however, represents the expected configuration **360(c)** of the same aircraft at 6 minutes, 15 seconds past time zero. According to expected configuration **360(c)**, aircraft **200(b)** should be at an altitude of 4,500 ft AGL, be pitched 5° above horizontal, have a ground speed of 120 kts with 80% power output, and gain 2,000 feet per minute: the performance of aircraft **200(b)** should correspond to a climbing segment. Trajectory evaluator module **314** may evaluate the difference between current configuration **360(b)** and expected configuration **360(c)**. FIG. **34** is a diagrammatic view of the environment of trajectory evaluator module **314** of the present invention. Trajectory evaluator module may receive an aircraft configuration **360** from aircraft configuration module **310** and an expected configuration from database **304** (or an onboard subset thereof). Trajectory evaluator module may then determine whether the difference between current configuration **360** and expected configuration is significant or severe (or the rate of change in magnitude is significant, i.e., the difference between current and expected configurations is accelerating), according to preselected parameters. For example, for a given flight segment insufficient climb rate may be assigned a higher priority than groundspeed or physical position. If the difference in configurations is significant or severe, trajectory evaluator module **314** may determine if there is an explanation for this difference; for example, aircraft **200** may be several miles away from its original flight path, but this difference in position may be explainable by a deviation from flight plan for weather-related reasons. If there is a severe difference in trajectory with no explanation, trajectory evaluator module **314** may then inform the pilot of the difference in configuration via display unit **702**, and signal alternative landing module **316** of a possible emergency condition.

FIG. **35** is a diagrammatic view of the environment of configuration evaluator module **312** of the present invention. Configuration evaluator module **312** may attempt to ascertain if current configuration **360** of aircraft **200** as provided by aircraft configuration module **310** differs significantly from the expected configuration of aircraft **200**, and to ascertain solutions for specific configuration errors. For example, given a particular flight segment and aircraft acoustic profile, configuration evaluator module **312** may determine if a particular configuration pattern corresponds to this acoustic profile, e.g., via a lookup table in database **304**. As a given aircraft performs a certain way through a given flight segment (i.e., according to its expected configuration), the corresponding combination of engine output, flap position, landing gear, etc. will produce a certain acoustic profile. A seasoned pilot will know if his or her aircraft is not producing the proper acoustic profile, i.e., if the aircraft does not “sound right”. Configuration evaluator module **312** may analyze the sounds recorded by at least one onboard sensor; any significant difference in frequency or consistency, in concert with additional differences in configuration data (speed too low, climb rate too low) may indicate a specific configuration error, e.g., flaps extended when they should be retracted. If this difference is significant, configuration evaluator module **312** may inform the pilot via display unit **702**, displaying any associated solutions to the identified configuration error per aircraft data provided by database **304**. As the system certainty of a particular configuration error increases, configuration evaluator module **312** may display increasingly insistent or definite notifications (e.g., “check configuration” to “check gear down”). If a configuration error is significant or severe and there is no associated

solution, configuration evaluator module **312** may so inform the pilot and signal alternative landing module **316** of a possible emergency condition.

FIG. **36** is a diagrammatic view of the environment of alternative landing module **316** of the present invention. In a preferred embodiment of the present invention alternative landing module **316** may, prior to departure, define an operating zone based on a particular flight plan, including the flight path and a pre-determined surrounding area. Within this operating zone, alternative landing module **316** may look for and indicate any potential alternative landing site, from a known full-service airport to a body of water, resulting in a working database **526** of all potential alternative landing sites within the operating zone.

On receiving current aircraft configuration information from aircraft configuration module **310**, alternative landing module **316** may continually generate a hierarchy of alternative landing sites most easily and safely reachable by a given aircraft of configuration **360** within a given effective range. As a default, this effective range may represent the area within which aircraft **200** may land when practicable (R_{P2}). An aircraft at higher altitude, for example, will consequently have more potential energy, a greater glide range, and therefore a broader effective range. Within the effective range, alternative landing sites may be prioritized according to pilot preferences, system preferences, pilot proficiency or currency, or the existence of an emergency. For example, the system of the present invention may be programmed to assign a full-service airport higher priority than a smaller airport but a pilot may prefer to divert to a smaller airport closer to his current position, or may need to divert to the smaller airport due to engine loss. Once the pilot selects an alternative landing site and elects to divert there, alternative landing module **316** may then load and display (via display unit **702**) an emergency procedure associated with landing at the chosen site. An emergency procedure may include any necessary steps for configuring the aircraft properly, notifying the ground, positioning the aircraft for approach to the landing site, etc.

FIG. **37A** is a highly diagrammatic top view of a condition under which alternative landing module **316** evaluates potential alternative landing sites. Flight path **240** represents a flight from originating airport **224** to land at runway **208(d)**. Open field **228** is situated close to originating airport **224** and flight path **240**. Alternative landing module **316** may identify an open area large enough to provide aircraft **200** with a safe landing on a gently upsloping grass surface, and may therefore indicate alternative landing site **208(a)** and touchdown point **212(a)**. In the event aircraft **200** experiences engine failure after takeoff, its pilot may be forced to decide, in a matter of minutes or seconds, whether to attempt to turn back for landing at airport **224** or look around for a forward emergency landing area. As alternative landing site **208(a)** is partially obscured by a stand of trees, the pilot may not immediately recognize site **208(a)** as suitable for landing or may not have sufficient time to assess the site’s suitability from the air. However, alternative landing module **316** may have recognized area **208(a)** as an alternative landing site, and may provide the pilot with an emergency procedure for landing there if necessary. Alternative landing module **316** may also identify landing sites **208(b)** and **208(c)**, located on a road and lake respectively.

FIG. **37B** is a diagrammatic view of the parameters used by alternative landing module **316** to determine adequate safe landing distance. A given aircraft **200** may define in its specifications a minimum safe landing distance (or the distance required for aircraft **200**, descending at a standard

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angle, to clear a 50-foot obstacle and touch down at point 270(a) plus a minimum ground-roll distance (the distance required for aircraft 200, touching down at point 270(a), to come to a full stop at point 270(b)). To this distance alternative landing module 316 may add an additional safety margin 272 to account for real-world conditions. In practice, the required landing distance for an aircraft will increase due to some factors (wet surfaces, downslope, tailwinds, less proficient pilots) and decrease due to others (upslope, headwinds) in addition to varying with temperature and air pressure. Therefore an aircraft landing on runway 208 should be able to safely touch down anywhere in range 212. A touchdown past point 270(c), however, may not provide sufficient stopping distance, and the pilot may want to consider going around for a second approach if circumstances allow it.

FIG. 38 is a diagrammatic view of the data components of a preferred embodiment of the present invention. In a space-based embodiment of the present invention, system 300 may download from main database 304 a dataset specific to a particular aircraft, pilot, and flight plan. Inflight, system 300 may record all dynamic flight data received by aircraft configuration module 310 and the resulting processing of that flight data by other components of the system: performance compared to flight plan or historical performance, tracking of alternative landing sites, any emergency procedures accessed, etc. Once the flight has ended, the resulting dataset representing the completed flight may be uploaded back to the main database, updating any parameters relevant to the pilot, aircraft or flight plan.

FIG. 39 is a diagrammatic view of the relationships between aircraft energy states. Aircraft 200 uses its fuel to accelerate, thereby converting chemical energy (fuel) into kinetic energy (speed and momentum). Similarly, aircraft 200 may convert kinetic energy into potential energy by climbing, trading speed for altitude. Conversely, aircraft 200 may convert potential energy into kinetic energy by descending, losing altitude but gaining speed. As a result, an aircraft without fuel or engines at high altitude still possesses energy that may be used to reach an alternative landing site. In addition, an aircraft at low altitude and high speed has the potential energy to climb, gaining altitude and increasing its effective reachable range.

FIG. 40 is a diagrammatic view of the effective glide range of a given aircraft 200 in an engine-out state. Aircraft 200 may be able to glide nearly 16 NM under optimal conditions (ideal temperature and pressure, best glide speed, proficient pilot). Real-world conditions may affect the effective range in practice, however: headwinds will reduce the gliding distance, and a less proficient pilot may not be able to maintain optimal glide angle and speed.

FIG. 41 is a highly diagrammatic top view of a condition whereby a preferred embodiment of the present invention directs aircraft 200 away from atmospheric disturbance 274. Flight path 240 directs aircraft 200 over waypoint 268(a) and then over 268(e), but disturbance 274 may be observed along flight path 240. Trajectory evaluator module 312 may then suggest to the pilot a diversion along path 202(a), over waypoint 268(b), or along path 202(b) and over waypoint 268(c), in either case resuming course over waypoint 268(e). Trajectory evaluator module 314 may observe a selectable buffer zone around identified weather disturbances, here observing a minimum 20 NM radius around disturbance 274. In the alternative aircraft 200(a), having passed through disturbance 274, may instead be unable to reach its intended destination. Trajectory evaluator module 314 may then

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signal alternative landing module 316 to suggest a diversion along path 202(c) to runway 208 and touchdown zone 212.

FIG. 42 is a pilot view of display unit 702 of a portable computing device onboard aircraft 200, currently following flight path 240 to arrival at KMLE Runway 30 208(a) and touchdown zone 212(a). Current atmospheric conditions at KMLE 276(a) are displayed and updated as new information becomes available (e.g., new METARs or TAFs are issued). Alternative landing module 316 is tracking two nearly alternative landing sites, KOMA 208(b) and KOFF 208(c), within the selected range of aircraft 200. Alternative landing module 316 may further display, for alternative landing sites 208(a) and 208(b), current weather conditions 278(b) and 278(c) at each site. Alternative landing module 316 may also display temperature and dewpoint conditions 278 at the current position of aircraft 200 as well as the projected flight path 240(a) of aircraft 200(a), also currently projected to land at KMLE. Sensitivity parameters may be adjusted by the pilot to track more or less air traffic depending on, for example, whether said traffic is projected to cross flight path 240 or whether said traffic is projected to land at a destination airport within a certain time window of aircraft 200. Note that specific runway and touchdown zone information is not displayed for KOMA or KOFF; this information may be displayed if either is selected as an alternative landing site or by request. As aircraft 200 approaches KMLE, the system of the present invention may display more precise approach and landing information, e.g., approach charts or a more detailed flight path, and inform the pilot if the currently projected approach should change due to shifting winds.

CONTEMPLATIONS OF THE INVENTION

1. A method for determining aircraft flight configuration during a flight, comprising:
 - (a) establishing aircraft position, altitude, and flight trajectory based upon aircraft position and altitude over time;
 - (b) establishing aircraft flight segment based on at least one of time since departure, position, altitude, and flight trajectory;
 - (c) determining aircraft attitude from sensing at least one of angle and rate for at least one of pitch, bank, and yaw;
 - (d) sensing aircraft acoustics from at least one of airframe vibration and aircraft sound;
 - (e) determining an expected aircraft configuration based at least in part on said established aircraft flight segment;
 - (f) determining aircraft configuration from said sensed aircraft attitude and said aircraft acoustics for aircraft configurations attributable to said sensed attitude and acoustics for said flight segment; and
 - (g) announcing an aircraft configuration different from said expected aircraft configuration.
2. The method of contemplation 1, wherein establishing aircraft position, altitude, and flight trajectory is based on at least one of: GNSS, VOR, VORTAC, ADF, LORAN, ADS-B, inertial navigation, radar, and pilot input.
3. The method of contemplation 1, wherein establishing aircraft flight segment is based on at least one of: flight plan, flight profile history, departure and destination profile history, aircraft attitude, aircraft vibration, aircraft acoustics, acoustic history, and consistent acoustic period.

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4. The method of contemplation 1, wherein sensing aircraft attitude is based on at least one of: accelerometer, magnetometer, flight control position, and angle and rate of at least one of pitch, bank, and yaw.

5. The method of contemplation 1, wherein sensing aircraft acoustics is based on at least one of: microphone, seismometer and other vibration sensor, velocity sensor, and engine instrumentation.

6. The method of contemplation 1, wherein sensing aircraft acoustics is based at least in part on identifying acoustic profiles associated with at least one of: gear extension, flap extension and retraction, airspeed, percent power, engine out, pressurization loss, aircraft damage, flight control position, and pitch angle.

7. The method of contemplation 1, wherein determining an expected aircraft configuration is based in part on at least one of: technical order, aircraft flight manual, pilot operating handbook, pilot input, flight envelope, lift to drag ratio, aircraft position, aircraft altitude, aircraft flight segment, time since departure, estimated time enroute, estimated time to arrival, and flight plan adjusted for environmental conditions determined from at least one of ground wind speed and direction, aloft wind speed and direction, thrust level, aircraft acoustics, angle of attack, drag, weather, traffic, air traffic control instructions, pilot response times, aircraft system condition, company instructions, pilot currency, pilot proficiency, pilot experience, pilot route experience, mission plan, and hostiles.

8. The method of contemplation 1, wherein determining aircraft configuration is based in part on at least one of aircraft vibration, aircraft acoustics, gear up, gear down, percent power, energy state, engine status, pressurization loss, aircraft damage, flight control position, flap extension and retraction, and aircraft attitude.

9. The method of contemplation 1, wherein announcing an aircraft configuration different from said expected aircraft configuration is based on a user-selectable hierarchy of likely unexpected configurations.

10. The method of contemplation 9, wherein an aircraft configuration different from said expected aircraft configuration includes an unexpected change in aircraft acoustics.

11. The method of contemplation 9, wherein announcing an aircraft configuration different from said expected aircraft configuration is at least one of increasingly insistent and definite as system certainty increases.

12. The method of contemplation 9, wherein announcing an aircraft configuration different from said expected aircraft configuration becomes more specific as system certainty of a specific difference in configuration increases.

13. The method of contemplation 9, wherein announcing an aircraft configuration different from said expected configuration is accomplished using at least one of: voice notification, graphic display, audio tone, and haptic notification.

14. A method for directing a pilot flying an aircraft to a reachable alternative landing site, comprising:

- (a) determining at least one of current aircraft trajectory, anticipated future aircraft position and altitude, and anticipated future trajectory;
- (b) determining at least one of expected aircraft position, altitude, and trajectory based on at least one of time since departure, position, altitude, groundspeed and heading;
- (c) determining a difference between:
 - (i) said current aircraft trajectory, said future position and altitude, and said future trajectory and
 - (ii) said expected position, altitude and trajectory;
- (d) determining the magnitude of difference between:

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- (i) at least one of said current aircraft trajectory and said future position, altitude, and trajectory; and
- (ii) at least one of said expected position, altitude, and trajectory;

(e) determining the rate of change of said magnitude of difference;

(f) determining whether a difference between said current aircraft trajectory and said future position, altitude, and trajectory and said expected position, altitude, and trajectory is the result of an emergency at least in part from said magnitude of difference and at least one of traffic and weather deviation, flight plan change, air traffic control requirement, and arrival change;

(g) determining a level of emergency, where the emergency has been determined, from a hierarchy of emergencies, selected from at least one of land immediately, land as soon as possible, and land as soon as practicable;

(h) continually selecting from a hierarchy of selectable landing site preferences and said selected level of emergency an alternative landing site at least one of reachable by said current trajectory and reachable by an available configuration change;

(i) preparing a procedure for safely positioning said aircraft in a landable configuration at the approach of said reachable alternative landing site; and

(j) announcing at least one of said prepared procedure and in seriatim the elements of said prepared procedure.

15. The method of contemplation 14, wherein determining at least one of current aircraft trajectory, anticipated future aircraft position and altitude, and anticipated future trajectory is based on at least one of GNSS, VOR, VORTAC, ADF, LORAN, ADS-B, inertial navigation, radar, and pilot input.

16. The method of contemplation 14, wherein determining at least one of expected aircraft position, altitude, and trajectory is based on at least one of: flight plan, flight profile history, departure and destination profile history, aircraft attitude, aircraft vibration, aircraft acoustics, aircraft acoustic history, and consistent acoustic period.

17. The method of contemplation 14, wherein determining a difference between said current aircraft trajectory, said future position, altitude, and trajectory and said expected position, altitude, and trajectory is based on a user-selectable hierarchy of likely unexpected configurations.

18. The method of contemplation 14, wherein determining the magnitude of difference between said current trajectory and said future position, altitude, and trajectory and said expected position, altitude, and trajectory is based on a user-selectable hierarchy of likely unexpected configurations.

19. The method of contemplation 14, wherein determining whether a difference between said current trajectory and said future position, altitude, and trajectory and said expected position, altitude, and trajectory is the result of an emergency is based on user-selectable hierarchy of likely unexpected configurations.

20. The method of contemplation 14, wherein said level of emergency is received from at least one of: current position, current altitude, current trajectory, and manual selection by pilot.

21. The method of contemplation 14, wherein said hierarchy of selectable landing site preferences includes:

- (a) a full service airport with on-site emergency services and sufficient accommodations for passenger manifest;
- (b) an airport with suitable runway and some on-site services;

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- (c) an airport with hard surface runway of sufficient width and length;
 - (d) an airport with an unpaved runway;
 - (e) a highway and other road;
 - (f) a field and other open area;
 - (g) sloping and rough terrain with guidance between rocks, trees, and other ground obstructions;
 - (h) bodies of water; and
 - (i) other mixed consistency surfaces.
22. The method of contemplation 21, wherein the attributes of selectable landing sites are periodically updated to reflect at least one of current season, crop type, road and water traffic, pilot proficiency, current pilot condition, and air traffic control requests, and aircraft condition.
23. The method of contemplation 14, wherein preparing a procedure for safely positioning said aircraft in a landable configuration includes compliance with at least one of emergency level procedures, standard operating procedure, drift-down procedure, and obstacle avoidance procedure.
24. The method of contemplation 14, wherein announcing at least one of said prepared procedure further comprises displaying at least one alternative landing site range at least one of reachable by said current trajectory and reachable by an available configuration change, in the form of an ellipse corresponding to the selected emergency level.
25. The method of contemplation 24, wherein displaying at least one alternative landing site range at least one of reachable by said current trajectory and reachable by an available configuration change further comprises displaying the at least one alternative landing site located within the said range.
26. A method of announcing appropriate available alternative landing sites during a flight, comprising:
- (a) receiving an emergency level selectable at least from the set including: (i) land as soon as practicable, (ii) land as soon as possible, and (iii) land immediately;
 - (b) determining flight environment from ground speed and at least one of: above ground level altitude, air-speed, descent rate, descent angle, ground wind speed and direction, aloft wind speed and direction, potential energy level, thrust level, sound level, angle of attack, drag, weather, traffic, air traffic control instructions, pilot response times, aircraft system condition, company instructions, pilot proficiency, pilot experience, pilot currency, pilot route experience, and flight segment;
 - (c) continually selecting from a hierarchy of selectable landing site preferences, said selected level of emergency, and said determined flight environment at least one alternative landing site reachable by at least one of a zero-thrust, partial-thrust, normal-thrust standard and emergency operating procedure, or available configuration change; and
 - (d) displaying at least one alternative landing site range reachable by the said aircraft, in the form of an ellipse corresponding to the selected emergency level.
27. The method of contemplation 26, wherein said emergency level is received from at least one of current position and altitude, current trajectory, and manual selection by pilot.
28. The method of contemplation 26, wherein continually selecting at least one alternative landing site is based at least in part on at least one service available at said alternative landing site.
29. The method of contemplation 26, wherein displaying at least one alternative landing site range reachable by the said

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- aircraft further comprises displaying the at least one alternative landing site located within the said range.
30. A method of assisting a pilot in an emergency, comprising:
- (a) determining at least one of current aircraft position and altitude, current aircraft trajectory, anticipated future aircraft position and altitude, and aircraft performance from at least one of position and altitude over time and at least one sensor;
 - (b) determining at least one of expected aircraft position and altitude, expected aircraft trajectory, expected future aircraft position and altitude, and expected aircraft performance from at least one of a lookup register, position and altitude on a flight plan, time since departure, arrival weather, arrival traffic, estimated time enroute, and estimated time of arrival;
 - (c) determining at least one of magnitude, expected magnitude, and rate of change of magnitude of difference between
 - (i) at least one of said current position and altitude, said current trajectory, said anticipated future position and altitude, and said aircraft performance; and
 - (ii) at least one of said expected position and altitude, said expected trajectory, said expected future position and altitude, and said expected performance;
 - (d) determining whether at least one of an aircraft configuration error and an emergency exists based at least on one of said magnitude, said expected magnitude, and said rate of change;
 - (e) determining whether said magnitude is the result of at least one of traffic and weather deviation, flight plan change, air traffic control requirement, and arrival change;
 - (f) determining a level of emergency, where the emergency has been determined, from a hierarchy of emergencies, selected from at least one of land as soon as practicable, land as soon as possible, and land immediately;
 - (g) continually selecting, from a hierarchy of selectable landing site preferences and said selected level of emergency, at least one alternative landing site at least one of reachable by said current trajectory and reachable by an available configuration change;
 - (h) preparing a procedure for safely positioning said aircraft in a landable configuration at the approach of said alternative landing site; and
 - (i) announcing at least one of said prepared procedure and in seriatim the elements of said prepared procedure.
31. The method of contemplation 30, wherein said sensor is at least one of a ground-based sensor, satellite-based sensor, space-based sensor, and aircraft sensor.
32. The method of contemplation 30, wherein continually selecting at least one alternative landing site at least one of reachable by said current trajectory and reachable by an available configuration change further comprises continually selecting at least one alternative landing site reachable by at least one of a zero-thrust, partial-thrust, and normal-thrust standard and emergency operating procedure.
33. The method of contemplation 30, wherein announcing at least one of said prepared procedure further comprises displaying at least one alternative landing site range reachable by the said aircraft, in the form of an ellipse corresponding to the selected emergency level.
34. The method of contemplation 30, wherein said level of emergency is received from at least one of: current position and altitude, current trajectory, and manual selection by pilot.

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35. The method of contemplation 30, wherein continually selecting at least one alternative landing site is based at least in part on at least one service available at said alternative landing site.

36. The method of contemplation 33, wherein displaying at least one alternative landing site range reachable by the said aircraft further comprises displaying the at least one alternative landing site located within the said range.

37. A system for determining aircraft flight configuration during a flight, implemented by at least one computing device, comprising:

- (a) an aircraft state module configured to:
 - (i) establish aircraft position, altitude, and trajectory based upon aircraft position and altitude over time;
 - (ii) establish aircraft flight segment based on at least one of time since departure, position and altitude, and trajectory;
 - (iii) determine aircraft attitude from sensing at least one of angle and rate for at least one of pitch, bank, and yaw;
 - (iv) sense aircraft acoustics from at least one of airframe vibration and aircraft sound;
- (b) a configuration evaluator module configured to:
 - (i) determine an expected aircraft configuration based at least in part on said established aircraft flight segment;
 - (ii) determine aircraft configuration from said sensed aircraft attitude and said aircraft acoustics for aircraft configurations attributable to said sensed attitude and acoustics for said flight segment; and
- (c) at least one display unit configured to announce an aircraft configuration different from said expected aircraft configuration.

38. The system of contemplation 37, wherein establishing aircraft position, altitude, and flight trajectory is based on at least one of GNSS, VOR, VORTAC, ADF, LORAN, ADS-B, inertial navigation, radar, and pilot input.

39. The system of contemplation 37, wherein establishing aircraft flight segment is based on at least one of flight plan, flight profile history, departure and destination profile history, aircraft attitude, aircraft vibration, aircraft acoustics, aircraft acoustic history, and consistent acoustic period.

40. The system of contemplation 37, wherein sensing aircraft attitude is based on at least one of: accelerometer, magnetometer, flight control position, and angle and rate of at least one of pitch, bank, and yaw.

41. The system of contemplation 37, wherein sensing aircraft acoustics is based on at least one of: microphone, seismometer and other like vibration sensor, velocity sensor, and engine instrumentation.

42. The system of contemplation 37, wherein sensing aircraft acoustics is based at least in part on identifying acoustic profiles associated with at least one of: gear extension, flap extension and retraction, airspeed, percent power, engine status, pressurization loss, aircraft damage, flight control position, and pitch angle.

43. The system of contemplation 37, wherein determining an expected aircraft configuration is based in part on at least one of: technical order, aircraft flight manual, pilot operating handbook, pilot input, flight envelope, lift to drag ratio, aircraft position, aircraft altitude, aircraft flight segment, time since departure, estimated time enroute, estimated time to arrival, and flight plan adjusted for environmental conditions determined from at least one of ground wind speed and direction, aloft wind speed and direction, thrust level, aircraft acoustics, angle of attack, drag, weather, traffic, air traffic control instructions, pilot response times, aircraft

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system condition, company instructions, pilot currency, pilot proficiency, pilot experience, pilot route experience, mission plan, and hostiles.

44. The system of contemplation 37, wherein determining aircraft configuration is based in part on at least one of aircraft vibration and aircraft acoustics (gear up, gear down, percent power, engine out, pressurization loss, aircraft damage, flight control position, flap extension and retraction, and pitch angle and against and with pitch angle).

45. The system of contemplation 37, wherein announcing an aircraft configuration different from said expected aircraft configuration is based on a user-selectable hierarchy of likely unexpected configurations.

46. The system of contemplation 45, wherein an aircraft configuration different from said expected aircraft configuration includes an unexpected change in aircraft acoustics.

47. The system of contemplation 45, wherein announcing an aircraft configuration different from said expected aircraft configuration is at least one of increasingly insistent and definite as system certainty increases.

48. The system of contemplation 45, wherein announcing an aircraft configuration different from said expected aircraft configuration becomes more specific as system certainty of a specific difference in configuration.

49. The system of contemplation 45, wherein announcing an aircraft configuration different from said expected configuration is accomplished using at least one of: voice notification, graphic display, audio tone, and haptic notification.

50. A system for directing a pilot flying an aircraft to a reachable alternative landing site, implemented by at least one computing device, comprising:

- (a) an aircraft state module configured to determine at least one of current aircraft trajectory, anticipated future aircraft position and altitude, and anticipated future trajectory;
- (b) a trajectory evaluator module configured to:
 - (i) determine at least one of expected aircraft position, altitude, and trajectory based on at least one of time since departure, position, altitude, groundspeed, and heading;
 - (ii) determine a difference between:
 - (1) at least one of said current aircraft trajectory, said future position and altitude, and said future trajectory; and
 - (2) at least one of said expected position, altitude, and trajectory;
 - (iii) determine the magnitude of difference between said current aircraft trajectory and said future position, altitude, and trajectory and said expected position, altitude, and trajectory;
 - (iv) determine the rate of change of said magnitude of difference;
 - (v) determine whether a difference between said current aircraft trajectory and said future position, altitude, and trajectory and said expected position, altitude, and trajectory is the result of an emergency at least in part from said magnitude of difference and at least one of traffic and weather deviation, flight plan change, air traffic control requirement, and arrival change;
- (c) an alternative landing module configured to:
 - (i) determine a level of emergency, where the emergency has been determined, from a hierarchy of emergencies, selected from at least one of land immediately, land as soon as possible, and land as soon as practicable;

(ii) continually select from a hierarchy of selectable landing site preferences and said selected level of emergency at least one alternative landing site at least one of reachable by said current trajectory and reachable by an available configuration change;

(iii) prepare a procedure for safely positioning said aircraft in a landable configuration at the approach of said reachable alternative landing site; and

(d) at least one display unit configured to announce at least one of said prepared procedure and in seriatim the elements of said prepared procedure.

51. The system of contemplation 50, wherein determining at least one of current aircraft trajectory, anticipated future aircraft position and altitude, and anticipated future trajectory is based on at least one of GNSS, VOR, VORTAC, ADF, LORAN, ADS-B, inertial navigation, radar, and pilot input.

52. The system of contemplation 50, wherein determining at least one of expected aircraft position, altitude, and trajectory is based on at least one of: flight plan, flight profile history, departure and destination profile history, aircraft attitude, aircraft vibration, aircraft acoustics, aircraft acoustic history, and consistent acoustic period.

53. The system of contemplation 50, wherein determining said difference between said current aircraft trajectory, said future position and altitude, and said future trajectory and said expected position, altitude, and trajectory is based on a user-selectable hierarchy of likely unexpected configurations.

54. The system of contemplation 50, wherein determining the magnitude of difference between said current aircraft trajectory and said future position, altitude, and trajectory and said expected position, altitude, and trajectory is based on a user-selectable hierarchy of likely unexpected configurations.

55. The system of contemplation 50, wherein determining whether said difference between said current aircraft trajectory and said future position, altitude, and trajectory and said expected position, altitude, and trajectory is the result of an emergency is based on a user-selectable hierarchy of likely unexpected configurations.

56. The system of contemplation 50, wherein said level of emergency is received from at least one of current position and altitude, current trajectory, and manual selection by pilot.

57. The system of contemplation 50, wherein said hierarchy of selectable landing site preferences includes:

- (a) a full service airport with on-site emergency services and sufficient accommodations for passenger manifest;
- (b) an airport with suitable runway and some on-site services;
- (c) an airport with hard surface runway of sufficient width and length;
- (d) an airport with an unpaved runway;
- (e) a highway and other road;
- (f) a field and other open area;
- (g) sloping and rough terrain with guidance between rocks, trees, and other ground obstructions;
- (h) bodies of water; and
- (i) other mixed consistency surfaces.

58. The system of contemplation 57, wherein the attributes of selectable landing sites are periodically updated to reflect at least one of: current season, crop type, road and water traffic, pilot proficiency, current pilot condition, air traffic control requests, and aircraft condition.

59. The system of contemplation 50, wherein preparing a procedure for safely positioning said aircraft in a landable

configuration includes compliance with at least one of emergency level procedures, standard operating procedure, drift-down procedure, and obstacle avoidance procedure.

60. The system of contemplation 50, wherein said display unit is further configured to announce at least one alternative landing site range reachable by the said aircraft, in the form of an ellipse corresponding to the selected emergency level.

61. The system of contemplation 50, wherein said display unit is further configured to announce the at least one alternative landing site located within the said range.

62. A system for announcing appropriate available alternative landing sites during a flight, implemented by at least one computing device, comprising:

- (a) an aircraft state module configured to determine flight environment from ground speed and at least one of: above ground level altitude, airspeed, descent rate, descent angle, ground wind speed and direction, aloft wind speed and direction, potential energy level, thrust level, sound level, angle of attack, drag, weather, traffic, air traffic control instructions, pilot response times, aircraft system condition, company instructions, pilot proficiency, pilot experience, pilot currency, pilot route experience, and flight segment;
- (b) an alternative landing module configured to:
 - (i) receive an emergency level selectable at least from the set including: (1) land as soon as practicable, (2) land as soon as possible, and (3) land immediately;
 - (ii) continually select from a hierarchy of selectable landing site preferences, said selected level of emergency, and said determined flight environment at least one alternative landing site reachable by at least one of a zero-thrust, partial-thrust, and normal-thrust standard and emergency operating procedure; and
- (c) at least one display unit configured to announce at least one alternative landing site range reachable by the said aircraft, in the form of an ellipse corresponding to the selected emergency level.

63. The system of contemplation 62, wherein said emergency level is received from at least one of current position, current altitude, current trajectory, and manual selection by pilot.

64. The system of contemplation 62, wherein continual selection of at least one alternative landing site is based at least in part on at least one service available at said alternative landing site.

65. The system of contemplation 62, wherein said display unit is further configured to announce the at least one alternative landing site located within the said range.

66. A system for assisting a pilot in an emergency, implemented by at least one computing device, comprising:

- (a) an aircraft state module configured to determine at least one of current aircraft position and altitude, current aircraft trajectory, anticipated future aircraft position and altitude, and aircraft performance from at least one of position and altitude over time and a sensor;
- (b) a trajectory evaluator module configured to:
 - (i) determine at least one of expected aircraft position and altitude, expected aircraft trajectory, expected future aircraft position and altitude, and expected aircraft performance from at least one of a lookup register, position and altitude on a flight plan, time since departure, estimated time enroute, and estimated time of arrival;
 - (ii) determine at least one of magnitude, expected magnitude, and rate of change of magnitude of difference between:

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- (1) at least one of said current position and altitude, said current aircraft trajectory, said anticipated future position and altitude, and said aircraft performance; and
 - (2) at least one of said expected position and altitude, said expected trajectory, said expected future position and altitude, and said expected performance;
 - (iii) determine whether said magnitude is the result of at least one of traffic and weather deviation, flight plan change, air traffic control requirement, and arrival change;
 - (c) a configuration evaluator module configured to determine whether at least one of an aircraft configuration error and an emergency exists based at least on one of said magnitude, said expected magnitude, and said rate of change of magnitude;
 - (d) an alternative landing module configured to:
 - (i) determine a level of emergency, where the emergency has been determined, from a hierarchy of emergencies, selected from at least one of land as soon as practicable, land as soon as possible, and land immediately;
 - (ii) continually select, from a hierarchy of selectable landing site preferences and said selected level of emergency, at least one alternative landing site at least one of reachable by said current trajectory and reachable by an available configuration change;
 - (iii) prepare a procedure for safely positioning said aircraft in a landable configuration at the approach of said alternative landing site; and
 - (e) at least one display unit configured to announce at least one of said prepared procedure and in seriatim the elements of said prepared procedure.
67. The system of contemplation 66, wherein said sensor is at least one of a ground-based sensor, satellite-based sensor, space-based sensor, and aircraft sensor.
68. The system of contemplation 66, wherein said alternative landing module is further configured to continually select at least one alternative landing site reachable by at least one of a zero-thrust, partial-thrust, and normal-thrust standard operating procedure.
69. The system of contemplation 66, wherein said display unit is further configured to announce at least one alternative landing site range reachable by the said aircraft, in the form of an ellipse corresponding to the selected emergency level.
70. The system of contemplation 66, wherein said level of emergency is received from at least one of current position, current altitude, current trajectory, and manual selection by pilot.
71. The system of contemplation 66, wherein continual selection of at least one alternative landing site is based at least in part on at least one service available at said alternative landing site.
72. The system of contemplation 69, wherein said display unit is further configured to announce the at least one alternative landing site located within the said range.
73. A flight assistant for determining aircraft flight configuration during a flight, comprising:
- (a) at least one sensor;
 - (b) at least one data repository operably coupled with a computing device;
 - (c) a control system, said control system communicatively connected to said sensor, including at least one processor configured to:
 - (i) determine at least one of the current position, altitude, and configuration of an aircraft in flight;

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- (ii) determine an expected aircraft configuration, and whether an aircraft configuration error exists;
 - (d) at least one flight recorder, configured to collect data determined by said control system;
 - (e) at least one display unit, communicatively connected to said control system, configured to announce an aircraft configuration different from said expected aircraft configuration.
74. The flight assistant of contemplation 73, wherein said data repository is operably coupled to a portable computing device, and is configured to receive preflight data from a second data repository operably coupled to a ground-based computing device.
75. The flight assistant of contemplation 73, wherein said sensor is at least one of a ground-based sensor, satellite-based sensor, space-based sensor, aircraft sensor, and sensor operably coupled to a portable computing device.
76. The flight assistant of contemplation 74, wherein said aircraft sensor is at least one of a GNSS receiver, ADS-B receiver, XM satellite receiver and receiver communicatively connected to a portable computing device.
77. The flight assistant of contemplation 74, wherein said sensor operably coupled to a portable computing device is at least one of a microphone, an accelerometer, a magnetometer, a barometer, and a GNSS receiver.
78. The flight assistant of contemplation 73, further comprising:
- (a) at least one ground-based computing device to which said control system and said data repository are operably coupled;
 - (b) at least one aircraft containing a portable computing device, said portable device having at least one display unit and at least one sensor operably coupled thereto, and said portable device communicatively connected to said ground-based device by stable data link.
79. The flight assistant of contemplation 78, wherein said stable data link includes at least one of: broadband air-to-ground Internet link; satellite-based data link; and high frequency, VHF, UHF, and other ground-based data link.
80. The flight assistant of contemplation 78, wherein said sensor operably coupled is at least one of a microphone, an accelerometer, a magnetometer, and a barometer.
81. A flight assistant for directing a pilot flying an aircraft to a reachable alternative landing site, comprising
- (a) at least one sensor;
 - (b) at least one data repository operably coupled with a computing device;
 - (c) a control system, said control system communicatively connected to said sensor, including at least one processor configured to:
 - (i) determine at least one of the current position, altitude, and trajectory of an aircraft in flight;
 - (ii) determine at least one of an expected position, altitude, and trajectory;
 - (iii) determine the magnitude of difference between said current position, altitude, and trajectory and said expected position, altitude, and trajectory;
 - (iv) determine the possible cause of said difference;
 - (v) continually select at least one alternative landing site reachable by said aircraft, and prepare a procedure for safely positioning said aircraft in landable configuration to approach said site;
 - (d) at least one flight recorder, configured to collect data determined by said control system;

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(e) at least one display unit, communicatively connected to said control system, configured to announce said procedure and in seriatim the elements of said procedure.

82. The flight assistant of contemplation 81, wherein said data repository is operably coupled to a portable computing device, and is configured to receive preflight data from a second data repository operably coupled to a ground-based computing device.

83. The flight assistant of contemplation 81, wherein said sensor is at least one of a ground-based sensor, satellite-based sensor, space-based sensor, aircraft sensor, and sensor operably coupled to a portable computing device.

84. The flight assistant of contemplation 82, wherein said aircraft sensor is at least one of a GNSS receiver, ADS-B receiver, XM satellite receiver, and receiver communicatively connected to a portable computing device.

85. The flight assistant of contemplation 82, wherein said sensor operably coupled to a portable computing device is a GNSS receiver.

86. The flight assistant of contemplation 81, further comprising:

(a) a ground-based computing device to which the said control system and said data repository are operably coupled;

(b) at least one aircraft containing a portable computing device, said portable device having at least one display unit and at least one sensor operably coupled thereto, and said portable device communicatively connected to said ground-based device by stable data link.

87. The flight assistant of contemplation 86, wherein said stable data link includes at least one of: broadband air-to-ground Internet link; satellite-based data link; and high frequency, VHF, UHF, and other ground-based data link.

88. A flight assistant for announcing appropriate available alternative landing sites during a flight, comprising

(a) at least one sensor;

(b) at least one data repository operably coupled with a computing device;

(c) a control system, said control system communicatively connected to said sensor, including at least one processor configured to:

(i) determine at least one of current position, altitude, and trajectory of an aircraft in flight;

(ii) continually select at least one alternative landing site reachable by said aircraft, and prepare a procedure for safely positioning said aircraft in landable configuration to approach said site;

(d) at least one flight recorder, configured to collect data determined by said control system;

(e) at least one display unit, communicatively connected to said control system, configured to announce said procedure and in seriatim the elements of said procedure.

89. The flight assistant of contemplation 88, wherein said data repository is operably coupled to a portable computing device, and is configured to receive preflight data from a second data repository operably coupled to a ground-based computing device.

90. The flight assistant of contemplation 88, wherein said sensor is at least one of a ground-based sensor, satellite-based sensor, space-based sensor, aircraft sensor, and sensor operably coupled to a portable computing device.

91. The flight assistant of contemplation 89, wherein said aircraft sensor is at least one of a GNSS receiver, ADS-B receiver, XM satellite receiver, and receiver communicatively connected to a portable computing device.

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92. The flight assistant of contemplation 89, wherein said sensor operably coupled to a portable computing device is at least one of a microphone, an accelerometer, a magnetometer, a barometer, and a GNSS receiver.

93. The flight assistant of contemplation 88, further comprising:

(a) a ground-based computing device to which the said control system and said data repository are operably coupled;

(b) at least one aircraft containing a portable computing device, said portable device having at least one display unit and at least one sensor operably coupled thereto, and said portable device communicatively connected to said ground-based device by stable data link.

94. The flight assistant of contemplation 93, wherein said stable data link includes at least one of: broadband air-to-ground Internet link; satellite-based data link; and high frequency, VHF, UHF, and other ground-based data link.

95. A flight assistant, comprising:

(a) at least one sensor;

(b) at least one data repository operably coupled with a computing device;

(c) a control system, said control system communicatively connected to said sensor, including one and more processors configured to:

(i) determine at least one of current position, altitude, and trajectory of an aircraft in flight;

(ii) determine at least one of expected position, altitude, and trajectory;

(iii) determine the magnitude of difference between said current position, altitude, and trajectory and said expected position, altitude, and trajectory, and the possible cause of said difference;

(iv) determine an expected aircraft configuration, and whether an aircraft configuration error exists; and

(v) continually select at least one alternative landing site reachable by said aircraft, and prepare a procedure for safely positioning said aircraft in landable configuration to approach said site;

(d) at least one flight recorder, configured to collect data determined by said control system;

(e) at least one display unit, communicatively connected to said control system, configured to announce said procedure and in seriatim the elements of said procedure.

96. The flight assistant of contemplation 95, wherein said display unit is further configured to announce an aircraft configuration different from said expected aircraft configuration.

97. The flight assistant of contemplation 95, wherein said data repository is operably coupled to a portable computing device, and is configured to receive preflight data from a second data repository operably coupled to a ground-based computing device.

98. The flight assistant of contemplation 95, wherein said sensor is at least one of a ground-based sensor, satellite-based sensor, space-based sensor, aircraft sensor, and sensor operably coupled to a portable computing device.

99. The flight assistant of contemplation 98, wherein said aircraft sensor is at least one of a GNSS receiver, ADS-B receiver, XM satellite receiver, and receiver communicatively connected to a portable computing device.

100. The flight assistant of contemplation 98, wherein said sensor operably coupled to a portable computing device is at least one of a microphone, an accelerometer, a magnetometer, a barometer, and a GNSS receiver.

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101. The flight assistant of contemplation 95, further comprising:

- (a) a ground-based computing device to which the said control system and said data repository are operably coupled;
- (b) at least one aircraft containing a portable computing device, said portable device having at least one display unit and at least one sensor operably coupled thereto, and said portable device communicatively connected to said ground-based device by stable data link.

102. The flight assistant of contemplation 101, wherein said stable data link includes at least one of: broadband air-to-ground Internet link; satellite-based data link; and high frequency, VHF, UHF, and other ground-based data link.

103. The flight assistant of contemplation 101, wherein said sensor operably coupled is at least one of a microphone, an accelerometer, a magnetometer, and a barometer.

The invention claimed is:

1. An aircraft, comprising:

- (a) a bus;
- (b) at least one sensor connected to said bus to report one or more of a position, airspeed, attitude, or altitude of the aircraft;
- (c) an autopilot connected to said bus and one or more aircraft controls, the autopilot capable of directing adjustment of one or more of the position, airspeed, attitude, or altitude via the one or more aircraft controls;
- (d) a wireless data-link connected to said bus to receive at least one expected flight path for the aircraft from a portable computing device;
- (e) a memory connected to said bus for storing the received expected flight path;
- (f) at least one processor connected to said bus and coupled with one or more of the sensor, autopilot, wireless data-link, or memory, the at least one processor programmed to:
 - receive at least one of a position, airspeed, altitude, or attitude of the aircraft from the at least one sensor;
 - receive, via a user interface, at least one pilot input including:

- (1) a flight control input to adjust one or more of the position, airspeed, altitude, or attitude, the flight control input associated with a deviation of the aircraft from at least one performance norm;

wherein, when the at least one pilot input is a flight control input, the at least one processor is further programmed to:

correct the deviation via the aircraft controls after one or more notifications are provided and the one or more notifications are unacknowledged, wherein the one or more notifications are more insistent as the deviation at least one of persists or increases;

or

- (2) a pilot request to correct the deviation of the aircraft; wherein, when the at least one pilot input is a pilot request, the at least one processor is further programmed to correct the deviation via the aircraft controls by initiating at least one emergency protocol;

and

create an alternate landing site (ALS) hierarchy by determining a suitability value for each of a plurality of alternate landing sites, wherein each assigned

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suitability value is based on weighting a plurality of factors related to the alternate landing site's suitability for landing;

and

- (g) a display connected to said bus for displaying flight information.

2. The aircraft of claim 1, wherein said deviation correction is a procedure selected from said memory, said procedure wirelessly updated to said memory via the portable computing device.

3. The aircraft of claim 2, wherein said memory is a data repository of a portable computing device configured to receive preflight data via said wireless data-link from a second data repository operably coupled to a ground-based computing device.

4. The aircraft of claim 3, wherein said portable computing device includes at least one of a tablet or a smartphone.

5. The aircraft of claim 1, wherein: said memory further comprises an onboard dataset to receive one or more updates wirelessly from a portable computing device, the one or more updates associated with at least one of a flight plan parameter, aircraft parameter, pilot parameter, or system parameter.

6. The aircraft of claim 1, further comprising:

at least one additional processor associated with at least one of said at least one sensor, weather sensor, traffic sensor, wireless data-link, aircraft performance control, autopilot, or annunciator;

wherein said initiation of the deviation correction is distributed between said at least one processor and said at least one additional processor,

and

wherein said annunciator is at least one of a voice notification, graphic display, audio tone, or haptic notification.

7. The aircraft of claim 1, wherein said display is capable of displaying a turn back for re-landing procedure and the processor is capable of instructing the display to display a re-land procedure when the processor determines the aircraft may re-land based at least in part on at least one of position, airspeed, attitude, or altitude.

8. The aircraft of claim 6, wherein said aircraft performance control includes at least two of a throttle, elevator, rudder, ailerons, flaps, spoiler, or gear.

9. The aircraft of claim 8, wherein said display is capable of displaying at least one of angle of attack or aircraft total energy.

10. The aircraft of claim 9, wherein said display is capable of displaying at least one of aircraft kinetic energy available to convert to potential energy, or aircraft potential energy available to convert to kinetic energy.

11. The aircraft of claim 10, wherein said processor includes a total energy flight controller.

12. The aircraft of claim 11, wherein said display at least one of textually or graphically displays information related to said initiated deviation correction.

13. The aircraft of claim 2, wherein said display at least one of textually or graphically displays one or more items associated with said procedure in serial.

14. The aircraft of claim 13, wherein said display at least one of textually or graphically displays said procedure as a flight director on an artificial horizon.

15. The aircraft of claim 1, wherein said wireless data-link communicatively connects said portable computing device to an aircraft receiver.

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16. The aircraft of claim 1, wherein said wireless data-link is a wireless data-link to aircraft avionics, and at least one of surface-based or space-based radio.

17. The aircraft of claim 2, wherein said wireless data-link receives said expected flight path of the aircraft from said portable computing device during at least one of a preflight, a flight, or a postflight of the aircraft.

18. The aircraft of claim 4, wherein said aircraft parameter further includes at least one of a maintenance status, a catering status or requirement, a potable water and waste status, a temperature or lighting status, or an aircraft requirement.

19. The aircraft of claim 4, wherein said flight plan parameter further includes a path of connected planned flight fixes including at least one of a navigational aid (navaid), a waypoint, or an airway.

20. The aircraft of claim 4, wherein said flight plan parameter further includes at least one of a schedule, a route, a load, a fuel plan, a fix list, an overflight authorization, or a notice to air missions (NOTAM).

21. The aircraft of claim 4, wherein said pilot parameter further includes at least one of a crew member list, a pilot monitor including a flight time limitation, a duty time limitation, a latest allowable takeoff time (LATT), or a pilot schedule.

22. The aircraft of claim 1, wherein said at least one processor is further programmed for communicating with maintenance at least one of before, during, or after a flight segment.

23. The aircraft of claim 4, wherein said at least one processor is further programmed for querying at least one of a ground-to-air, air-to-air, or space-based satellite database for said at least one flight plan parameter, aircraft parameter, pilot parameter, or system parameter.

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24. The aircraft of claim 1, wherein said at least one processor is further programmed for sending a notification to the pilot or ground personnel during a preflight assessment.

25. The aircraft of claim 1, wherein said at least one processor is programmed for informing the pilot of a configuration of the aircraft different from an expected configuration.

26. The aircraft of claim 1, wherein the weighted factors include one or more of

the current position of the aircraft;
an airport landing procedure;

or

at least one additional selection criteria selected from the group including at least one of a weather condition, wind condition, or runway length.

27. The aircraft of claim 1, wherein said processor selects an alternate landing site from the ALS hierarchy based on the determined suitability values.

28. The aircraft of claim 27, wherein said processor creates a flight path from a current position of the aircraft to a landing associated with the selected alternate landing site via at least one of a waypoint or an initial approach fix.

29. The aircraft of claim 28, wherein said processor reports to air traffic control an aircraft diversion.

30. The aircraft of claim 29, wherein said processor is programmed to:

determine an estimated time of arrival at the selected alternate landing site;

display the selected alternate landing site and a route to the selected alternate landing site on the display;

and

continue to determine the suitability of at least the selected alternate landing site, and reroute the aircraft to a new alternate landing site, based on changes to the determined suitability.

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